Electrodynamics

FY549

Lecture Notes for 3rd semester physics and physics and engineering students

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Based on D.J. Griffiths, Introduction to Electrodynamics
and D. Tong, Lectures on Electromagnetism

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How to read these notes:

- Sections in blue boxes contain extra material that is helpful or interesting but not strictly necessary.
- Sections in purple boxes are for the interested reader who wants to go deeper.
- Sections in red boxes are in-class exercises.
- Key equations are marked with orange boxes.
- Text marked in teal refers to slides shown in the lecture.
- You have plenty of space in the left margin for your own comments.

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1 Maxwell's Equations

Literature: Griffiths Chapter 7.3

Maxwell's equations describe how electric and magnetic fields are generated from charges and currents. Historically, they are the culmination of theory to describe experimental results obtained by Faraday, Ampère, and many others. They form the axioms of classical electrodynamics.

Once we have written them down, we could stop right there — they encode (almost) everything in electrodynamics. We will study their consequences throughout this course, with emphasis on theory. Let us first review them and some of the vector calculus we need to understand them.

We write them in differential form and derive the integral form that you should be familiar with from your electrostatics course.

Consider the electric field $\boldsymbol{E} = \left(E_x, E_y, E_z\right)^T$ and the magnetic field $\boldsymbol{B} = \left(B_x, B_y, B_z\right)^T$.

1.1 Gauss's Law

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{x}, t) = \frac{\rho(\boldsymbol{x}, t)}{\varepsilon_0} \tag{1.1}$$

Here, the left-hand side is the *divergence* of $\boldsymbol{E}$, with the nabla operator

$$\boldsymbol{\nabla} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \tag{1.2}$$

so that

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \sum_i \partial_i E_i \equiv \partial_i E_i \tag{1.3}$$

where in the last step we have used the *Einstein summation convention*: repeated indices are summed over.

> **Note on index notation**
>
> We will frequently use index notation where $\partial_i \equiv \frac{\partial}{\partial x_i}$ with $x_1 = x$, $x_2 = y$, $x_3 = z$. Repeated indices (one upper, one lower, or two of the same in this context) are implicitly summed over: $\partial_i E_i \equiv \sum_{i=1}^{3} \partial_i E_i$.
>
> In the second part of the course, it will also become important if we write the indices as subscrits or as superscripts, but this will not bother us now.

Note that $\boldsymbol{E}$ is a *vector field*: its strength and direction may vary with position and time. The divergence measures whether $\boldsymbol{E}$ has sources or sinks at a given point.

An example of a vector field with divergence is the electric field of a point charge, see the slides for an example. Another example is the velocity field of a fluid flowing out of a faucet. We will explore this further in the exercises.

Gauss's law states: the source of divergence of $\boldsymbol{E}$ is the charge density:

- ρ: charge density (charge per unit volume)
- ε_0: vacuum permittivity (ability of $\boldsymbol{E}$ to permeate vacuum)

$$\varepsilon_0 = 8.85 \times 10^{-12}\mathrm{C}^2\mathrm{N}^{-1}\mathrm{m}^{-2} \quad \left(\varepsilon_0 = \frac{10^7}{4\pi c^2}\right) \tag{1.4}$$

1.1.1 Physical Interpretation: Integral Form

Integrate both sides of Eq. (1.1) over a volume V:

$$\underbrace{\int_V (\boldsymbol{\nabla}\cdot\boldsymbol{E})\mathrm{d}V}_{\text{Gauss' theorem}} = \frac{1}{\varepsilon_0}\int_V \rho\,\mathrm{d}V \tag{1.5}$$

$$\text{Gauss' theorem} \rightarrow = \oint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{a}$$

The right-hand side gives the total charge Q enclosed in V.

> **Note on notation for volume element $\mathrm{d}V$**
>
> Throughout the course, we will use $\mathrm{d}V$ interchangeably with d^3x to denote the volume element (Griffith uses $\mathrm{d}\tau$). In Cartesian coordinates, this is $\mathrm{d}^3x = \mathrm{d}V = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$.

Applying the *divergence theorem* (Gauss's theorem) to the left-hand side:

$$\oint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{a} = \frac{Q_{\mathrm{enc}}}{\varepsilon_0} \tag{1.6}$$

where S is the closed surface enclosing V and $\mathrm{d}\boldsymbol{a}$ points outward. The left-hand side is the *electric flux* through S.

The larger the charge inside V, the larger the flux of $\boldsymbol{E}$ through the closed surface S.

> **Note on flux**
>
> Flux is not flow. Flow is the average normal component of the field times the surface area.

Importantly, the *shape* of S does not matter — only the total charge enclosed in S does, see Figure 1.1.

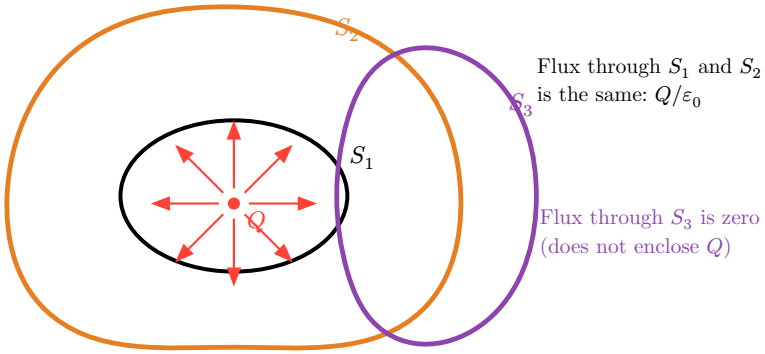


Figure 1.1: The flux through S_1 and S_2 is the same ($= Q/\varepsilon_0$), since both enclose Q. The flux through S_3 is zero: as much flux enters V_3 as leaves it, since S_3 does not enclose the charge.

1.2 Gauss's Law for Magnetism

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0 \tag{1.7}$$

This is analogous to Gauss's law in Eq. (1.1), but for the $\boldsymbol{B}$ field. It states that there are no magnetic charges (no magnetic monopoles — at least, we have not found them yet).

The net magnetic flux through any closed surface is zero: any outward flux is balanced by inward flux. If you like to visualize the fields with field lines: they are always *closed* for magnetic fields.

This is also true in the static case (magnetostatics).

1.3 Faraday's Law of Induction

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{1.8}$$

This is Faraday's law of induction. Note that this is a *vector* equation (in contrast to Gauss's law), so both left- and right-hand sides have three components.

The curl is computed as

$$\boldsymbol{\nabla} \times \boldsymbol{E} = \det \begin{pmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{pmatrix} = \begin{pmatrix} \partial_y E_z - \partial_z E_y \\ -\partial_x E_z + \partial_z E_x \\ \partial_x E_y - \partial_y E_x \end{pmatrix} \tag{1.9}$$

The Levi-Civita tensor

In index notation, using the Levi-Civita tensor:

$$(\boldsymbol{\nabla} \times \boldsymbol{E})_i = \varepsilon_{ijk} \partial_j E_k \tag{1.10}$$

The Levi-Civita tensor ε_{ijk} is defined as

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{even permutation of } 1,2,3 \\ -1 & \text{odd permutation of } 1,2,3 \\ 0 & \text{for repeated indices} \end{cases} \tag{1.11}$$

An *even* permutation is one obtained by cyclic rotation: $123 \to 231 \to 312$. You can think of a cyclic rotation as a rotation of a triangle with the numbers 1, 2, 3 at the corners. An *odd* permutation is obtained by swapping two indices: $123 \to 213 \to 321 \to 132 \to 231 \to 312$.

1.3.1 Physical Interpretation: Integral Form

Integrate both sides over a surface S:

Holding S constant in time

$$\underbrace{\int_S (\boldsymbol{\nabla} \times \boldsymbol{E}) \cdot \mathrm{d}\boldsymbol{a}}_{\text{Stokes' theorem}} = -\int_S \frac{\partial \boldsymbol{B}}{\partial t} \cdot \mathrm{d}\boldsymbol{a} = \underbrace{-\frac{\mathrm{d}}{\mathrm{d}t} \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{a}}_{\text{Magnetic flux } \Phi_B \text{ through surface } S} \tag{1.12}$$

$$= \oint_C \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l}$$

where C is the closed curve bounding S, and $\Phi_B = \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{a}$ is the magnetic flux through the surface. The last equality only holds when S is fixed in time, i.e. the surface does not move or deform as time progresses. In that case, the time derivative can be taken outside of the integral.

Collecting results gives the integral form of Faraday's law:

$$\oint_C \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} \tag{1.13}$$

Geometrical interpretation of Stokes' theorem

The curl measures the "twist" of a vector field (a whirlpool is a region with high curl). The integral of the flux of the curl through a surface represents the total "amount of swirl". Stokes' theorem says this equals the circulation — the flow along the boundary. This is illustrated in Figure 1.2. Crucially, the integral depends only on the boundary C, not on the surface S itself.

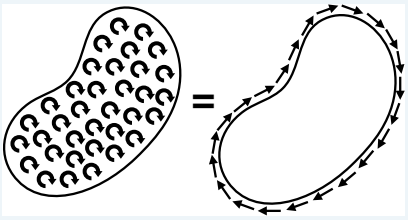


Figure 1.2: Small circulation loops covering the surface cancel along shared interior edges, leaving only the circulation along the outer boundary C.

The physical content: a time-varying magnetic flux through a surface induces an electric field along the closed path around the surface.

If you place a wire along this path, current will flow as $\boldsymbol{E}$ exerts force on the charge carriers inside the wire. This is the *electromotive force* (emf):

$$\mathcal{E} = \oint \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{l} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} \tag{1.14}$$

where $\boldsymbol{f}$ is the force per unit charge responsible for pushing the charge around the circuit.

Lenz's law

The minus sign in Faraday's law encodes Lenz's law: the emf opposes the change in magnetic flux.

This was Faraday's experimental breakthrough: three experiments involving a wire and a magnetic field all yielded a current through the wire, see the slides for the experimental setups. In experiment (a), the wire moves — this is a case of *motional emf*, where the relevant force is the Lorentz force $\boldsymbol{f} = \boldsymbol{v} \times \boldsymbol{B}$. Experiment (b) gives the same result — only relative motion matters. In experiment (c), the wire is not moving ($\boldsymbol{v} = 0$), so who does the work? The answer: the electric field that is *induced* by the changing magnetic field. This was Faraday's great insight.

1.4 Ampère's Law with Maxwell's Addition

$$\underbrace{\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J}}_{\text{Ampère's law}} + \underbrace{\mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}}_{\text{Maxwell's addition}} \tag{1.15}$$

The first term is *Ampère's law*, and the second is *Maxwell's addition* (the displacement current term).

Here:

- $\boldsymbol{J}$: current density — the amount of current passing through a surface per unit time. The number of charges passing per second through an area element $\mathrm{d}\boldsymbol{S}$ is $\boldsymbol{J} \cdot \hat{\boldsymbol{n}}\, \mathrm{d}S$. Units: $[\boldsymbol{J}] = \mathrm{A} \cdot \mathrm{m}^{-2}$.
- μ_0: permeability of free space, $\mu_0 \approx 4\pi \times 10^{-7}$ N A^{-2}.

Definition of μ_0

μ_0 is defined through the force of attraction between two infinite wires with infinitesimal cross section, 1 m apart, carrying 1 A current in the same direction. In natural units: $\mu_0/(4\pi) = \alpha/(2\pi) \cdot \hbar/(e^2 c)$. Here, α is the fine-structure constant, $\hbar$ is the reduced Planck constant (see your quantum mechanics class!), e is the electric charge and c is the speed of light.

1.4.1 Physical Interpretation: Integral Form (Static Case)

In the static case ($\frac{\partial \boldsymbol{E}}{\partial t} = 0$), integrate over a surface S and apply Stokes' theorem:

$$\int_S (\boldsymbol{\nabla} \times \boldsymbol{B}) \cdot \mathrm{d}\boldsymbol{a} = \oint_C \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = \mu_0 \int_S \boldsymbol{J} \cdot \mathrm{d}\boldsymbol{a} \tag{1.16}$$

The right-hand side is μ_0 times the total current through S. The direction of C is determined by the right-hand rule with respect to the surface normal $\hat{\boldsymbol{n}}$.

The magnetic field along C is determined by the total current passing through S.

1.4.2 Why Is Maxwell's Addition Necessary?

The displacement current term ensures the *consistency* of Maxwell's equations.

Take the divergence of Faraday's law Eq. (1.8):

$$\underbrace{\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{E})}_{=0 \text{ (divergence of curl always zero)}} = -\frac{\partial}{\partial t} \left(\underbrace{\boldsymbol{\nabla} \cdot \boldsymbol{B}}_{=0 \text{ (2nd Maxwell equation)}} \right) = 0 \quad \checkmark \tag{1.17}$$

Now take the divergence of Ampère's law in Eq. (1.15) *without* Maxwell's addition:

$$\underbrace{\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B})}_{=0} = \mu_0 \boldsymbol{\nabla} \cdot \boldsymbol{J} \tag{1.18}$$

This would require $\boldsymbol{\nabla} \cdot \boldsymbol{J} = 0$ always, which is not true in general (e.g. when charges leave a volume).

With Maxwell's addition:

$$0 = \mu_0 \boldsymbol{\nabla} \cdot \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} (\boldsymbol{\nabla} \cdot \boldsymbol{E}) = \mu_0 \boldsymbol{\nabla} \cdot \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \frac{\rho}{\varepsilon_0} \tag{1.19}$$

which gives the *continuity equation*:

$$\boldsymbol{\nabla} \cdot \boldsymbol{J} + \frac{\partial \rho}{\partial t} = 0 \tag{1.20}$$

This is an important result that we will use often. It expresses *conservation of charge*: a change in charge density in time is associated with a divergence in the current (charges flowing in or out of the volume).

In integral form:

$$\oint_S \boldsymbol{J} \cdot \mathrm{d}\boldsymbol{a} = -\frac{\mathrm{d}Q}{\mathrm{d}t} \tag{1.21}$$

1.5 Summary: Maxwell's Equations

These are Maxwell's *microscopic* equations: ρ and $\boldsymbol{J}$ include *all* charges and currents, also "internal" ones like bound charges in atoms and molecules. They are a system of coupled first-order partial differential equations.

In-class exercise

- How many equations do we have (counting components)?
- Why didn't we write them in Cartesian coordinates?

Here they are in one place (combining Eq. (1.1), Eq. (1.7), Eq. (1.8), and Eq. (1.15)):

$$\begin{aligned} \boldsymbol{\nabla} \cdot \boldsymbol{E} &= \frac{\rho}{\varepsilon_0} && \text{(i)} \\ \boldsymbol{\nabla} \cdot \boldsymbol{B} &= 0 && \text{(ii)} \\ \boldsymbol{\nabla} \times \boldsymbol{E} &= -\frac{\partial \boldsymbol{B}}{\partial t} && \text{(iii)} \\ \boldsymbol{\nabla} \times \boldsymbol{B} &= \mu_0 \boldsymbol{J} + + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} && \text{(iv)} \end{aligned} \tag{1.22}$$

1.5.1 Consequences

Suppose you switch on a current $\boldsymbol{J}$:

- $\boldsymbol{J}$ varies in time
- generates a time-variable $\boldsymbol{B}$ field through Ampère's law (iv)
- generates a time-variable $\boldsymbol{E}$ field through Faraday's law (iii)
- which in turn modifies $\boldsymbol{B}$, and so on…

This sounds like an oscillation that propagates — Maxwell's equations are the source of *electromagnetic waves*. We will get to that in the coming weeks.

2 Maxwell's Equations in Matter

Literature: Griffiths Ch. 7.3 (cont'd)

Maxwell's equations Eq. (1.22) (i)–(iv) are also valid in the presence of matter. However, ρ and $\boldsymbol{J}$ then include *all* charges and currents, also those that are "bound" in matter — over which we have no direct influence.

It is convenient to rewrite Maxwell's equations in terms of new fields $\boldsymbol{D}$ and $\boldsymbol{H}$ that distinguish between *free* charges and currents (ρ_f, $\boldsymbol{J}_f$) that we can experimentally control, and *bound* charges and currents (ρ_b, $\boldsymbol{J}_b$).

The bound charges ρ_b and currents $\boldsymbol{J}_b$ are produced as $\boldsymbol{E}$ and $\boldsymbol{B}$ *polarize* and *magnetize* a medium (e.g. electrons and protons in a material get slightly shifted against each other).

At the microscopic level, $\boldsymbol{E}$ and $\boldsymbol{B}$ are extremely complex: close to particles, fields become extremely large; further away, closer to another particle, the field might flip direction; at finite temperature, particles undergo thermal fluctuations and fields change continuously.

We focus on the *macroscopic* level — averaged over large enough regions such that we are not affected by microscopic effects. This approach of finding relevant information using an *effective* (averaged) description instead of accounting for all microscopic physics is ubiquitous in physics.

2.1 Dielectrics and Polarization

In-class exercise

What happens inside matter when you apply an electric field?

Answer: it depends on the material.

In a **conductor**, there are lots of free electrons. Any electric field is compensated very fast, so $\boldsymbol{E} = 0$ inside the conductor. (There will still be a current when you hook up a battery, but inside the wire $\boldsymbol{E} = 0$. We will look at conductors later.)

In **dielectrics** (non-conducting materials), there are no free charge carriers — electrons are bound in atoms. When you apply an electric field, the positions of electrons (negatively charged) and nuclei (positively charged) shift (in which direction will the particles shift with respect to each other?), leading to small *dipole moments*:

$$\boldsymbol{p} = q \cdot \boldsymbol{d} = \alpha \cdot \boldsymbol{E} \tag{2.1}$$

where α is the *polarizability* (a scalar or tensor).

Molecules like water that already have a dipole moment will experience a torque and align with the $\boldsymbol{E}$ field.

The result: lots of dipoles pointing along $\boldsymbol{E}$ — the material gets *polarized*. A convenient measure is the **polarization**:

$\boldsymbol{P}$ = dipole moment per unit volume

2.1.1 Bound Charges

We can also understand polarization in terms of *bound charges* or the bound charge density ρ_b.

When the induced dipoles all align, the interior charges cancel in pairs, as depected in Figure Figure 2.1. The net effect is a surface charge — what we call *bound charge*.

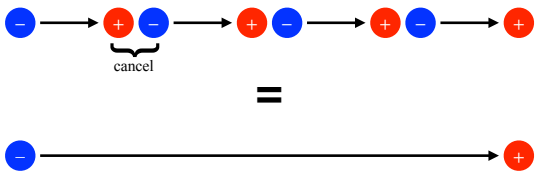


Figure 2.1: Illustration of aligned dipoles inside a material. The inner charges cancel leaving effectively a bound surface charge.

One can show (Griffiths Ch. 4) that bound charges and polarization are connected through

$$-\boldsymbol{\nabla} \cdot \boldsymbol{P} = \rho_b \tag{2.2}$$

The total charge density is

$$\rho = \rho_b + \rho_f \tag{2.3}$$

where ρ_f is the *free* charge density that we can manipulate in the lab.

Example on Polarzation

Imagine a sphere with a uniform distribution of positive and negative charges, as shown in Figure 2.2.

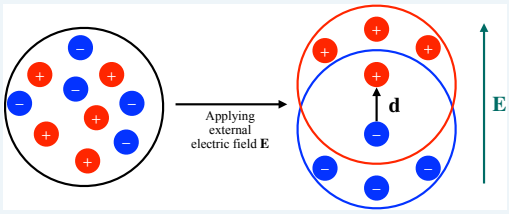


Figure 2.2: When an electric field is applied, the positive charges move in the direction of $\boldsymbol{E}$ while the negative charges move in the opposite direction. As a result, the spheres are shifted against each other.

The electric field in the overlaping region is given by

$$\boldsymbol{E}_{\text{overlap}} = -\frac{1}{4\pi\varepsilon_0}\frac{q \cdot \boldsymbol{d}}{R^3}, \tag{2.4}$$

where R is the radius of the sphere (believe me or check in Griffiths Ch. 2 and 4).

The electric field can be expressed in terms of the polarization $\boldsymbol{P}$ replacing the dipole moment $\boldsymbol{p}$ (with charge q the net positive charge) with the polarization $\boldsymbol{P}$:

$$\boldsymbol{p} = q \cdot \boldsymbol{d} = \boldsymbol{P} \cdot V = \boldsymbol{P} \cdot \left(\frac{4}{3}\pi R^3\right), \tag{2.5}$$

which is uniform accross the sphere. Plugging in:

$$\Rightarrow \boldsymbol{E}_{\text{overlap}} = -\frac{1}{3\varepsilon_0}\boldsymbol{P}. \tag{2.6}$$

Outside the sphere, for distances $|\boldsymbol{r}| \gg |\boldsymbol{d}|$ we have a dipole with potential

$$U = \frac{1}{4\pi\varepsilon_0}\frac{\boldsymbol{p}\cdot\boldsymbol{r}}{|\boldsymbol{r}|^3} \tag{2.7}$$

The total electric field is then the sum of the field that caused the polarization in the first place and the electric field produced by the polarizatio, i.e. the shifting of the spheres. The result is shown in Figure 2.3.

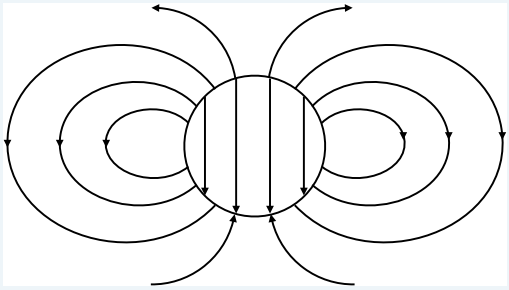


Figure 2.3: The total electric field from the uniform polarization. The field is uniform inside the sphere and behaves like a dipole outside.

2.1.2 The Electric Displacement $\boldsymbol{D}$

We define the *electric displacement* $\boldsymbol{D}$ whose divergence equals the free charge density:

$$\boldsymbol{\nabla} \cdot \boldsymbol{D} = \rho_f \tag{2.8}$$

This is like Gauss's law but only for free charges and without ε_0. Starting from the divergence of the total $\boldsymbol{E}$ field:

$$\begin{aligned} \boldsymbol{\nabla} \cdot \boldsymbol{E} &= \frac{1}{\varepsilon_0}(\rho_f + \rho_b) \\ &= \frac{1}{\varepsilon_0}\boldsymbol{\nabla}\cdot\boldsymbol{D} - \frac{1}{\varepsilon_0}\boldsymbol{\nabla}\cdot\boldsymbol{P} \\ &= \frac{1}{\varepsilon_0}\boldsymbol{\nabla}\cdot(\boldsymbol{D} - \boldsymbol{P}) \end{aligned} \tag{2.9}$$

This gives us the definition of $\boldsymbol{D}$:

$$\boldsymbol{D} = \varepsilon_0\boldsymbol{E} + \boldsymbol{P} \tag{2.10}$$

This is actually the definition of $\boldsymbol{D}$, motivated here from consistency with Maxwell's equations.

Next lecture: we will see what happens in matter when you apply a magnetic field, and rewrite Maxwell's equations fully in terms of free charges and currents.

2.2 Magnetization

Last lecture we looked at the electric field in dielectrics and introduced the *electric displacement*

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P}, \quad \nabla \cdot \boldsymbol{D} = \rho_f. \tag{2.11}$$

We saw that dielectrics become polarized when an external field is applied (note that $\boldsymbol{E}$ above is the *total* field).

We now ask the analogous question for magnetic fields.

> **In-class exercise**
>
> What happens when you place a piece of material in a magnetic field?

Again, it depends on the type of material.

Some materials, like iron, become magnets themselves and can even stay magnetized once the external field is switched off. Such substances (e.g., iron, nickel, cobalt) are called **ferromagnets** — we will not discuss them further in this course.

But other materials are also affected by external fields, and are divided into:

- **Paramagnets**: acquire a magnetization *parallel* to $\boldsymbol{B}$.
- **Diamagnets**: acquire a magnetization *antiparallel* to $\boldsymbol{B}$.

Compared to ferromagnets, the magnetization is much smaller (you can't pick up wood with a magnet).

Let's look at paramagnets and diamagnets more closely.

2.2.1 Microscopic Origin: Magnetic Dipoles

In fact, all magnetic phenomena are caused by tiny currents:

- electrons that have *spin* (imagine them as spinning charged spheres, even though this picture is not correct in a quantum-mechanical sense),
- electrons that *orbit* around the nucleus.

Both phenomena generate current loops that we can treat as magnetic dipoles, sketched in Figure 2.4. The dipole moment is given by

$$\boldsymbol{m} = I\boldsymbol{A} \tag{2.12}$$

where I is the current circulating around the loop of area A, and $\boldsymbol{A} = A\hat{\boldsymbol{n}}$ points along the loop's normal (direction set by the right-hand rule).

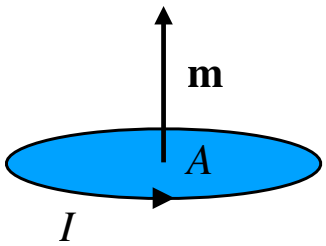


Figure 2.4: A magnetic dipole moment $\boldsymbol{m}$ is generated by a current loop. The direction of $\boldsymbol{m}$ is given by the right-hand rule.

2.2.2 Paramagnetism

Without an external field, these dipoles point in random directions. Once you apply a field, the dipoles experience a torque that will line them up parallel to the field. This torque accounts for paramagnetism.

You might expect this to be a universal phenomenon, since every spinning electron constitutes a dipole.

However, in quantum mechanics we have the **Pauli exclusion principle**, which states that two electrons (or more generally, *fermions* — particles with half-integer spin) cannot occupy the same quantum state. This tends to lock electrons with opposite spin together in atoms, so that their torques neutralize.

As a result, paramagnetism occurs mostly in atoms or molecules with an *odd* number of electrons. But even in such materials, random thermal fluctuations compete with the ordering of the dipoles.

2.2.3 Diamagnetism

Orbits of electrons also constitute tiny current loops, but these are randomly oriented. In the presence of a field, an orbiting electron experiences an additional Lorentz force that will either speed it up or slow it down. This changes its dipole moment *antiparallel* to the field. This is diamagnetism.

Diamagnetism is typically weaker than paramagnetism, and mainly observed in atoms/molecules with an *even* number of electrons (where paramagnetism is largely cancelled by the Pauli exclusion principle).

2.2.4 The Magnetization $\boldsymbol{M}$

Similarly to polarization, we can thus define the **magnetization**:

$$\boldsymbol{M} = \text{magnetic dipoles per unit volume}$$

The magnetization is connected to a **bound current density** $\boldsymbol{J}_b$ (produced by the electron spins and orbits) through

$$\boldsymbol{J}_b = \boldsymbol{\nabla} \times \boldsymbol{M} \tag{2.13}$$

See the slides for an illustration.

2.2.5 The Auxiliary Field $\boldsymbol{H}$

The currents we control directly — e.g. by increasing a voltage — are called **free currents**, and they produce a field $\boldsymbol{H}$, analogous to the electric displacement $\boldsymbol{D}$ in the electrostatic case:

$$\boldsymbol{\nabla} \times \boldsymbol{H} = \boldsymbol{J}_f. \tag{2.14}$$

In practice, $\boldsymbol{H}$ is used far more frequently than $\boldsymbol{D}$. Usually $\boldsymbol{H}$ is defined through

$$\boldsymbol{H} = \frac{1}{\mu_0}\boldsymbol{B} - \boldsymbol{M} \tag{2.15}$$

Note

Unlike $\boldsymbol{B}$, the field $\boldsymbol{H}$ is not generally divergence-free: $\boldsymbol{\nabla} \cdot \boldsymbol{H} = -\boldsymbol{\nabla} \cdot \boldsymbol{M}$, which need not vanish.

2.2.6 The Polarization Current

So far, the relation $\boldsymbol{J}_b = \boldsymbol{\nabla} \times \boldsymbol{M}$ was only established for the electro-/magnetostatic case (only in this case we have $\boldsymbol{J} = \boldsymbol{J}_f + \boldsymbol{J}_b$). In the time-dependent case, we get an additional contribution to the current density: a change in polarization also drives a **polarization current** $\boldsymbol{J}_p$:

$$\boldsymbol{J}_p = \frac{\partial \boldsymbol{P}}{\partial t} \tag{2.16}$$

What's the physical meaning of this current? It has nothing to do with the bound current $\boldsymbol{J}_b$; $\boldsymbol{J}_b$ is associated with the magnetization of the material which involves the the spin and orbital motion of electrons. In contrast, the polarization current $\boldsymbol{J}_p$ is the result of the linear motion of charge as the polarization changes. For example: if $\boldsymbol{P}$ points in the $+x$ direction and increases, positive charges will move slightly in the $+x$ direction while negative charges move slightly in the $-x$ direction. The cumulative effect of these tiny shifts is the current $\boldsymbol{J}_p$.

This current is essential to ensure conservation of bound charge. We can check this against the continuity equation (Eq. (1.20)):

$$\boldsymbol{\nabla} \cdot \boldsymbol{J}_p = \boldsymbol{\nabla} \cdot \frac{\partial \boldsymbol{P}}{\partial t} = \frac{\partial}{\partial t}(\boldsymbol{\nabla} \cdot \boldsymbol{P}) = -\frac{\partial \rho_b}{\partial t}, \tag{2.17}$$

where we used the definition of the polarization current (Eq. (2.16)). In the last step we used the relation between polarization and bound charge, Eq. (2.2). This is exactly the continuity equation for the bound charge density ρ_b, as required.

Does a changing magnetization lead to a charge accumulation?

No — a changing magnetization changes the bound current, but

$$\boldsymbol{\nabla} \cdot \boldsymbol{J}_b = \underbrace{\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{M})}_{=0 \text{ (divergence of curl always zero)}}, \tag{2.18}$$

so a changing $\boldsymbol{M}$ redistributes current but never accumulates net charge anywhere. (Note that this is not to be confused with the polarization current $\boldsymbol{J}_p$ above, which is associated with the polarization $\boldsymbol{P}$, not the magnetization.)

2.2.7 Ampère's Law in Matter

The total current density now has three contributions:

$$\begin{aligned} \boldsymbol{J} &= \boldsymbol{J}_f + \boldsymbol{J}_b + \boldsymbol{J}_p \\ &= \boldsymbol{J}_f + \boldsymbol{\nabla} \times \boldsymbol{M} + \frac{\partial \boldsymbol{P}}{\partial t} \end{aligned} \tag{2.19}$$

Let's write Ampère's law with Maxwell's addition, Eq. (1.22) (iii), in terms of $\boldsymbol{H}$ and $\boldsymbol{D}$ instead of $\boldsymbol{B}$ and $\boldsymbol{J}$. Starting from

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}, \tag{2.20}$$

we substitute $\boldsymbol{B} = \mu_0(\boldsymbol{H} + \boldsymbol{M})$ (from Eq. (2.15)) on the left-hand side, and Eq. (2.19) on the right-hand side:

$$\underbrace{\mu_0 \nabla \times (\boldsymbol{H} + \boldsymbol{M})}_{\text{using definition of } \boldsymbol{H}} = \underbrace{\mu_0 \left(\boldsymbol{J}_f + \nabla \times \boldsymbol{M} + \frac{\partial \boldsymbol{P}}{\partial t} \right)}_{\text{using definition of } \boldsymbol{J}} + \underbrace{\mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}}_{\text{will combine into } \boldsymbol{D}} \tag{2.21}$$

The $\mu_0 \nabla \times \boldsymbol{M}$ term appears on both sides and cancels. Dividing by μ_0:

$$\nabla \times \boldsymbol{H} = \boldsymbol{J}_f + \frac{\partial \boldsymbol{P}}{\partial t} + \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}. \tag{2.22}$$

Using the definition of $\boldsymbol{D}$, Eq. (2.10), the last two terms combine into $\frac{\partial \boldsymbol{D}}{\partial t}$:

$$\nabla \times \boldsymbol{H} = \boldsymbol{J}_f + \frac{\partial \boldsymbol{D}}{\partial t} \tag{2.23}$$

Faraday's law Eq. (1.22) (iii) and Gauss's law for magnetism Eq. (1.22) (ii) stay unchanged — they don't involve currents or a free/bound charge distinction.

2.3 Macroscopic Maxwell's Equations

We arrive at an alternative formulation of Maxwell's equations, valid inside matter, sometimes called the **macroscopic** Maxwell's equations:

$$\begin{aligned} \nabla \cdot \boldsymbol{D} &= \rho_f && \text{(i)} \\ \nabla \cdot \boldsymbol{B} &= 0 && \text{(ii)} \\ \nabla \times \boldsymbol{E} &= -\frac{\partial \boldsymbol{B}}{\partial t} && \text{(iii)} \\ \nabla \times \boldsymbol{H} &= \boldsymbol{J}_f + \frac{\partial \boldsymbol{D}}{\partial t} && \text{(iv)} \end{aligned} \tag{2.24}$$

plus the constitutive relations

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P} \quad \text{and} \quad \boldsymbol{H} = \frac{1}{\mu_0} \boldsymbol{B} - \boldsymbol{M}. \tag{2.25}$$

Alternatively, one can stick with the *microscopic* Maxwell's equations (Eq. (1.22)) in terms of $\boldsymbol{E}$ and $\boldsymbol{B}$ only, as long as the sources ρ and $\boldsymbol{J}$ include *all* charges and currents — free and bound.

2.4 Linear Media and Susceptibility

Often, the displacement and magnetizing fields are written in terms of the **permittivity** ε and **permeability** μ of the material:

$$\boldsymbol{D} = \varepsilon \boldsymbol{E}, \quad \boldsymbol{H} = \frac{1}{\mu} \boldsymbol{B}. \tag{2.26}$$

ε and μ can be expressed in terms of the electric and magnetic **susceptibilities** χ_e, χ_m:

$$\varepsilon = \varepsilon_0(1+\chi_e), \quad \mu = \mu_0(1+\chi_m). \tag{2.27}$$

The dielectric constant

The (relative) **dielectric constant** is

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = 1 + \chi_e. \tag{2.28}$$

χ_e and χ_m are dimensionless numbers, see the slides for typical values.

Linear media have a linear relation between polarization/magnetization and the fields:

$$\boldsymbol{P} = \varepsilon_0 \chi_e \boldsymbol{E}, \quad \boldsymbol{M} = \chi_m \boldsymbol{H}. \tag{2.29}$$

- For **homogeneous** materials, ε and μ are constants; for **inhomogeneous** materials, they are functions of position.
- For **isotropic** materials, ε and μ are scalars; for **anisotropic** materials (like crystals), they are tensors.

Interlude: Connection to my research on axions

Axions could solve two pressing problems in particle physics and cosmology:

- The nature of *dark matter*
- The non-observation of an electric dipole moment of the neutron (the *strong CP problem*)

If axions existed, they could couple to photons in the presence of $\boldsymbol{B}$-fields. If axions are dark matter, they would surround us and *modify Maxwell's equations*. Gauss's law would get an additional source term from the spatial gradient of the axion field:

$$\rho \to \rho - \frac{g_{a\gamma}}{\mu_0 c} \boldsymbol{B} \cdot \boldsymbol{\nabla} a \tag{2.30}$$

where $g_{a\gamma}$ is the axion-photon coupling constant and a the axion field. Ampère's law similarly gets modified:

$$\boldsymbol{J}_f + \frac{\partial \boldsymbol{D}}{\partial t} \to \boldsymbol{J}_f + \frac{\partial \boldsymbol{D}}{\partial t} + \frac{g_{a\gamma}}{\mu_0 c}\left[\frac{\partial a}{\partial t}\boldsymbol{B} - \boldsymbol{E} \times \boldsymbol{\nabla} a\right] \tag{2.31}$$

Even without charges and currents, you could generate electromagnetic fields through axion dark matter! Typically, the spatial variations of the axion field are negligible, so the gradient term can be dropped leaving the axion current as the new dominating source term,

$$\boldsymbol{J}_a = \frac{g_{a\gamma}}{\mu_0 c}\frac{\partial a}{\partial t}\boldsymbol{B}. \tag{2.32}$$

The axion field oscillates at a frequency $\omega_a = m_a \frac{c^2}{\hbar}$ set by the axion mass m_a, which is unknown but expected to be very small. This leads to an oscillating

axion current in the presence of a static magnetic field, which is the basis for many axion dark matter searches also here at SDU.

2.5 Boundary Conditions

Starting from Maxwell's equations in matter, we now ask: what happens at the boundary between two media with different permittivities ε_1, ε_2 and permeabilities μ_1, μ_2? For now we only look at the general boundary conditions that the electric and magnetic fields must fulfil.

We will soon see that electromagnetic fields are propagating waves, and that these boundary conditions lead to reflection, transmission, and refraction (e.g. light passing from air into glass or water).

In general, $\boldsymbol{E}$, $\boldsymbol{D}$, $\boldsymbol{B}$, and $\boldsymbol{H}$ are all discontinuous at the boundary between two media.

2.5.1 Perpendicular Components: $\boldsymbol{D}$ and $\boldsymbol{B}$

The form of the discontinuity can be derived from Maxwell's equations in integral form. Starting from Gauss's law, (i) in Eq. (2.24), and applying the divergence theorem:

$$\int_V (\boldsymbol{\nabla} \cdot \boldsymbol{D})\, \mathrm{d}V = \oint_S \boldsymbol{D} \cdot \mathrm{d}\boldsymbol{a} = Q_{\text{f, enc}}, \tag{2.33}$$

where $Q_{\text{f, enc}}$ is the total free charge enclosed by the surface S. This holds for *any* closed surface — including a thin "pillbox" straddling the boundary between the two media, see Figure 2.5.

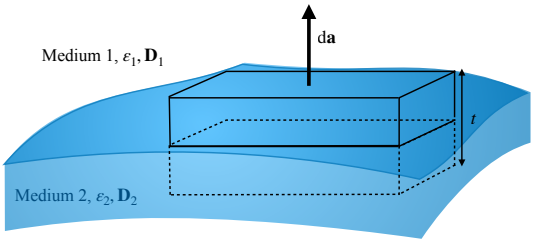


Figure 2.5: A thin pillbox straddling the boundary between two media. The outward normal vectors $\hat{\boldsymbol{n}}$ point into medium 1 and medium 2, respectively.

Consider the limit where the pillbox thickness $t \to 0$: the integral only gets a contribution from the top and bottom faces (each of area a, with outward normal vectors $\pm\hat{\boldsymbol{n}}$ pointing into medium 1 and medium 2 respectively):

$$\oint_S \boldsymbol{D} \cdot \mathrm{d}\boldsymbol{a} = (\boldsymbol{D}_1 \cdot \hat{\boldsymbol{n}})a - (\boldsymbol{D}_2 \cdot \hat{\boldsymbol{n}})a = Q_{\text{f, enc}} = \sigma_f \cdot a, \tag{2.34}$$

where σ_f is the free surface charge density. Dividing by a:

$$D_1^{\perp} - D_2^{\perp} = \sigma_f \tag{2.35}$$

where $\perp$ denotes the component perpendicular to the boundary. For linear media (Eq. (2.26)) with no free surface charge:

$$\varepsilon_1 E_1^{\perp} = \varepsilon_2 E_2^{\perp}. \tag{2.36}$$

Similarly, from Gauss's law for magnetism, (ii) in Eq. (2.24), we have $\oint_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{a} = 0$. The same pillbox argument then gives:

$$B_1^\perp = B_2^\perp \tag{2.37}$$

The perpendicular component of $\boldsymbol{B}$ is always continuous.

2.5.2 Parallel Components: $\boldsymbol{E}$ and $\boldsymbol{H}$

Now we turn to the two Maxwell equations that involve curls. Using Stokes' theorem, for a closed path C around S, we find the integral forms of (iii) and (iv) in Eq. (2.24),

$$\int_S (\boldsymbol{\nabla} \times \boldsymbol{E}) \cdot \mathrm{d}\boldsymbol{a} = \oint_C \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l}, \tag{2.38}$$

and similarly

$$\int_S (\boldsymbol{\nabla} \times \boldsymbol{H}) \cdot \mathrm{d}\boldsymbol{a} = \oint_C \boldsymbol{H} \cdot \mathrm{d}\boldsymbol{l} = I_{f,\,\mathrm{enc}} + \frac{\mathrm{d}}{\mathrm{d}t} \int_S \boldsymbol{D} \cdot \mathrm{d}\boldsymbol{a}, \tag{2.39}$$

where $I_{f,\,\mathrm{enc}}$ is the total free current enclosed by the loop.

Consider a thin rectangular Amperian loop with sides of length Δl straddling the boundary, with height h, see Figure Figure 2.6. In the limit $h \to 0$, the area S enclosed by the loop vanishes, so the flux term on the right-hand side of Faraday's law vanishes:

$$\boldsymbol{E}_1 \cdot \Delta\boldsymbol{l} - \boldsymbol{E}_2 \cdot \Delta\boldsymbol{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{a} \overset{h \to 0}{\to} 0, \tag{2.40}$$

so

$$\boldsymbol{E}_1^\parallel = \boldsymbol{E}_2^\parallel \tag{2.41}$$

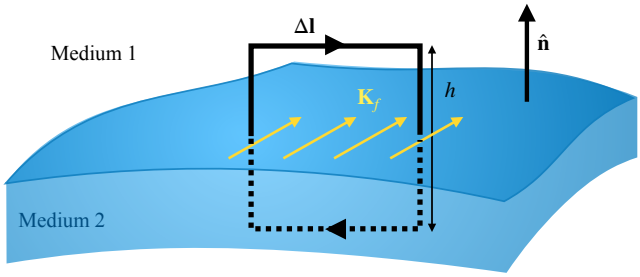


Figure 2.6: A thin rectangular Amperian loop straddling the boundary between two media.

The parallel component of $\boldsymbol{E}$ is always continuous across the boundary (note that $\boldsymbol{E}^\parallel$ is a two-dimensional vector, living in the plane of the boundary).

For Ampère's law (Eq. (2.39)), the total free current enclosed by the loop can be written in terms of a free surface current density $\boldsymbol{K}_f$:

$$I_{\mathrm{f,\,enc}} = \boldsymbol{K}_f \cdot (\hat{\boldsymbol{n}} \times \Delta\boldsymbol{l}), \tag{2.42}$$

where $\hat{\boldsymbol{n}} \times \Delta\boldsymbol{l}$ is the vector perpendicular to the loop, so that $\boldsymbol{K}_f \cdot (\hat{\boldsymbol{n}} \times \Delta\boldsymbol{l})$ picks out the component of $\boldsymbol{K}_f$ passing through the loop. Using the cyclic property of the scalar triple product, $\boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B} \cdot (\boldsymbol{C} \times \boldsymbol{A}) = \boldsymbol{C} \cdot (\boldsymbol{A} \times \boldsymbol{B})$:

$$I_{\text{f, enc}} = \Delta\boldsymbol{l} \cdot \left(\boldsymbol{K}_f \times \hat{\boldsymbol{n}}\right), \tag{2.43}$$

so that

$$\boldsymbol{H}_1 \cdot \Delta\boldsymbol{l} - \boldsymbol{H}_2 \cdot \Delta\boldsymbol{l} = I_{\text{f, enc}} = \Delta\boldsymbol{l} \cdot \boldsymbol{K}_f \times \hat{\boldsymbol{n}}. \tag{2.44}$$

Pulling out the common factor $\Delta\boldsymbol{l}$, we find free surface current density

$$\boldsymbol{H}_1^{\parallel} - \boldsymbol{H}_2^{\parallel} = \boldsymbol{K}_f \times \hat{\boldsymbol{n}}. \tag{2.45}$$

The parallel component of $\boldsymbol{H}$ has a discontinuity proportional to the free surface current density. For linear media (Eq. (2.26)) with no surface currents:

$$\frac{1}{\mu_1}\boldsymbol{B}_1^{\parallel} = \frac{1}{\mu_2}\boldsymbol{B}_2^{\parallel}. \tag{2.46}$$

Summary: boundary conditions

Collecting the four results derived above (Eq. (2.35), Eq. (2.37), Eq. (2.41), and Eq. (2.45) respectively), the fields on either side of a boundary between two media are related by:

$$\begin{aligned} D_1^{\perp} - D_2^{\perp} &= \sigma_f && \text{(i, perpendicular)} \\ B_1^{\perp} &= B_2^{\perp} && \text{(ii, perpendicular)} \\ \boldsymbol{E}_1^{\parallel} &= \boldsymbol{E}_2^{\parallel} && \text{(iii, parallel)} \\ \boldsymbol{H}_1^{\parallel} - \boldsymbol{H}_2^{\parallel} &= \boldsymbol{K}_f \times \hat{\boldsymbol{n}} && \text{(iv, parallel)} \end{aligned} \tag{2.47}$$

For **linear media** (Eq. (2.26)) with no free surface charge or current ($\sigma_f = 0$, $\boldsymbol{K}_f = \boldsymbol{0}$), the perpendicular and parallel conditions on $\boldsymbol{D}$ and $\boldsymbol{H}$ simplify to relations purely between $\boldsymbol{E}$- and $\boldsymbol{B}$-fields (Eq. (2.36), Eq. (2.37), Eq. (2.41), and Eq. (2.46) respectively):

$$\begin{aligned} \varepsilon_1 E_1^{\perp} &= \varepsilon_2 E_2^{\perp} && \text{(i, perpendicular)} \\ B_1^{\perp} &= B_2^{\perp} && \text{(ii, perpendicular)} \\ \boldsymbol{E}_1^{\parallel} &= \boldsymbol{E}_2^{\parallel} && \text{(iii, parallel)} \\ \frac{1}{\mu_1}\boldsymbol{B}_1^{\parallel} &= \frac{1}{\mu_2}\boldsymbol{B}_2^{\parallel} && \text{(iv, parallel)} \end{aligned} \tag{2.48}$$

Note that $\boldsymbol{B}^{\perp}$ and $\boldsymbol{E}^{\parallel}$ are always continuous — they don't depend on the medium at all.

2.6 Worked Example: Capacitor and Coil

To bring together everything from this chapter, consider two simple circuits — a battery-driven capacitor and a DC-driven coil — first in vacuum, then with a dielectric or diamagnetic material inserted.

In-class exercise

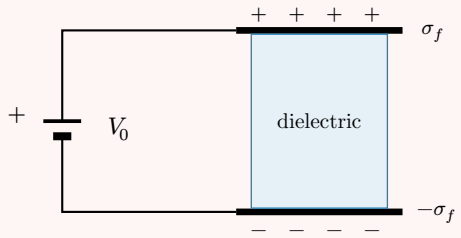


Figure 2.7: A battery charges a parallel-plate capacitor with a dielectric slab (e.g. glass) between the plates.

1. Vacuum case. Ignore the dielectric slab in Figure 2.7 for now. How would you determine the electric displacement $\boldsymbol{D}$ between the plates? Which formula/ equation would you use? How are $\boldsymbol{D}$ and $\boldsymbol{E}$ connected in this case?

2. Adding a dielectric. Now insert a dielectric — e.g. a glass cell — between the plates, as shown in Figure 2.7. What happens to the glass at the microscopic level? How would you determine the polarization $\boldsymbol{P}$ and the field $\boldsymbol{E}$ in this case?

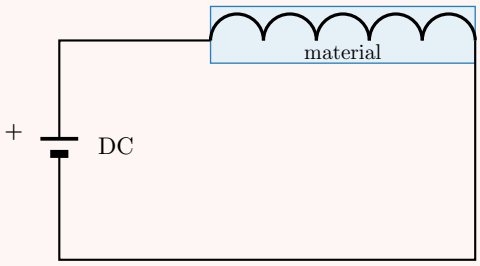


Figure 2.8: A DC source drives a current through a coil with a material core (e.g. glass).

3. Vacuum case. Ignore the material core in Figure 2.8 for now. Consider the coil connected to a DC source. After the current has been ramped up and is constant, a magnetic field has been generated. Which field is it exactly, and from which equation would you determine it?

4. Adding a material core. Now insert a glass cell inside the coil, as shown in Figure 2.8. What happens to the glass at the microscopic level?

5. DC → AC. What happens when we change the power source from DC to AC?

Qualitative answers

1. Capacitor in vacuum. $\boldsymbol{D}$ is fixed directly by the **free** charge on the plates via Gauss's law for $\boldsymbol{D}$, (i) in Eq. (2.24) — exactly the pillbox argument from Section 2.5. For plates much larger than their separation, symmetry gives a uniform $\boldsymbol{D}$ perpendicular to the plates, so a pillbox straddling one plate (Eq. (2.35)) gives $D = \sigma_f$. In vacuum, $\boldsymbol{D} = \varepsilon_0 \boldsymbol{E}$, so $\boldsymbol{D}$ and $\boldsymbol{E}$ are really the same field, just in different units.

2. Capacitor with a dielectric. In the field, the bound electrons and nuclei in the glass shift slightly relative to each other, producing small induced dipole moments $\boldsymbol{p} = \alpha \boldsymbol{E}$ (the same polarizability relation from the start of this chapter) that add up to a macroscopic polarization $\boldsymbol{P}$, aligned with $\boldsymbol{E}$. As in Figure 2.1, the aligned dipoles cancel in the bulk of the glass but leave behind an uncompensated bound surface charge $\sigma_b = \boldsymbol{P} \cdot \hat{\boldsymbol{n}}$: negative on the face touching the + plate and positive on the face touching the − plate; opposite in sign to the free charge on that same adjacent plate, since each dipole's own − end faces the + plate (and vice versa). As a result, the free charge on the plates is partially screened reducing $\boldsymbol{E}$ inside the dielectric. Crucially, the pillbox argument for $\boldsymbol{D}$ is **unchanged**: it only encloses free charge (the bound charge sits on the dielectric's surface, outside the pillbox), so $D = \sigma_f$ still holds. Only $\boldsymbol{E} = (\boldsymbol{D} - \boldsymbol{P})/\varepsilon_0$ changes — for a linear dielectric, $E = D/(\varepsilon_0 \varepsilon_r) = \sigma_f/(\varepsilon_0 \varepsilon_r)$, smaller than the vacuum value by a factor ε_r. *Extra: which quantity actually changes depends on the rest of the circuit — with the battery still connected (V_0 fixed), $E = V_0/d$ stays the same and the battery supplies extra free charge so that D (and σ_f) grow by ε_r; with the battery disconnected (Q fixed), D stays the same and E (and V) drop by ε_r.*

3. Coil in vacuum. By the same logic, the field fixed directly by the **free** current is $\boldsymbol{H}$, via Ampère's law, (iv) in Eq. (2.24) (with $\partial_t \boldsymbol{D} = 0$ in the static case):

$$\oint_C \boldsymbol{H} \cdot \mathrm{d}\boldsymbol{l} = I_{\mathrm{f,\,enc}}. \tag{2.49}$$

For a long solenoid this gives $H = nI$, **independent of the core material** — the direct analogue of the pillbox argument for $\boldsymbol{D}$. In vacuum, $\boldsymbol{B} = \mu_0 \boldsymbol{H}$.

4. Coil with a material core. Glass has no permanent atomic magnetic moments (all electrons are paired), so — unlike the capacitor case — nothing aligns with the field. Instead, the changing flux as the current is switched on induces tiny orbital currents in the electron clouds that **oppose** it (the atomic-scale analogue of Lenz's law): glass, like most everyday materials, is weakly **diamagnetic**, with $\boldsymbol{M}$ antiparallel to $\boldsymbol{H}$ and χ_m of order -10^{-5}. As before, $\boldsymbol{H} = nI$ is unaffected by the core; only $\boldsymbol{B} = \mu_0(\boldsymbol{H} + \boldsymbol{M})$ changes, by about one part in 10^5. The bound surface current $\boldsymbol{K}_b = \boldsymbol{M} \times \hat{\boldsymbol{n}}$ on the glass surface is the magnetic analogue of the bound surface charge $\sigma_b = \boldsymbol{P} \cdot \hat{\boldsymbol{n}}$ from the capacitor.

5. From DC to AC. Once the source is AC, every field becomes time-dependent, and two new things happen. First, the changing polarization in the capacitor drives a real polarization current, $\boldsymbol{J}_p = \frac{\partial \boldsymbol{P}}{\partial t}$ (Eq. (2.16)), on top of the free current. Second — and more importantly — the electric and magnetic sectors, which were completely independent in the static examples above, become **coupled**: a changing $\boldsymbol{E}$ (through the $\partial_t \boldsymbol{D}$ term) generates $\boldsymbol{B}$, and a changing $\boldsymbol{B}$ generates $\boldsymbol{E}$ (Faraday's law). This is exactly the interplay we study next: in Chapter 3 we will see that this coupling between $\boldsymbol{E}$ and $\boldsymbol{B}$ gives rise to propagating **electromagnetic waves**.

3 Electromagnetic Waves

Literature: Griffiths Ch. 9.1 and 9.2

Last lectures we reviewed Maxwell's equations and derived the boundary conditions at the interface between two media. We will now study how electromagnetic waves arise from Maxwell's equations. Let's first review waves in general.

3.1 Waves: A Review

In-class exercise

What is a wave?

In simplest form: a **disturbance that propagates with a fixed shape at constant velocity**. This neglects, among other things:

- *absorption* (the wave amplitude will diminish),
- *dispersive effects* (different frequencies travel at different speeds).

In one dimension, a wave is a solution to the **wave equation**:

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} \tag{3.1}$$

where v is the velocity of the wave.

The solution is any function of the form

$$f(z,t) = g(z - vt) + h(z + vt) \tag{3.2}$$

In-class exercise

What is the physical interpretation of g and h?

$g(z - vt)$ is a wave traveling to the *right*: the displacement found at z at time t is found again at $z + v\Delta t$ at the later time $t + \Delta t$, see Figure 3.1. Likewise, $h(z + vt)$ is a wave traveling to the *left*.

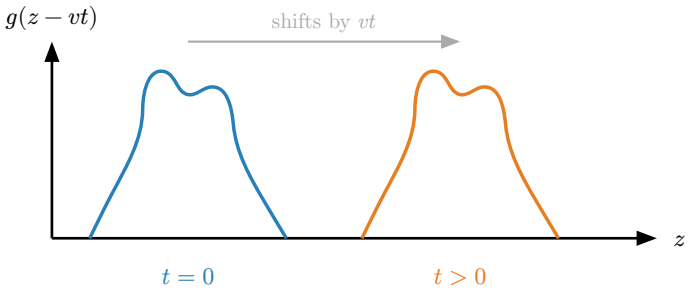


Figure 3.1: A wave $g(z - vt)$ traveling to the right: the same shape reappears, shifted by vt, at a later time.

3.1.1 Sinusoidal Waves

A particularly important special case is the **sinusoidal wave**:

$$f(z,t) = A\cos(k(z-vt)+\delta) \tag{3.3}$$

Amplitude, wave number, phase

To see that this is a wave: think of it at fixed t at different z, and at fixed z at different times t, then consider what happens if you change both z and t.

Terminology

- **Wavelength** $\lambda = 2\pi/k$: distance between two maxima.
- **Period** $T = 2\pi/(kv)$: in one period, a point goes through a full cycle (e.g. from $+A$ to $-A$ and back to $+A$).
- **Frequency** $\nu = 1/T = (kv)/(2\pi) = v/\lambda$: number of oscillations per unit time.
- **Angular frequency** $\omega = 2\pi\nu = kv$: number of radians per unit time.

Using $\omega = kv$, we can write the sinusoidal wave as

$$f(z,t) = A\cos(kz - \omega t + \delta) \tag{3.4}$$

We can use Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$, to write a wave in complex notation:

$$f(z,t) = \mathrm{Re}\left(Ae^{i(kz-\omega t+\delta)}\right) \tag{3.5}$$

This notation is very useful for solving the wave equation.

The wave equation Eq. (3.1) is *linear*, so that the sum of two solutions is also a solution. This is the **superposition principle**, and it is essential for combining waves (e.g. via Fourier analysis).

3.2 Waves in 3D

In 3D, there are different possibilities for the direction of oscillation. If we take the z-direction as the direction of propagation, we can have:

1. **Longitudinal waves**: oscillation in the z-direction. Example: sound waves, as density fluctuations in a medium, see Figure 3.2.
2. **Transverse waves**: oscillations in directions orthogonal to z.

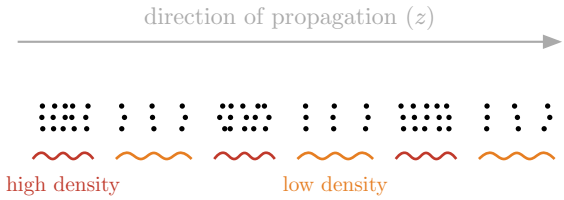
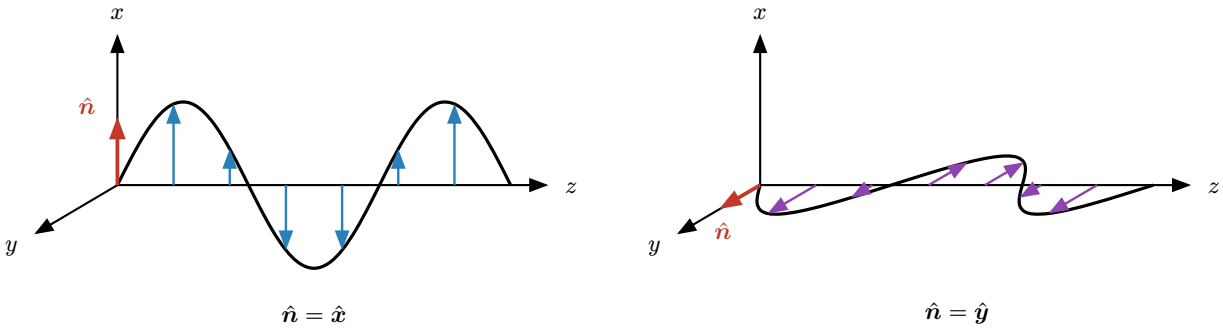


Figure 3.2: A longitudinal wave: regions of high and low density (compressions and rarefactions) propagate along the direction of the wave.

3.2.1 Polarization

Transverse waves in 3D have **two independent states of polarization**, because there are two directions orthogonal to the propagation direction.

The **polarization vector** $\hat{\boldsymbol{n}}$ (with $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}} = 1$) denotes the plane of oscillation, and $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{z}} = 0$ for transverse waves. Figure 3.3 shows two examples for fixed time t, with $\hat{\boldsymbol{n}} = \hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{n}} = \hat{\boldsymbol{y}}$ respectively.



$\hat{\boldsymbol{n}} = \hat{\boldsymbol{x}}$ $\hat{\boldsymbol{n}} = \hat{\boldsymbol{y}}$

Figure 3.3: A transverse wave for two linear polarizations: oscillation along $\hat{\boldsymbol{x}}$ (left) and along $\hat{\boldsymbol{y}}$ (right). Both are orthogonal to the direction of propagation $\hat{\boldsymbol{z}}$.

In general, for **linear polarization**:

$$\hat{\boldsymbol{n}} = \cos\theta\, \hat{\boldsymbol{x}} + \sin\theta\, \hat{\boldsymbol{y}} \tag{3.6}$$

with **polarization angle** θ.

Circular polarization

$\hat{\boldsymbol{n}}$ can also depend on time. If it rotates clockwise around the $\hat{\boldsymbol{z}}$ axis (as seen looking toward the source), we speak of **right circular polarization**; if it rotates counter-clockwise, of **left circular polarization**, see the slides for an example animation.

3.3 Electromagnetic Waves

Recall Maxwell's equations in vacuum, Eq. (1.22):

$$\begin{aligned}
\boldsymbol{\nabla} \cdot \boldsymbol{E} &= \frac{\rho}{\varepsilon_0} && \text{(i)} \\
\boldsymbol{\nabla} \cdot \boldsymbol{B} &= 0 && \text{(ii)} \\
\boldsymbol{\nabla} \times \boldsymbol{E} &= -\frac{\partial \boldsymbol{B}}{\partial t} && \text{(iii)} \\
\boldsymbol{\nabla} \times \boldsymbol{B} &= \mu_0 \boldsymbol{J} + + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} && \text{(iv)}
\end{aligned} \tag{3.7}$$

What are the consequences of these equations?

Suppose that you switch on a current $\boldsymbol{J}(t)$:

- $\boldsymbol{J}$ will vary in time,
- $\Rightarrow$ generates a $\boldsymbol{B}$ field, variable in time,
- $\Rightarrow$ generates an $\boldsymbol{E}$ field, variable in time,
- $\Rightarrow$ will again generate $\boldsymbol{B}$ through (iv),
- $\Rightarrow$ perpetual interplay between $\boldsymbol{E}$ and $\boldsymbol{B}$ through (iii) and (iv).

This already feels like something is oscillating. Let's play a trick to transform Maxwell's equations — first-order differential equations — into *second-order* differential equations, with the hope that they decouple.

We do this by taking the curl of (iii):

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) = \boldsymbol{\nabla} \times \left(-\frac{\partial \boldsymbol{B}}{\partial t} \right) = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \boldsymbol{B}) = -\mu_0 \frac{\partial \boldsymbol{J}}{\partial t} - \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2}, \quad (3.8)$$

where in the last step we exchanged the order of the spatial curl and temporal derivative (coordinates do not depend on time), and then used (iv).

What is the left-hand side? In Section 3.7 we derive the vector calculus identity

$$[\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E})] = [\boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{E}) - \nabla^2 \boldsymbol{E}], \quad (3.9)$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the **Laplacian operator**, acting on all three components of $\boldsymbol{E}$ separately.

Using this identity and $\boldsymbol{\nabla} \cdot \boldsymbol{E} = \rho/\varepsilon_0$ from (i):

$$\boldsymbol{\nabla}\left(\frac{\rho}{\varepsilon_0} \right) - \nabla^2 \boldsymbol{E} = -\mu_0 \frac{\partial \boldsymbol{J}}{\partial t} - \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2}. \quad (3.10)$$

Rearranging, with all $\boldsymbol{E}$-terms on the left:

$$\underbrace{\mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} - \nabla^2 \boldsymbol{E}}_{\Box \boldsymbol{E}} = -\boldsymbol{\nabla} \frac{\rho}{\varepsilon_0} - \mu_0 \frac{\partial \boldsymbol{J}}{\partial t} \quad (3.11)$$

The operator $\Box \equiv \mu_0 \varepsilon_0 \frac{\partial^2}{\partial t^2} - \nabla^2$ is the **d'Alembert operator** (or wave operator).

Let's do the same for Maxwell's equation (iv). Due to the high symmetry between $\boldsymbol{E}$ and $\boldsymbol{B}$ in the vacuum Maxwell equations, we suspect that we will also arrive at a wave equation:

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{B}) = \boldsymbol{\nabla}\left(\underbrace{\boldsymbol{\nabla} \cdot \boldsymbol{B}}_{=0} \right) - \nabla^2 \boldsymbol{B} = \mu_0 \boldsymbol{\nabla} \times \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\underbrace{\boldsymbol{\nabla} \times \boldsymbol{E}}_{=-\frac{\partial \boldsymbol{B}}{\partial t}} \right) \quad (3.12)$$

where we used (ii) and (iv) on the left, and (iv) then (iii) on the right. Rearranging:

$$\mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{B}}{\partial t^2} - \nabla^2 \boldsymbol{B} = \mu_0 \boldsymbol{\nabla} \times \boldsymbol{J} \quad (3.13)$$

We have transformed Maxwell's equations into decoupled second-order differential equations for regions with charges and currents. (They are not *always* decoupled — e.g. Ohm's law $\boldsymbol{J} = \sigma \boldsymbol{E}$ couples them back together, as is relevant for conductors like radio antennas or the plasma in the sun's corona.)

3.4 Electromagnetic Waves in Vacuum

Let's consider $\boldsymbol{E}$ and $\boldsymbol{B}$ fields generated in some region (e.g. by an antenna). How will the fields propagate in vacuum (free space), i.e. away from the region with charges and currents?

There, Eq. (3.11) and Eq. (3.13) become:

$$\mu_0\varepsilon_0\frac{\partial^2 \boldsymbol{E}}{\partial t^2} - \nabla^2 \boldsymbol{E} = 0, \quad \mu_0\varepsilon_0\frac{\partial^2 \boldsymbol{B}}{\partial t^2} - \nabla^2 \boldsymbol{B} = 0 \tag{3.14}$$

These are 3D wave equations! (Eq. (3.11) and Eq. (3.13), with sources on the right-hand side, are called *inhomogeneous* wave equations.)

The velocity of the wave is

$$c^2 = \frac{1}{\mu_0\varepsilon_0} \tag{3.15}$$

μ_0 and ε_0 can be measured in the lab, and we find $c \approx 2.9979 \times 10^8$ m/s — the speed of light! (Check yourself that $1/\sqrt{\mu_0\varepsilon_0}$ indeed has units of m/s.)

We have stumbled onto the nature of light: it is, in fact, a disturbance in the $\boldsymbol{E}$ and $\boldsymbol{B}$ fields. With this, Maxwell's equations unify electricity, magnetism, *and* light. This is one of the great triumphs of 19th century physics.

Note that Maxwell's equations are 8 equations (1 from (i), 1 from (ii), 3 each from (iii) and (iv)), but Eq. (3.14) only give us 6 (3 components each for $\boldsymbol{E}$ and $\boldsymbol{B}$). Not every solution of the wave equation is a solution of Maxwell's equations — it must fulfill some extra constraints, as we will see next.

3.4.1 Plane Wave Solutions

Consider the plane wave ansatz

$$\boldsymbol{E}(z,t) = \boldsymbol{E}_0 e^{i(kz-\omega t)}, \tag{3.16}$$

with constant $\boldsymbol{E}_0$. In vacuum, $\boldsymbol{\nabla}\cdot\boldsymbol{E} = 0$, and since $\boldsymbol{E}(z,t)$ is independent of x, y:

$$\partial_x E_x + \partial_y E_y + \partial_z E_z = 0 \Rightarrow E_{0z}(ik)e^{i(kz-\omega t)} = 0, \tag{3.17}$$

which only holds if $E_{0_z} = 0$. Hence:

EM waves are **transversal**: $\boldsymbol{E}$ and $\boldsymbol{B}$ are perpendicular to the direction of propagation.

Take also $\boldsymbol{B}(z,t) = \boldsymbol{B}_0 e^{i(kz-\omega t)}$ as a plane wave. From Faraday's law, $\boldsymbol{\nabla}\times\boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$, one finds (check this yourself!) the compact relation

$$\boldsymbol{B}_0 = \frac{k}{\omega}(\hat{\boldsymbol{z}}\times\boldsymbol{E}_0) \tag{3.18}$$

This tells us:

- $\boldsymbol{E}$ and $\boldsymbol{B}$ are both perpendicular to each other (and to $\hat{\boldsymbol{z}}$), and must both be non-zero.
- $\boldsymbol{E}$ and $\boldsymbol{B}$ are *in phase* — they reach their maximum at the same time, since there is no extra phase factor in Eq. (3.18).
- There are only *electromagnetic* waves (not waves of $\boldsymbol{E}$ or $\boldsymbol{B}$ alone).

3.4.2 General Monochromatic Plane Waves

With $\hat{\boldsymbol{k}}$ the unit wave vector in the direction of propagation, and the dispersion relation $k = \omega/c$, we have for monochromatic plane waves:

$$\begin{aligned} \boldsymbol{E}(\boldsymbol{r}, t) &= E_0 e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\hat{\boldsymbol{n}}, \\ \boldsymbol{B}(\boldsymbol{r}, t) &= \frac{1}{c} E_0 e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}\left(\hat{\boldsymbol{k}} \times \hat{\boldsymbol{n}}\right) = \frac{1}{c}\left(\hat{\boldsymbol{k}} \times \boldsymbol{E}\right) \end{aligned} \tag{3.19}$$

with $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{k}} = 0$.

3.5 Energy and Momentum in EM Waves

The energy per unit volume stored in the fields is

$$u = \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) \tag{3.20}$$

(check for yourself through dimensional analysis that the units work out).

From Eq. (3.18) we see that $B^2 = E^2/c^2$, so that

$$u = \varepsilon_0 E^2 = \frac{1}{\mu_0} B^2 \tag{3.21}$$

— $\boldsymbol{E}$ and $\boldsymbol{B}$ contribute equally to u — and

$$u = \varepsilon_0 E_0^2 \cos^2(kz - \omega t + \delta). \tag{3.22}$$

As the wave propagates, it transports energy. The **energy flux density** (amount of energy transported through a surface per unit time) is given by the Poynting vector:

$$\boldsymbol{S} = \frac{1}{\mu_0}(\boldsymbol{E} \times \boldsymbol{B}) \tag{3.23}$$

$\boldsymbol{S}$ points in the direction of propagation, with $|\boldsymbol{S}| = cu$.

The momentum density is

$$\boldsymbol{g} = \frac{1}{c^2}\boldsymbol{S}, \quad |\boldsymbol{g}| = \frac{u}{c}. \tag{3.24}$$

3.5.1 Time-Averaged Quantities

> **Scale of visible light**
>
> Visible light has a wavelength between $\sim 10^2$ and 10^3 nm, which corresponds to a frequency $\nu \sim 10^{15}$ Hz, i.e. a period $T \sim 10^{-15}$ s: very fast oscillations. Usually, we measure for durations much longer than that.

We therefore consider the **time-averaged energy density** over one period $T = 2\pi/\omega$:

$$\langle u \rangle = \varepsilon_0 \frac{E_0^2}{T} \int_0^T \cos^2(kz - \omega t + \delta)\, \mathrm{d}t \tag{3.25}$$

(we will look at time averages like this in the tutorials).

Since the time average of $\cos^2$ over a full period is $1/2$, we find the result

$$\langle u \rangle = \frac{1}{2}\varepsilon_0 E_0^2 \tag{3.26}$$

This immediately gives the time averages for $\boldsymbol{S}$ and $\boldsymbol{g}$ as well.

A historical puzzle

Note that the average energy density is *independent of frequency*. This led to an experimental puzzle — the photoelectric effect — which could not be explained classically, and required quantum mechanics to describe.

3.6 Electromagnetic Waves in Matter

In-class exercise

What are the wave equations for $\boldsymbol{E}$ and $\boldsymbol{B}$ in a linear and homogeneous medium, without free currents or charges? Do EM waves propagate slower than in vacuum?

In this case, the macroscopic Maxwell's equations, Eq. (2.24), read:

$$\begin{aligned} &\text{(i)} \quad \boldsymbol{\nabla} \cdot \boldsymbol{D} = 0, && \text{(iii)} \quad \boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \\ &\text{(ii)} \quad \boldsymbol{\nabla} \cdot \boldsymbol{B} = 0, && \text{(iv)} \quad \boldsymbol{\nabla} \times \boldsymbol{H} = \frac{\partial \boldsymbol{D}}{\partial t}. \end{aligned} \tag{3.27}$$

For **linear media** (Eq. (2.26)), $\boldsymbol{D} = \varepsilon \boldsymbol{E}$ and $\boldsymbol{H} = \frac{1}{\mu}\boldsymbol{B}$; for **homogeneous** media, ε and μ are constant (they don't vary from point to point). Plugging in, we find:

$$\begin{aligned} &\text{(i)} \quad \boldsymbol{\nabla} \cdot \boldsymbol{E} = 0, && \text{(iii)} \quad \boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \\ &\text{(ii)} \quad \boldsymbol{\nabla} \cdot \boldsymbol{B} = 0, && \text{(iv)} \quad \boldsymbol{\nabla} \times \boldsymbol{B} = \mu\varepsilon \frac{\partial \boldsymbol{E}}{\partial t}. \end{aligned} \tag{3.28}$$

The only difference to the vacuum case, Eq. (1.22), is that $\varepsilon_0 \to \varepsilon$ and $\mu_0 \to \mu$. Thus, we find EM waves that propagate at speed

$$v = \frac{1}{\sqrt{\varepsilon\mu}} = \frac{c}{n} \tag{3.29}$$

where

$$n \equiv \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \tag{3.30}$$

is the **index of refraction**.

All previous results carry over to the case of a medium, with $\varepsilon_0 \to \varepsilon$ and $\mu_0 \to \mu$. For most materials, $\mu \approx \mu_0$, so that

$$n \approx \sqrt{\varepsilon_r} = \sqrt{\varepsilon/\varepsilon_0}, \tag{3.31}$$

where ε_r is called the **dielectric constant**, which might depend on frequency. Almost always $\varepsilon_r > 1$, so waves travel *slower* in media.

The energy density and Poynting vector become

$$u = \frac{1}{2}\left(\varepsilon E^2 + \frac{1}{\mu}B^2\right), \quad \boldsymbol{S} = \frac{1}{\mu}(\boldsymbol{E} \times \boldsymbol{B}). \tag{3.32}$$

Why does light slow down in matter?

Mathematically, replacing $\varepsilon_0 \to \varepsilon$ and $\mu_0 \to \mu$ is a pretty trivial result — but physically, it is quite astonishing. As waves pass through matter, they polarize and magnetize the molecules, producing oscillating dipoles that create their own fields. These new fields combine with the original wave to give a wave at the *same* frequency, but a *different* speed.

This is a non-trivial consequence of linearity, and is the reason materials like glass or water are transparent: the wave that emerges is still a single coherent wave at the original frequency, just slowed down.

3.7 Deriving the curl of a curl identity

The "BAC − CAB" rule for the curl of a curl

Similarly to the vector identity

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) \tag{3.33}$$

("BAC − CAB"), one can show an analogous identity for the differential operator $\boldsymbol{\nabla}$:

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{C}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{C}) - \nabla^2 \boldsymbol{C}, \tag{3.34}$$

where ∇^2 is the **Laplacian operator**,

$$\nabla^2 f = \partial_x^2 f + \partial_y^2 f + \partial_z^2 f = \sum_i \partial_i^2 f \equiv \partial_i \partial_i f, \tag{3.35}$$

acting on each component of $\boldsymbol{C}$ separately.

Proof (in Einstein notation): look at the i-th component,

$$[\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})]_i = \varepsilon_{ijk} A_j (\boldsymbol{B} \times \boldsymbol{C})_k = \varepsilon_{ijk} A_j \varepsilon_{klm} B_l C_m. \tag{3.36}$$

Using the cyclic property $\varepsilon_{ijk} = \varepsilon_{kij}$ and the identity $\varepsilon_{kij}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ (which follows from the definition of ε_{ijk}, in the same way as in the note on the Levi-Civita tensor in Chapter 1), this becomes

$$\begin{aligned} \varepsilon_{ijk} A_j \varepsilon_{klm} B_l C_m &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) A_j B_l C_m \\ &= B_i(A_j C_j) - C_i(A_j B_j) = [\boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B})]_i. \checkmark \end{aligned} \tag{3.37}$$

The same computation with $\boldsymbol{A} = \boldsymbol{B} = \boldsymbol{\nabla}$ (so that $A_j \to \partial_j$) and $\boldsymbol{C} \to \boldsymbol{C}$ gives

$$
\begin{aligned}
[\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{C})]_i &= \varepsilon_{ijk} \partial_j \varepsilon_{klm} \partial_l C_m \\
&= \partial_i \big(\partial_j C_j\big) - \partial_j \partial_j C_i = \big[\boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{C}) - \nabla^2 \boldsymbol{C}\big]_i . \checkmark
\end{aligned} \tag{3.38}
$$

4 Reflection and Transmission

Literature: Griffiths Ch. 9.3

Last lecture we saw that Maxwell's equations lead to EM waves. Today we continue our discussion of EM waves in matter. In particular, we want to see what happens when EM waves traverse the boundary between two media, and use the boundary conditions we derived previously (Section 2.5).

4.1 Normal Incidence

In-class exercise

What happens when a wave passes an interface, e.g. air → water or water → air? (Think of your personal experience!)

For an incident wave, there will be a **reflected** and a **transmitted** wave, both for $\boldsymbol{E}$ and $\boldsymbol{B}$, which we call $\boldsymbol{E}_I, \boldsymbol{E}_R, \boldsymbol{E}_T$ and $\boldsymbol{B}_I, \boldsymbol{B}_R, \boldsymbol{B}_T$ respectively. The reflected and transmitted fields are determined from the boundary conditions we already derived (Section 2.5).

Let's first look at the case of **normal incidence**: the incident $\boldsymbol{E}$ field is a plane wave traveling in the $+z$ direction, polarized along $\hat{\boldsymbol{x}}$, with the boundary in the xy plane at $z = 0$, see Figure 4.1.

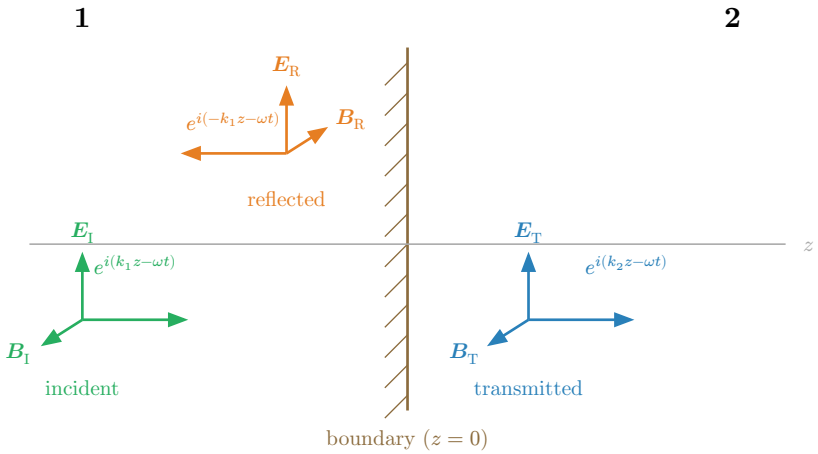


Figure 4.1: Normal incidence: the incident wave (medium 1) generates a reflected wave (medium 1) and a transmitted wave (medium 2) at the boundary $z = 0$.

The incoming wave is

$$\boldsymbol{E}_I(z,t) = E_{0I} e^{i(k_1 z-\omega t)} \hat{\boldsymbol{x}}, \quad \boldsymbol{B}_I(z,t) = B_{0I} e^{i(k_1 z-\omega t)} \hat{\boldsymbol{y}}, \tag{4.1}$$

where, as we saw last lecture, $\boldsymbol{B} = \frac{1}{v}\left(\hat{\boldsymbol{k}} \times \boldsymbol{E}\right)$, so that

$$B_{0I} = \frac{E_{0I}}{v_1} \tag{4.2}$$

wave velocity in medium (1)

with

$$v_i = \frac{1}{\sqrt{\varepsilon_i \mu_i}} = \frac{c}{n_i}, \quad n_i > 1, \quad i = 1, 2. \tag{4.3}$$

A **reflected** wave is generated,

$$\boldsymbol{E}_R(z,t) = E_{0R} e^{i(-k_1 z - \omega t)} \hat{\boldsymbol{x}}, \quad \boldsymbol{B}_R(z,t) = -\frac{E_{0R}}{v_1} e^{i(-k_1 z - \omega t)} \hat{\boldsymbol{y}}, \tag{4.4}$$

note the minus sign!

required so $\boldsymbol{S}$ points along propagation direction $-z$

And a **transmitted** wave is generated,

$$\boldsymbol{E}_T(z,t) = E_{0T} e^{i(k_2 z - \omega t)} \hat{\boldsymbol{x}}, \quad \boldsymbol{B}_T(z,t) = \frac{E_{0T}}{v_2} e^{i(k_2 z - \omega t)} \hat{\boldsymbol{y}}. \tag{4.5}$$

We show the incident, reflected, and transmitted waves in Figure 4.1.

Now, given E_{0I}, we would like to determine E_{0R} and E_{0T}. For this we use the boundary conditions (Section 2.5 and a recap on the slides), noting that the boundary is a surface in the xy plane at $z = 0$. We assume linear media without surface charges or currents.

4.1.1 Boundary Conditions and Amplitudes

In general, the field on the left, $\boldsymbol{E}_I + \boldsymbol{E}_R$ and $\boldsymbol{B}_I + \boldsymbol{B}_R$, must join with the fields on the right, $\boldsymbol{E}_T$ and $\boldsymbol{B}_T$, so that the boundary conditions are satisfied. The fields have no components perpendicular to the surface (no z component), so boundary conditions for the perpendicular compoents, (i) and (ii) in Eq. (2.48), are trivial.

From (iii), stating that $\boldsymbol{E}^{\parallel}$ is continuous across the boundary, we have:

$$\underbrace{\boldsymbol{E}_R + \boldsymbol{E}_I}_{\boldsymbol{E}_1^{\parallel}} = \underbrace{\boldsymbol{E}_T}_{\boldsymbol{E}_2^{\parallel}} \tag{4.6}$$

and since the boudary is at $z = 0$ we find

$$\Rightarrow \quad E_{0R} + E_{0I} = E_{0T}, \tag{4.7}$$

where we have divided by $e^{-i\omega t}$.

With boundary condition (iv) gives

$$\underbrace{\frac{1}{\mu_1}\left(-\frac{E_{0R}}{v_1} + \frac{E_{0I}}{v_1}\right)}_{B_1^{\parallel}/\mu_1} = \underbrace{\frac{E_{0T}}{\mu_2 v_2}}_{B_2^{\parallel}/\mu_2} \tag{4.8}$$

$$\Rightarrow \quad E_{0I} - E_{0R} = \underbrace{\frac{\mu_1 v_1}{\mu_2 v_2}}_{\beta = \frac{\mu_1 n_2}{\mu_2 n_1}} E_{0T} \tag{4.9}$$

We have two unknowns, E_{0R} and E_{0T}, and two equations, Eq. (4.7) and Eq. (4.9), which we can solve, e.g. by solving Eq. (4.9) for E_{0T} and plugging this into Eq. (4.7):

$$E_{0R} + E_{0I} = \frac{1}{\beta}(E_{0I} - E_{0R}). \tag{4.10}$$

Solving for E_{0R} we find

$$E_{0R} = \frac{1-\beta}{1+\beta} E_{0I} \tag{4.11}$$

and plugging this back into Eq. (4.9) we get

$$E_{0T} = \frac{2}{1+\beta} E_{0I} \tag{4.12}$$

with

$$\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1} \tag{4.13}$$

The special case $\mu_1 \approx \mu_2 \approx \mu_0$

As mentioned before, for most media $\mu \approx \mu_0$. Under this assumption, $\beta = v_1/v_2 = n_2/n_1$, and Eq. (4.11) and Eq. (4.12) become

$$E_{0R} = \frac{n_1 - n_2}{n_1 + n_2} E_{0I}, \quad E_{0T} = \frac{2n_1}{n_1 + n_2} E_{0I}. \tag{4.14}$$

(Check this yourself!)

4.1.2 Reflection and Transmission Coefficients

What fraction of the incident energy is transmitted, and what fraction is reflected? For this we look at the incoming intensity, i.e. the time-averaged Poynting vector (cf. Tutorial 3) for a plane wave in a linear medium:

$$I = \underbrace{\langle \boldsymbol{S} \rangle}_{\text{time avg. over one cycle}} = \underbrace{\frac{1}{2}\varepsilon v E_0^2}_{\text{for plane waves in linear media}} \tag{4.15}$$

With this, we can determine the reflection and transmission coefficients from Eq. (4.14):

$$R = \frac{I_R}{I_I} = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2, \tag{4.16}$$

where we have used that the reflected and incident waves are in the same medium, so v_1 and ε_1 cancel. And

$$T = \frac{I_T}{I_I} = \frac{\varepsilon_2}{\varepsilon_1} \frac{v_2}{v_1} \left(\frac{E_{0T}}{E_{0I}} \right)^2 = \frac{4 n_1 n_2}{(n_1 + n_2)^2} \tag{4.17}$$

$\frac{n_2^2}{n_1^2}$ since $n_i \sim \sqrt{\varepsilon_i/\varepsilon_0}$; Eq. (4.14)

And it's easy to verify that

$$R + T = 1, \tag{4.18}$$

as required from energy conservation. (Check this yourself!)

Connection to quantum mechanics

We will revisit reflection and transmission coefficients in quantum mechanics, when discussing phenomena like scattering and tunneling.

In-class exercise

Discuss with your neighbor: Is the derivation clear? What happens when $n_1 = n_2$? What happens when $n_1 \gg n_2$?

4.2 Reflection and Transmission at Oblique Incidence

Let's turn to the more general case of **oblique incidence**, i.e. where light is not coming in perpendicular to the boundary, but at an angle θ_I (think e.g. of light shining through a window). Normal incidence is just the special case $\theta_I = 0$.

We have the following situation, see Figure 4.2:

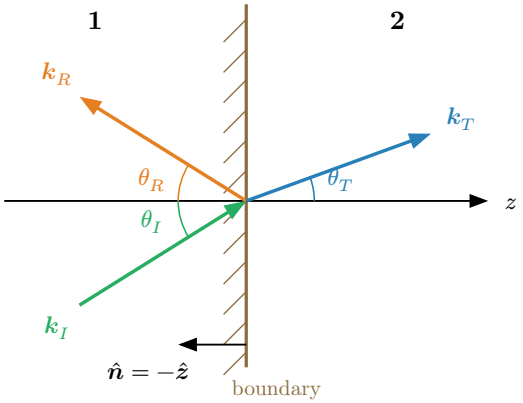


Figure 4.2: Oblique incidence: the incident, reflected, and transmitted wave vectors, together with the angles they make with the surface normal $\hat{\boldsymbol{n}} = -\hat{\boldsymbol{z}}$.

where the incoming wave is now

$$\boldsymbol{E}_I = \boldsymbol{E}_{0I} e^{i(\boldsymbol{k}_I \cdot \boldsymbol{r} - \omega t)}, \quad \boldsymbol{B}_I = \frac{1}{v_1}\left(\hat{\boldsymbol{k}}_I \times \boldsymbol{E}_I\right). \tag{4.19}$$

(Remember: the physical fields are the real part of these complex expressions and also note that we're still assuming monochromatic waves.)

Similarly, the transmitted and reflected waves are

$$\boldsymbol{E}_R = \boldsymbol{E}_{0R} e^{i(\boldsymbol{k}_R \cdot \boldsymbol{r} - \omega t)}, \quad \boldsymbol{B}_R = \frac{1}{v_1}\left(\hat{\boldsymbol{k}}_R \times \boldsymbol{E}_R\right), \tag{4.20}$$

$$\boldsymbol{E}_T = \boldsymbol{E}_{0T} e^{i(\boldsymbol{k}_T \cdot \boldsymbol{r} - \omega t)}, \quad \boldsymbol{B}_T = \frac{1}{v_2}\left(\hat{\boldsymbol{k}}_T \times \boldsymbol{E}_T\right). \tag{4.21}$$

All waves have the same frequency ω, determined e.g. by the light source. Remembering the **dispersion relation** for plane waves, $\omega = kv$, we find in our case:

$$\omega = k_I v_1 = k_R v_1 = k_T v_2, \tag{4.22}$$

or, in terms of just the magnitude of the wave vectors,

$$k_I = k_R = \frac{v_2}{v_1} k_T = \frac{n_1}{n_2} k_T. \tag{4.23}$$

4.2.1 Laws of Geometric Optics

We again use the boundary conditions, where

$$\boldsymbol{E}_1 = \boldsymbol{E}_I + \boldsymbol{E}_R, \quad \boldsymbol{B}_1 = \boldsymbol{B}_I + \boldsymbol{B}_R, \quad \boldsymbol{E}_2 = \boldsymbol{E}_T, \quad \boldsymbol{B}_2 = \boldsymbol{B}_T. \tag{4.24}$$

Parallel and perpendicular components

We can get the parallel ($\parallel$) and perpendicular ($\perp$) components with the help of the surface normal $\hat{\boldsymbol{n}}$, e.g.

$$E_1^{\perp} = \boldsymbol{E}_1 \cdot \hat{\boldsymbol{n}}, \quad \boldsymbol{E}_1^{\parallel} = \boldsymbol{E}_1 - (\boldsymbol{E}_1 \cdot \hat{\boldsymbol{n}})\hat{\boldsymbol{n}}. \tag{4.25}$$

From boundary condition (i) at $z = 0$, we have

$$\varepsilon_1 \left(\boldsymbol{E}_{0I}^{\parallel} e^{i(\boldsymbol{k}_I \cdot \boldsymbol{r} - \omega t)} + \boldsymbol{E}_{0R}^{\parallel} e^{i(\boldsymbol{k}_R \cdot \boldsymbol{r} - \omega t)} \right) = \varepsilon_2 \boldsymbol{E}_{0T}^{\parallel} e^{i(\boldsymbol{k}_T \cdot \boldsymbol{r} - \omega t)}. \tag{4.26}$$

With $\boldsymbol{r} = (x, y, 0)$. The x, y, t dependence is only in the exponents, so for any point on the boundary and any time t we must have

$$\boldsymbol{k}_I \cdot \boldsymbol{r} = \boldsymbol{k}_R \cdot \boldsymbol{r} = \boldsymbol{k}_T \cdot \boldsymbol{r}, \tag{4.27}$$

(simply divide both sides by the exponential factors $e^{-i\omega t}$ in Eq. (4.26) to see this). Multiplying out the dot products, we find

$$k_I^x x + k_I^y y = k_R^x x + k_R^y y = k_T^x x + k_T^y y, \tag{4.28}$$

which only holds if the components are separately equal — e.g. for $x = 0$: $k_I^y = k_R^y = k_T^y$ or $y = 0$:

$$k_I^x = k_R^x = k_T^x. \tag{4.29}$$

Let's choose a coordinate system where $\boldsymbol{k}_I$ lies in the xz plane like in Figure 4.2 ($k_I^y = 0$); then Eq. (4.29) forces $\boldsymbol{k}_R$ and $\boldsymbol{k}_T$ to also lie in the xz plane. From this we find the **fundamental laws of geometric optics**:

1. **First Law**: The incident, reflected, and transmitted wave vectors form a plane — the *plane of incidence* — that also contains the normal vector $\hat{\boldsymbol{n}}$ of the interface.
2. **Second Law**: $\theta_I = \theta_R$: the angle of incidence equals the angle of reflection. This follows from Eq. (4.29),

$$k_I^x = k_I \sin\theta_I = k_R \sin\theta_R = k_R^x, \quad \text{and} \quad k_I = k_R. \tag{4.30}$$

3. **Third Law (Snell's law)**:

$$\frac{\sin(\theta_T)}{\sin(\theta_I)} = \frac{n_1}{n_2}, \tag{4.31}$$

which again follows from Eq. (4.29), $k_I \sin\theta_I = k_T \sin\theta_T$, together with $k_{I,T} = \omega / v_{1,2}$.

In-class exercise

Discuss with your neighbor: Is the derivation clear?

Now that we've taken care of the exponential factors in the boundary conditions, we can continue with finding the amplitudes. The exponential factors cancel, and we're left with:

$$\begin{aligned}
&\text{(i)} && \varepsilon_1(\boldsymbol{E}_{0I}+\boldsymbol{E}_{0R})^{\perp} = \varepsilon_2(\boldsymbol{E}_{0T})^{\perp} \\
&\text{(ii)} && (\boldsymbol{B}_{0I}+\boldsymbol{B}_{0R})^{\perp} = (\boldsymbol{B}_{0T})^{\perp} \\
&\text{(iii)} && (\boldsymbol{E}_{0I}+\boldsymbol{E}_{0R})^{\parallel} = (\boldsymbol{E}_{0T})^{\parallel} \\
&\text{(iv)} && \frac{1}{\mu_1}(\boldsymbol{B}_{0I}+\boldsymbol{B}_{0R})^{\parallel} = \frac{1}{\mu_2}(\boldsymbol{B}_{0T})^{\parallel}
\end{aligned} \tag{4.32}$$

In our coordinate system, the perpendicular component ($\perp$) is along $\hat{\boldsymbol{z}}$. Note that Eqs. (iii) and (iv) contain two equations for both the x and y components separately.

4.2.2 Amplitudes: Polarization in the Plane of Incidence

Let's assume that the polarization of the incident wave is in the plane of incidence, see Figure 4.3.

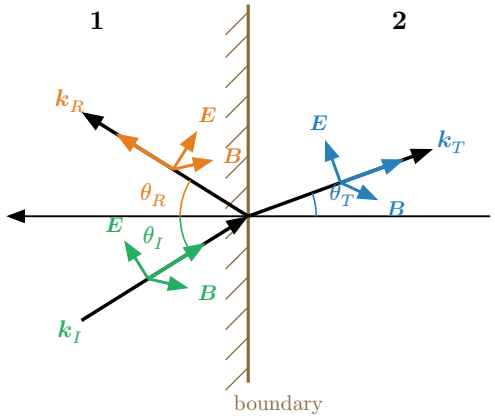


Figure 4.3: Oblique incidence with polarization in the plane of incidence: $\boldsymbol{E}$ lies in the plane of the page for all three waves, while $\boldsymbol{B}$ points out of the page.

Then (i), the perpendicular component, becomes

$$\begin{aligned}
\varepsilon_1 \Bigg(&\underbrace{-E_{0I}}_{\substack{\text{surface normal direction,}\\ \boldsymbol{E}_I\cdot\hat{\boldsymbol{n}}=-E_{0I}\sin\theta_I}} \sin\theta_I + E_{0R}\sin\theta_R \Bigg) \\
&= \varepsilon_2 \Bigg(\underbrace{-E_{0T}}_{\substack{\text{same direction}\\ \text{as incident wave}}} \sin\theta_T \Bigg),
\end{aligned} \tag{4.33}$$

using $\theta_I = \theta_R$ (law 2):

$$\begin{aligned}
&\varepsilon_1 \sin\theta_I(-E_{0I}+E_{0R}) = -\varepsilon_2 \sin\theta_T E_{0T} \\
\Leftrightarrow\quad &E_{0R}-E_{0I} = -\frac{\varepsilon_2}{\varepsilon_1}\frac{\sin(\theta_T)}{\sin(\theta_I)}E_{0T}.
\end{aligned} \tag{4.34}$$

Using Snell's law (Eq. (4.31)) and multiply with -1 to flip the sign, this becomes

$$\begin{aligned} E_{0I} - E_{0R} = \frac{\varepsilon_2}{\varepsilon_1}\frac{n_1}{n_2}E_{0T} = \underbrace{\frac{\mu_1 n_2}{\mu_2 n_1}}_{\equiv\beta} E_{0T} \\ = \beta E_{0T} \end{aligned} \tag{4.35}$$

— the same β as in Eq. (4.13) for which we used $v = 1/\sqrt{\varepsilon\mu} \Leftrightarrow \varepsilon = 1/(\mu v^2) = n^2/(\mu c^2)$.

Boundary condition (ii) does not add anything, since $\boldsymbol{B}$ has no z component.

Boundary condition (iii), the parallel component, becomes

$$\begin{aligned} E_{0I}\cos\theta_I + E_{0R}\cos\theta_R &= E_{0T}\cos\theta_T \\ \Leftrightarrow \quad E_{0I} + E_{0R} &= \underbrace{\frac{\cos(\theta_T)}{\cos(\theta_I)}}_{\equiv\alpha} E_{0T} \\ &= \alpha E_{0T} \end{aligned} \tag{4.36}$$

using law 2 again.

Boundary condition (iv) gives the same result as (i) by using $\boldsymbol{B} = \frac{1}{v}\left(\hat{\boldsymbol{k}} \times \boldsymbol{E}\right)$ (check this yourself!).

With Eq. (4.35) and Eq. (4.36), we again have two equations for two unknown amplitudes E_{0R} and E_{0T}.

Plugging in Eq. (4.36) into Eq. (4.35) gives

$$\begin{aligned} E_{0I} - E_{0R} &= \frac{\beta}{\alpha}(E_{0I} + E_{0R}) \\ \Leftrightarrow \quad E_{0R} &= \frac{1 - \frac{\beta}{\alpha}}{1 + \frac{\beta}{\alpha}} E_{0I} \\ \Leftrightarrow \quad E_{0R} &= \frac{\alpha - \beta}{\alpha + \frac{\beta}{\alpha}} E_{0I}. \end{aligned} \tag{4.37}$$

Plugging this back into Eq. (4.36) we find

$$E_{0R} = \frac{\alpha - \beta}{\alpha + \beta} E_{0I}, \quad E_{0T} = \frac{2}{\alpha + \beta} E_{0I} \tag{4.38}$$

with

$$\alpha \equiv \frac{\cos(\theta_T)}{\cos(\theta_I)}, \quad \beta \equiv \frac{\mu_1 n_2}{\mu_2 n_1} \tag{4.39}$$

These are the **Fresnel equations** for the case of polarization in the plane of incidence.

Using Snell's law (Eq. (4.31)) we can show that α and thus the amplitudes of the reflected and transmitted waves depend only on the angle of incidence θ_I and the refractive indices n_1 and n_2 of the two media:

$$\alpha = \frac{\sqrt{1-\sin^2\theta_T}}{\cos\theta_I} = \frac{\sqrt{1-\left(\frac{n_1}{n_2}\right)^2\sin^2\theta_I}}{\cos\theta_I}. \tag{4.40}$$

Consistency checks

- For $\theta_I = 0$, $\alpha = 1$, and we recover normal incidence, Eq. (4.11) and Eq. (4.12).
- For $\theta_I = 90°$ (grazing incidence), $\cos\theta_I \to 0$ so $\alpha \to \infty$, and the wave is totally reflected.

4.2.3 The Brewster Angle

There is also an intermediate angle, known as the **Brewster angle** θ_B, where E_{0R} vanishes. This occurs when θ_I is such that $\alpha = \beta$. Expressing α in terms of θ_I using Eq. (4.40) and $\cos^2\theta_I = 1 - \sin^2\theta_I$, ones finds

$$\sin^2\theta_B = \frac{1-\beta^2}{(n_1/n_2)^2 - \beta^2}. \tag{4.41}$$

For $\mu_1 \approx \mu_2 \approx \mu_0$ we have $\beta = \frac{n_2}{n_1}$, and this simplifies to

$$\sin^2\theta_B = \frac{\beta^2}{1+\beta^2}. \tag{4.42}$$

Using trigonometric identities, we can also express this in terms of the tangent:

$$\tan\theta_B = \frac{n_2}{n_1}. \tag{4.43}$$

This only occurs for polarization in the plane of incidence — this is why polarized (sun)glasses reduce the glare off a horizontal surface: they block the (unpolarized) light component reflected at close to the Brewster angle.

4.2.4 Reflection and Transmission Coefficients (Oblique Incidence)

Lastly, we derive the transmission and reflection coefficients for the general case of oblique incidence. The intensity (power per unit area) of the incident wave hitting the interface is $I = \boldsymbol{S}\cdot\hat{\boldsymbol{z}}$, so that

$$I_I = \frac{1}{2}\varepsilon_1 v_1 E_{0I}^2 \cos\theta_I, \tag{4.44}$$

while the reflected and transmitted intensities are

$$I_R = \frac{1}{2}\varepsilon_1 v_1 E_{0R}^2 \cos\theta_R, \quad I_T = \frac{1}{2}\varepsilon_2 v_2 E_{0T}^2 \cos\theta_T. \tag{4.45}$$

This gives

$$\begin{aligned} R &= \frac{I_R}{I_I} = \left(\frac{E_{0R}}{E_{0I}}\right)^2 = \left(\frac{\alpha-\beta}{\alpha+\beta}\right)^2, \\ T &= \frac{I_T}{I_I} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1}\left(\frac{E_{0T}}{E_{0I}}\right)^2 \frac{\cos(\theta_T)}{\cos(\theta_I)} = \alpha\beta\left(\frac{2}{\alpha+\beta}\right)^2 \end{aligned} \tag{4.46}$$

Again, you may check that $R + T = 1$, as required by energy conservation.

Example: air to glass

Consider the interface between air ($n_1 = 1$) and glass ($n_2 = 1.5$), so that $\beta = n_2/n_1 = 1.5$. See the slides for plots of R and T as a function of θ_I.

Next lecture: we will see what happens when waves propagate in media with free charges and surface currents, such as conductors — this leads to the phenomena of absorption.

5 Absorption and Dispersion

Literature: Griffiths Ch. 9.4

Last lecture we considered the reflection and transmission of EM waves at the boundary between two **non-conducting** media, i.e. media without free surface charges. This simplified our boundary conditions for the parallel and perpendicular $\boldsymbol{E}$ and $\boldsymbol{B}$ components.

In this lecture, we'll study what happens now at the boundary of a non-conducting and a **conducting** medium.

In the second part, we will return to the propagation of EM waves in non-conducting media, but now allowing for a **frequency-dependent permittivity** $\varepsilon(\omega)$, which leads to **dispersion**.

5.1 Electromagnetic Waves in Conductors

In a conductor, the free current density is given by (the generalized form of) **Ohm's law**:

$$\boldsymbol{J}_f = \sigma \boldsymbol{E}, \tag{5.1}$$

where σ is called the **conductivity**.

Connection to $U = RI$

The more familiar form of Ohm's law, $U = RI$, is a special case of Eq. (5.1). You can retrieve it by integrating the electric field along a wire of length L and cross-sectional area A.

For a constant field $\boldsymbol{E}$ along the wire, the voltage is

$$U = \int_0^L \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l} = EL \tag{5.2}$$

and thus $E = \frac{V}{L}$. If $\boldsymbol{E}$ is uniform across the cross-section, the current is

$$I = \int_A \boldsymbol{J}_f \cdot \mathrm{d}\boldsymbol{A} = J_f A \tag{5.3}$$

or $J_f = \frac{I}{A}$.

Plugging everything into Eq. (5.1), we recover $U = RI$ with the resistance is $R = \frac{L}{\sigma A}$ and the current is $I = J_f A$.

Plugging this into Maxwell's equations (for a linear, homogeneous medium):

$$\begin{aligned} &\text{(i)} \quad \boldsymbol{\nabla} \cdot \boldsymbol{E} = \frac{\rho_f}{\varepsilon} \qquad \text{(iii)} \quad \boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \\ &\text{(ii)} \quad \boldsymbol{\nabla} \cdot \boldsymbol{B} = 0 \qquad \text{(iv)} \quad \boldsymbol{\nabla} \times \boldsymbol{B} = \mu\sigma\boldsymbol{E} + \mu\varepsilon\frac{\partial \boldsymbol{E}}{\partial t} \end{aligned} \tag{5.4}$$

You will show in the exercises that the free charge density is then a function of time and decays,

$$\rho_f(t) = \rho_f(0)e^{-\frac{\sigma}{\varepsilon}t}. \tag{5.5}$$

$\equiv 1/\tau$

Thus, if you put some free charge on a conductor, it will dissipate on a time scale $\tau = \varepsilon/\sigma$, and "flow out to the edges." If you wait sufficiently long, (i) in Eq. (5.4) becomes

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0. \tag{5.6}$$

5.1.1 Wave Equation with a Complex Wavenumber

In the exercises you will also show that this set of Maxwell's equations can be transformed into a wave equation (following the same steps as in Chapter 3), and that it is solved by plane waves with **complex wavenumbers**,

$$\tilde{k} = k + i\kappa. \tag{5.7}$$

The complex part leads to an **attenuation** of the wave. Writing this out explicitly — I'm now including the tilde on $\boldsymbol{E}$ and $\boldsymbol{B}$ to remind you that they are complex:

$$\tilde{\boldsymbol{X}}(z,t) = \tilde{\boldsymbol{X}}_0 e^{-\kappa z} e^{i(kz-\omega t)}, \quad \boldsymbol{X} = \boldsymbol{E} \text{ or } \boldsymbol{B}, \tag{5.8}$$

with

$$\begin{aligned} k &= \omega\sqrt{\frac{\varepsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2}+1\right]^{\frac{1}{2}}, \\ \kappa &= \omega\sqrt{\frac{\varepsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2}-1\right]^{\frac{1}{2}}. \end{aligned} \tag{5.9}$$

Consequence: waves only penetrate a finite amount into a conductor, characterized by the **skin depth**

$$d \equiv \frac{1}{\kappa}. \tag{5.10}$$

Example: skin depth of silver at optical frequencies

Metals like silver have high conductivity, $\sigma \sim 10^7(\Omega \cdot \mathrm{m})^{-1}$, so the $(\sigma/(\varepsilon\omega))^2$ term dominates in κ, Eq. (5.9):

$$\kappa \approx \omega\sqrt{\frac{\varepsilon\mu}{2}}\sqrt{\frac{\sigma}{\varepsilon\omega}} \approx \sqrt{\frac{\omega\mu_0\sigma}{2}}, \tag{5.11}$$

so that

$$d = \frac{1}{\kappa} \approx \sqrt{\frac{2}{\omega\mu_0\sigma}} = \sqrt{\frac{2}{10^{15} \cdot 4\pi \times 10^{-7} \cdot 10^{7}}}\ \text{m} \approx 1.3 \times 10^{-8}\ \text{m} = 13\ \text{nm}, \tag{5.12}$$

where we have used $\omega \approx 10^{15}$ Hz.

See also this skin-depth calculator.

The real part of $\tilde{k}$ determines the wavelength, propagation speed, and index of refraction, as before:

$$\lambda = \frac{2\pi}{k}, \quad v = \frac{\omega}{k}, \quad n = \frac{ck}{\omega}. \tag{5.13}$$

In-class exercise

Are the waves still transverse?

5.1.2 Polarization and Phase of $\boldsymbol{E}$ and $\boldsymbol{B}$

Let's choose the electric field to be polarized along the x-axis and travelling along the z-direction:

$$\tilde{\boldsymbol{E}} = \tilde{E}_0 e^{-\kappa z} e^{i(kz-\omega t)} \hat{\boldsymbol{x}}, \tag{5.14}$$

and from Maxwell's equation (iii) we find the same relation as for plane waves, Eq. (3.19),

$$\tilde{\boldsymbol{B}} = \frac{\tilde{k}}{\omega} \tilde{E}_0 e^{-\kappa z} e^{i(kz-\omega t)} \hat{\boldsymbol{y}}. \tag{5.15}$$

The complex number $\tilde{k}$ in the magnetic field will lead to a **phase difference** between the $\boldsymbol{E}$ and $\boldsymbol{B}$ fields.

How large is the phase difference? Remember that we can rewrite a complex number as

$$\tilde{k} = k + i\kappa = Ke^{i\phi}, \quad K = \sqrt{k^2 + \kappa^2}, \quad \tan\phi = \frac{\kappa}{k}. \tag{5.16}$$

Thus, the complex amplitudes of the fields are related by

$$\tilde{B}_0 = B_0 e^{i\delta_B} = \frac{Ke^{i\phi}}{\omega} E_0 e^{i\delta_E} \quad \Rightarrow \quad \delta_B - \delta_E = \phi : \tag{5.17}$$

the magnetic field **lags behind** the electric field (see the slides for a plot).

5.2 Reflection off a Conductor

Now we can carry out the analysis of what happens when an EM wave hits the boundary between a non-conducting medium and a conductor. For simplicity, let's look at the case of **normal incidence**: waves polarized in the $\hat{\boldsymbol{x}}$ direction, propagating along the z-axis, i.e. $\theta_I = 0$.

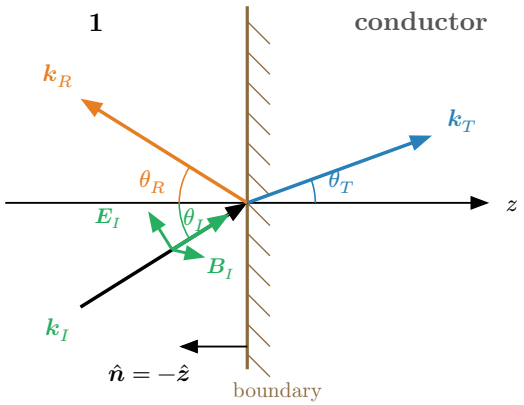


Figure 5.1: A wave hits the boundary between a non-conducting medium (1) and a conductor. For the derivation in the text, we specialize to normal incidence, $\theta_I = 0$.

Now, we use again the general boundary conditions, Eq. (2.47). (see the slides for a reminder). For ohmic conductors, there can be no free surface current $\boldsymbol{K}_f$, as this would require an infinite electric field at the boundary:

$$\boldsymbol{K}_f = 0. \tag{5.18}$$

With $\theta_I = 0$, we don't have a polarization perpendicular to the boundary (i.e. parallel to the surface normal $\hat{\boldsymbol{n}}$), so boundary condition (i) yields

$$\sigma_f = 0. \tag{5.19}$$

5.2.1 Parallel Components

For the parallel components, the analysis is the same as last lecture (Chapter 4). The only difference: the wave vector $\tilde{k}_2$, in the conductor, is now complex. Going through the same steps as before, we get

$$\tilde{E}_{0R} = \frac{1-\tilde{\beta}}{1+\tilde{\beta}}\tilde{E}_{0I}, \quad \tilde{E}_{0T} = \frac{2}{1+\tilde{\beta}}\tilde{E}_{0I}, \tag{5.20}$$

with

$$\tilde{\beta} = \frac{\mu_1 v_1}{\mu_2 \omega}\tilde{k}_2, \tag{5.21}$$

$\tilde{k}_2 \in \mathbb{C}$ (complex number)

so that $\tilde{\beta}$ is itself a complex number.

For a **perfect conductor**, $\sigma \to \infty$. From Eq. (5.9), both $k_2 \to \infty$ and $\kappa_2 \to \infty$, so $\tilde{k}_2 \to \infty$ and thus $\tilde{\beta} \to \infty$. As a result, we find

$$\tilde{E}_{0R} \to -\tilde{E}_{0I}, \quad \tilde{E}_{0T} \to 0. \tag{5.22}$$

There's no transmitted wave — only a reflected wave, which has a 180° phase shift compared to the incoming wave. **This is how a mirror works!**

Mirrors in practice

A household mirror usually has a thin layer of silver on the back of a glass plate — the glass is just there to support the thin silver layer.

In-class exercise

Discuss with your neighbor: are these derivations clear?

5.3 Frequency Dependence of the Permittivity

Literature: Griffiths Ch. 9.4.3

5.3.1 Recap

We have seen that the propagation of EM waves in media depends on three material properties.

In-class exercise

Can you name them?

Here's a summary:

1. **Permittivity** ε and **permeability** μ, giving

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{n\sqrt{\varepsilon_0\mu_0}} = \frac{c}{n}, \tag{5.23}$$

   with the **index of refraction**

$$n = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \approx \sqrt{\varepsilon_r}, \quad \varepsilon_r = 1 + \chi_e. \tag{5.24}$$

2. **Conductivity** σ ($= 1/\varrho$, where ϱ is the resistivity), which in general is a tensor σ_{ij}.

ε, μ, and σ all depend on the material. In addition, they depend on the **wavelength** (or energy) — and with them, the refractive index depends on energy. This is a well-known phenomenon in optics called **dispersion**.

Example: typical glass

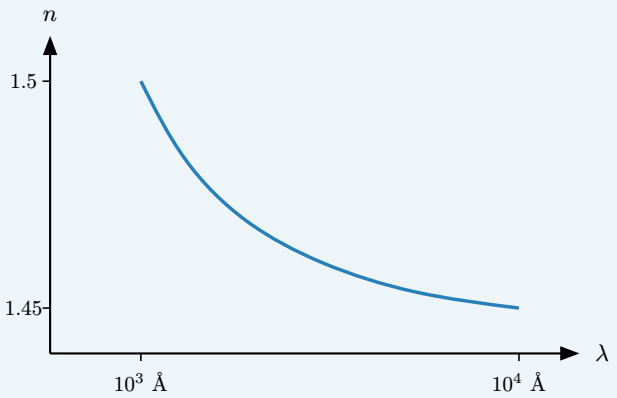


Figure 5.2: Index of refraction of typical glass as a function of wavelength.

This is how a prism works, and how a rainbow is formed by rain droplets.

We will also see **anomalous dispersion**, where the material experiences a resonance.

5.3.2 Phase and Group Velocity

In a dispersive medium, the wave velocity depends on the EM frequency, $v(\omega)$ (as it turns out, in quantum electrodynamics even the vacuum is dispersive!).

Each sinusoidal component travels at the **wave** (or **phase**) **velocity**

$$v = \frac{\omega}{k}, \tag{5.25}$$

whereas the envelope of a wave packet moves at the **group velocity**

$$v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k}. \tag{5.26}$$

5.3.3 A Model for the Frequency-Dependent Permittivity

We now want to develop a model for the frequency dependence of ε in **dielectrics**, i.e. non-conducting media in which the electrons are bound to atoms or molecules.

We describe the binding force with a **spring force**:

$$\boldsymbol{F}_{\text{binding}} = -k_{\text{spring}} x = -m\omega_0^2 x, \tag{5.27}$$

where x is the displacement of the electron from its equilibrium position relative to the nucleus, Figure 5.3, and

$$\omega_0 = \sqrt{\frac{k_{\text{spring}}}{m}}. \tag{5.28}$$

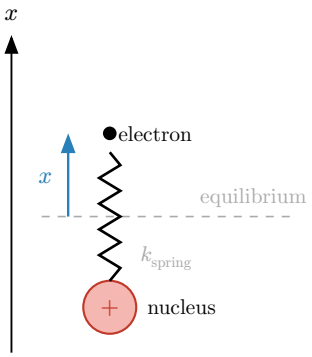


Figure 5.3: An electron bound to a nucleus by a spring force. The displacement x is measured from the equilibrium position.

Why a spring force is not a crazy model

If $U(x)$ was the potential of the (much more complicated) true binding force, we could expand it around the equilibrium point $x = 0$ with a Taylor series:

$$\begin{aligned} U(x) = & \underbrace{U(0)}_{\text{can be chosen arbitrarily}} + x \underbrace{U'(0)}_{=-F=0 \text{ (equilibrium)}} \\ & + \frac{1}{2}x^2 \underbrace{U''(0)}_{=k_{\text{spring}}>0 \text{ (stable equilibrium)}} + \ . \end{aligned} \tag{5.29}$$

Choosing $U(0) = 0$ and using $U'(0) = 0$, this becomes

$$U(x) = \frac{1}{2}U''(0)x^2 + O(x^3), \tag{5.30}$$

which is exactly the potential of a harmonic oscillator, with $k_{\text{spring}} \equiv U''(0) > 0$ (we will use the same line of argument in quantum mechanics).

We probably also have a **damping force** that acts against the velocity of the electron. The simplest model is

$$\boldsymbol{F}_{\text{damping}} = -m\gamma\frac{\mathrm{d}x}{\mathrm{d}t}. \tag{5.31}$$

And lastly, we have a **driving force** given by the electromagnetic wave, polarized in the x-direction:

$$\boldsymbol{F}_{\text{driving}} = q\boldsymbol{E} = qE_0\cos(\omega t). \tag{5.32}$$

The electron is then driven by the wave, Figure 5.4, which combines all three forces on the bound electron.

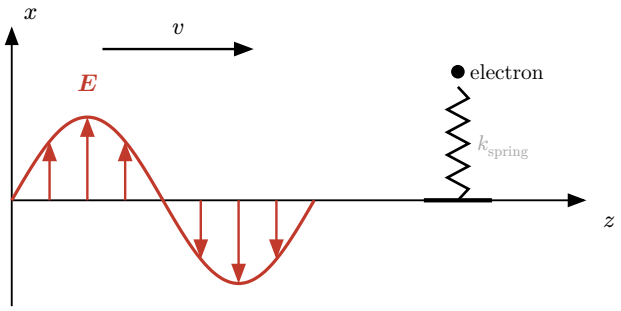


Figure 5.4: An electromagnetic wave (electric field $\boldsymbol{E}$ along x, propagating along z with velocity v) drives an electron bound to the axis by a spring.

Using Newton's second law,

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = F_{\text{binding}} + F_{\text{damping}} + F_{\text{driving}}, \tag{5.33}$$

we find

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + m\gamma\frac{\mathrm{d}x}{\mathrm{d}t} + m\omega_0^2 x = qE_0\cos(\omega t), \tag{5.34}$$

which is a **damped harmonic oscillator**, driven at frequency ω.

This is easier to solve in complex notation, i.e. Eq. (5.34) is the real part of

$$\frac{\mathrm{d}^2\tilde{x}}{\mathrm{d}t^2} + \gamma\frac{\mathrm{d}\tilde{x}}{\mathrm{d}t} + \omega_0^2\tilde{x} = \frac{q}{m}E_0 e^{-i\omega t}, \tag{5.35}$$

with $\tilde{x} = x + iy$.

In steady state, the system will oscillate at the driving frequency,

$$\tilde{x}(t) = \tilde{x}_0 e^{-i\omega t}. \tag{5.36}$$

Plugging this into Eq. (5.35) and solving for $\tilde{x}_0$, we find

$$\tilde{x}_0 = \frac{q/m}{\omega_0^2 - \omega^2 - i\gamma\omega}E_0. \tag{5.37}$$

In the end, we are interested in the polarization $\boldsymbol{P}$, which is the dipole moment per unit volume. Therefore, we calculate the dipole moment of one electron, which will be the real part of the following expression:

$$\tilde{p}(t) = q\tilde{x}(t) = \frac{q^2/m}{\omega_0^2 - \omega^2 - i\gamma\omega} E_0 e^{-i\omega t}. \tag{5.38}$$

The complex number in the denominator of Eq. (5.38) means that the dipole is **out of phase** with $\boldsymbol{E}$: it lags behind by an angle

$$\phi = \tan^{-1} \frac{\gamma\omega}{\omega_0^2 - \omega^2}. \tag{5.39}$$

Where does this phase come from?

$\tilde{p}$ has the form $\tilde{p} = \frac{A}{x - iy} e^{-i\omega t}$, with (comparing to Eq. (5.38)) $x = \omega_0^2 - \omega^2$, $y = \gamma\omega$, $A = q^2 E_0/m$.

Let's bring this to the form $\tilde{p} = \tilde{p}_0 e^{i(\phi - \omega t)}$ by multiplying numerator and denominator by the complex conjugate of the denominator:

$$\tilde{p} = \frac{A(x + iy)}{(x - iy)(x + iy)} e^{-i\omega t} = \frac{A}{x^2 + y^2}(x + iy) e^{-i\omega t}. \tag{5.40}$$

With $r^2 \equiv x^2 + y^2$ and $x + iy \equiv re^{i\phi}$ (as sketched in the complex plane, with $\tan\phi = y/x$), this becomes

$$\tilde{p} = \frac{A}{r} e^{i(\phi - \omega t)}. \tag{5.41}$$

So the dipole oscillates at the same frequency ω as the driving field, but with an extra phase ϕ. Using $y = \gamma\omega$ and $x = \omega_0^2 - \omega^2$ from Eq. (5.38), we recover

$$\tan\phi = \frac{y}{x} = \frac{\gamma\omega}{\omega_0^2 - \omega^2}, \tag{5.42}$$

i.e. Eq. (5.39). ■

5.3.4 From One Electron to the Polarization $\boldsymbol{P}$

Let's consider a medium made up of N molecules per unit volume, where each molecule has f_j electrons. Each electron may have a different (resonant) frequency ω_j and damping γ_j.

The polarization $\boldsymbol{P}$ is then the real part of

$$\tilde{\boldsymbol{P}}(t) = \frac{Nq^2}{m} \left(\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \right) \tilde{\boldsymbol{E}}. \tag{5.43}$$

In-class exercise

Previously, we considered the *real* case, where $\boldsymbol{E}$ and $\boldsymbol{P}$ were proportional to each other in linear media, $\boldsymbol{P} = \varepsilon_0 \chi_e \boldsymbol{E}$. Here, this is not the case — why?

The reason: $\tilde{\boldsymbol{P}}(t)$ and $\tilde{\boldsymbol{E}}(t)$ have a **phase difference** (Eq. (5.39)), so their *real parts* $\boldsymbol{P}(t)$ and $\boldsymbol{E}(t)$ are not proportional at every instant in time. But the *complex* $\tilde{\boldsymbol{P}}$ and $\tilde{\boldsymbol{E}}$ **are** proportional:

$$\tilde{\boldsymbol{P}} = \varepsilon_0 \tilde{\chi}_e \tilde{\boldsymbol{E}}, \tag{5.44}$$

with the **complex susceptibility** $\tilde{\chi}_e$.

All of our previous considerations for linear media carry over, and we can define the **complex permittivity** $\tilde{\varepsilon} = \varepsilon_0(1 + \tilde{\chi}_e)$ and the **complex dielectric constant**

$$\tilde{\varepsilon}_r = \frac{\tilde{\varepsilon}}{\varepsilon_0} = 1 + \frac{Nq^2}{m\varepsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j \omega}. \tag{5.45}$$

For $\gamma_j = 0$, we see that there's an infinity when $\omega \to \omega_j$ in $\tilde{\varepsilon}_r$. We call this a **resonance** (also in the case when $\gamma_j \neq 0$, where the divergence is tamed but there is still a pronounced peak).

The (complex) wave equation in a dispersive medium is (using $\mu \approx \mu_0$)

$$\nabla^2 \tilde{\boldsymbol{E}} = \tilde{\varepsilon} \mu_0 \frac{\partial^2 \tilde{\boldsymbol{E}}}{\partial t^2}, \tag{5.46}$$

which is solved by plane waves

$$\tilde{\boldsymbol{E}}(z,t) = \tilde{\boldsymbol{E}}_0 e^{i(\tilde{k}z - \omega t)}. \tag{5.47}$$

Just as with conductors, the wavenumber is complex, $\tilde{k} = k + i\kappa$, so

$$\tilde{\boldsymbol{E}}(z,t) = \tilde{\boldsymbol{E}}_0 e^{-\kappa z} e^{i(kz - \omega t)}, \tag{5.48}$$

but now the attenuation is not related to the conductivity, but to the *damping force* acting on the electrons.

The intensity is $I = \tilde{\boldsymbol{E}} \cdot \tilde{\boldsymbol{E}}^* \propto e^{-2\kappa z}$, and we call

$$\alpha \equiv 2\kappa \tag{5.49}$$

the **absorption coefficient**.

The last thing we want to do is find α and the index of refraction,

$$n = \frac{ck}{\omega} = \sqrt{\frac{\varepsilon \mu}{\varepsilon_0 \mu_0}} \approx \sqrt{\varepsilon_r} = \sqrt{1 + \chi_e}. \tag{5.50}$$

So we need the real and imaginary parts of

$$\tilde{k} = \frac{\omega}{c} \sqrt{\tilde{\varepsilon}_r} = \frac{\omega}{c} \sqrt{1 + \tilde{\chi}_e}. \tag{5.51}$$

Since $\tilde{\chi}_e$ is typically small, we Taylor-expand $\sqrt{1+\tilde{\chi}_e}$ around $\tilde{\chi}_e = 0$, keeping only the linear term:

$$\tilde{k} = \frac{\omega}{c}\sqrt{1+\tilde{\chi}_e} \approx \frac{\omega}{c}\left(1+\frac{1}{2}\tilde{\chi}_e\right), \tag{5.52}$$

$\sqrt{1+\tilde{\chi}_e} \approx 1+\frac{1}{2}\tilde{\chi}_e$

and using Eq. (5.45) for $\tilde{\chi}_e$:

$$\tilde{k} \approx \frac{\omega}{c}\left[1+\frac{Nq^2}{2m\varepsilon_0}\sum_j \frac{f_j}{\omega_j^2-\omega^2-i\gamma_j\omega}\right]. \tag{5.53}$$

The index of refraction is

$$n = \frac{c}{\omega}\,\mathrm{Re}(\tilde{k}) \approx \mathrm{Re}\left(1+\frac{1}{2}\tilde{\chi}_e\right). \tag{5.54}$$

To extract the real part, we rationalize the denominator of each term in Eq. (5.53) by multiplying by the complex conjugate:

$$\frac{1}{\omega_j^2-\omega^2-i\gamma_j\omega} = \frac{\omega_j^2-\omega^2+i\gamma_j\omega}{\left(\omega_j^2-\omega^2\right)^2+\gamma_j^2\omega^2}, \tag{5.55}$$

so that

$$n \approx 1+\frac{Nq^2}{2m\varepsilon_0}\sum_j \frac{f_j\left(\omega_j^2-\omega^2\right)}{\left(\omega_j^2-\omega^2\right)^2+\gamma_j^2\omega^2}. \tag{5.56}$$

And the **absorption coefficient** is

$$\alpha = 2\kappa = 2\,\mathrm{Im}(\tilde{k}) \approx 2\frac{\omega}{c}\,\mathrm{Im}\left(1+\frac{1}{2}\tilde{\chi}_e\right) = \frac{\omega}{c}\,\mathrm{Im}(\tilde{\chi}_e). \tag{5.57}$$

Using the imaginary part of the rationalized fraction above (i.e. $\gamma_j\omega$ in the numerator, instead of $\omega_j^2-\omega^2$):

$$\alpha \approx \frac{Nq^2\omega^2}{m\varepsilon_0 c}\sum_j \frac{f_j\gamma_j}{\left(\omega_j^2-\omega^2\right)^2+\gamma_j^2\omega^2}. \tag{5.58}$$

In-class exercise

Check this yourself: verify the rationalization step above, and that you recover Eq. (5.56) and Eq. (5.58).

Let's sketch α and n in the vicinity of one of the resonances ω_j, see Figure 5.5.

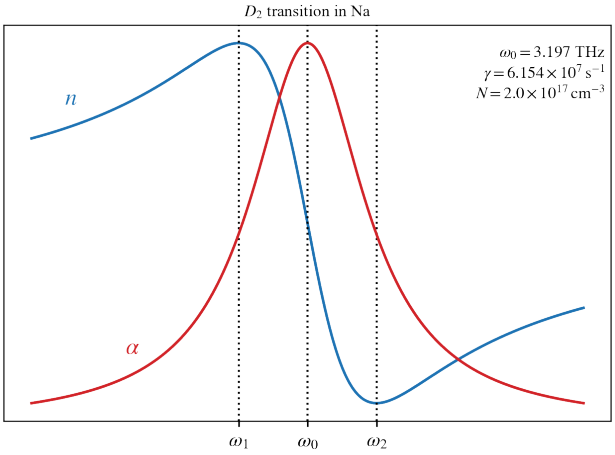


Figure 5.5: Index of refraction n and absorption coefficient α near the D_2 resonance of sodium.

Away from the resonance ($\omega < \omega_1$ or $\omega > \omega_2$), n increases slowly with ω — this is the behavior we saw in e.g. glass (Figure 5.2), called **normal dispersion**.

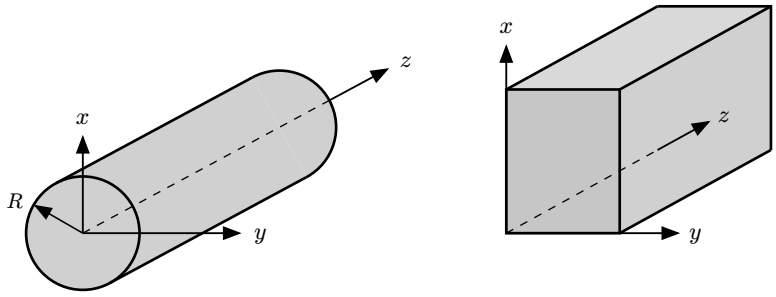
In the region $\omega_1 < \omega_j < \omega_2$ (surrounding ω_j), we instead see a rapid *drop* of n with increasing ω — called **anomalous dispersion** — together with a pronounced peak in the absorption α_j: the electrons are being driven at (close to) their "favorite" frequency ω_j, leading to oscillations with large amplitude, and hence a maximum amount of energy dissipated through damping. The material can become **opaque** in this region.

6 Guided Waves

Literature: Griffiths Ch. 9.5

In the previous lectures we saw how EM waves propagate in (linear) dielectrics, and what happens at the boundary between two types of media — we essentially looked at infinite planes. We also studied conductors.

Now we want to study how EM waves propagate inside **wave guides**, i.e. inside hollow pipes — e.g. an optical fiber, or a coaxial cable. Two common examples are the **cylindrical wave guide** (a hollow tube of radius R) and the **rectangular wave guide** (a hollow pipe of rectangular cross-section, with side lengths a and b along x and y, and the wave propagating along z). We will focus on the rectangular case.



cylindrical wave guide rectangular wave guide

Figure 6.1: The two example wave guide geometries: a cylindrical wave guide of radius R, and a rectangular wave guide with side lengths a and b. In both cases the wave propagates along z.

6.1 Motivation: Why Use Wave Guides?

Why should we use wave guides at all to transmit information (e.g. sound, light, ...)?

The intensity of a wave, measured at some distance d from its source, drops with d^2: if the source radiates into all directions, the power P it emits is spread out over a sphere of area $A = 4\pi R^2$ at distance R, and since power is intensity times area, $P = I \cdot A$, the intensity at the receiver falls off as $1/R^2$.

A receiver of area A_{receiver}, at distance R from the transmitter, only intercepts a fraction A_{receiver}/A of the total power, as illustrated in Figure 6.2.

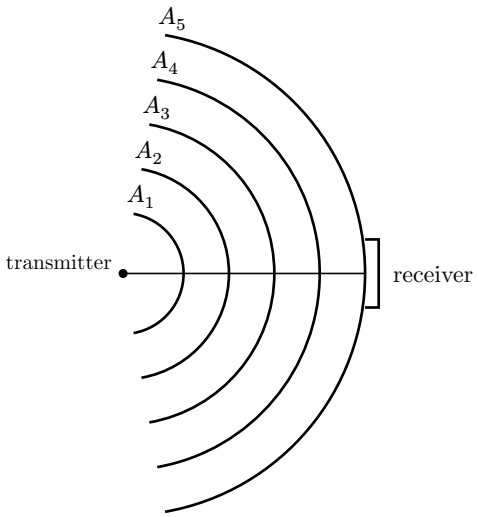


Figure 6.2: The wavefronts emitted by a point source grow in area ($A_1 < A_2 < ...$) as they spread out, so the power per unit area — the intensity — drops off as $1/R^2$.

The power received is

$$P_{\text{receiver}} = \frac{A_{\text{receiver}}}{A} P_{\text{transmitter}}, \quad A = 4\pi R^2. \tag{6.1}$$

Example: listening from 1 km away

Take a receiver with the area of a human ear, $A_{\text{receiver}} \sim 10\text{cm}^2$. Let's take a distance of $R = 1$ km to compute the area A_5 in Figure 6.2. Then, the received power is

$$P_{\text{receiver}} = \frac{10\text{cm}^2}{4\pi(1\text{ km})^2} P_{\text{transmitter}} \approx 10^{-10} P_{\text{transmitter}}. \tag{6.2}$$

Essentially all of the transmitted power is lost!

6.2 Waves in a Hollow Conductor

To avoid enormous losses when transmitting over long distances, we need a wave guide as depicted in Figure 6.3.

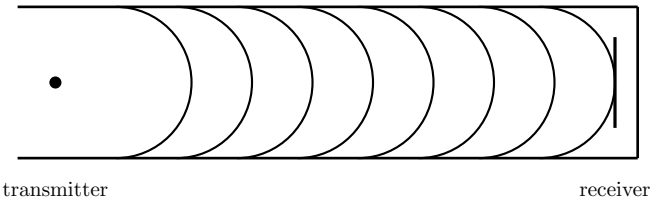


Figure 6.3: A wave guide carries a wave from a transmitter to a receiver, confined to the inside of the tube.

We'll make the simplifying assumption that the wave guide is a **perfect conductor**, i.e. $\sigma = \infty$. As a consequence, inside the material of the wave guide,

$$\boldsymbol{E} = \boldsymbol{B} = \boldsymbol{0}. \tag{6.3}$$

Why $\boldsymbol{E} = \boldsymbol{B} = \boldsymbol{0}$ inside a perfect conductor

1. $\boldsymbol{E} = \boldsymbol{0}$: inside a conductor, $\boldsymbol{J}_f = \sigma \boldsymbol{E}$ (Eq. (5.1)). For $\sigma = \infty$, $\boldsymbol{J}_f$ would be infinite unless $\boldsymbol{E} = \boldsymbol{0}$ — free charges immediately short out any electric field. Then, from Faraday's law,

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} = \boldsymbol{0}, \tag{6.4}$$

so if $\boldsymbol{B}(t = 0) = \boldsymbol{0}$, then $\boldsymbol{B}(t) = \boldsymbol{0}$ for all time.

2. $\boldsymbol{E} = \boldsymbol{0}$ in a perfect conductor is also what we found when EM waves impinge on a conductor last lecture: there, $\tilde{\boldsymbol{E}} \propto e^{-\kappa z}$, with $\kappa \to \infty$ as $\sigma \to \infty$ (Eq. (5.9)) — the field is attenuated to zero instantly.

6.2.1 Boundary Conditions

We again use the general boundary conditions from Section 2.5, now between the inside of the wave guide (medium 1) and the inside of the conducting wall (medium 2). Our geometry is shown in Figure 6.4.

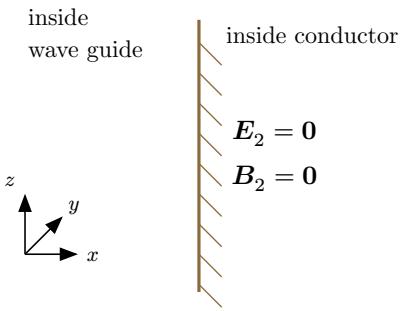


Figure 6.4: The boundary between the inside of the wave guide and the inside of the conducting wall.

Inside the conductor, $\boldsymbol{E}_2^{\parallel} = \boldsymbol{0}$ and $\boldsymbol{B}_2^{\perp} = 0$ (Eq. (6.3)), so continuity of the parallel $\boldsymbol{E}$ (Eq. (2.41)) and perpendicular $\boldsymbol{B}$ (Eq. (2.37)) across the boundary gives

$$\left.\begin{array}{l} \boldsymbol{E}_2^{\parallel} = \boldsymbol{0} \Rightarrow \boldsymbol{E}_1^{\parallel} = \boldsymbol{0} \\ B_2^{\perp} = 0 \Rightarrow B_1^{\perp} = 0 \end{array}\right\} \boldsymbol{E}^{\parallel} = \boldsymbol{0}, B^{\perp} = 0 \quad \text{inside the wave guide.} \tag{6.5}$$

Free charges and currents will form on the surface of the wall to enforce these conditions.

6.3 Monochromatic Waves in a Wave Guide

Let's consider monochromatic waves that propagate down the tube in the z-direction:

$$\tilde{\boldsymbol{E}}(x,y,z,t) = \tilde{\boldsymbol{E}}_0(x,y)e^{i(kz-\omega t)}, \quad \tilde{\boldsymbol{B}}(x,y,z,t) = \tilde{\boldsymbol{B}}_0(x,y)e^{i(kz-\omega t)}. \tag{6.6}$$

In-class exercise

Is the wave number k a real or a complex number in this case?

Two remarks on this ansatz:

1. The complex amplitudes $\tilde{\boldsymbol{E}}_0$ and $\tilde{\boldsymbol{B}}_0$ are only functions of x, y — not z — i.e. we look at harmonic waves in z of the form $e^{i(kz-\omega t)}$.
2. $\tilde{\boldsymbol{E}}$ and $\tilde{\boldsymbol{B}}$ are, in general, **not** plane waves: they are not constant on planes of constant z.

Our job is to solve Maxwell's equations, subject to the boundary conditions Eq. (6.5).

Let's write out the components of our ansatz Eq. (6.6) (leaving the subscript "0" and the tilde off the components for brevity):

$$\begin{aligned} \tilde{\boldsymbol{E}}_0 &= E_x(x,y)\hat{\boldsymbol{x}} + E_y(x,y)\hat{\boldsymbol{y}} + E_z(x,y)\hat{\boldsymbol{z}}, \\ \tilde{\boldsymbol{B}}_0 &= B_x(x,y)\hat{\boldsymbol{x}} + B_y(x,y)\hat{\boldsymbol{y}} + B_z(x,y)\hat{\boldsymbol{z}}. \end{aligned} \tag{6.7}$$

In particular, we are **not** assuming that the waves are transverse — i.e. we do not assume $E_z = B_z = 0$ as we did in the vacuum case for $\hat{\boldsymbol{k}} = (0,0,1)$. We will need this generality in order to satisfy the boundary conditions Eq. (6.5).

Proof that E and B cannot both be transverse

In fact, we can show that $\boldsymbol{E}$ and $\boldsymbol{B}$ cannot **both** be transverse.

Suppose $E_z = B_z = 0$. Since E_x, E_y, E_z have no explicit z-dependence (they only depend on x, y), it follows that $\boldsymbol{\nabla} \times \tilde{\boldsymbol{E}}_0 = \mathbf{0}$:

$$\underbrace{\frac{\partial E_z}{\partial y}}_{=0 \text{ by assumption, } E_z=0} - \underbrace{\frac{\partial E_y}{\partial z}}_{=0 \text{ since } E_y \text{ has no } z\text{-dependence}} = 0 - 0 = 0,$$

$$\underbrace{\frac{\partial E_x}{\partial z}}_{=0 \text{ since } E_x \text{ has no } z\text{-dependence}} - \underbrace{\frac{\partial E_z}{\partial x}}_{=0 \text{ by assumption, } E_z=0} = 0 - 0 = 0, \tag{6.8}$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -\frac{\partial B_z}{\partial t} = 0 \quad \text{(by assumption, } B_z = 0 \text{).}$$

Since $\boldsymbol{\nabla} \times \tilde{\boldsymbol{E}}_0 = \mathbf{0}$, we can write it as the gradient of a scalar function, $\tilde{\boldsymbol{E}}_0 = \boldsymbol{\nabla}\varphi$. Plugging this into Maxwell's equation (i), $\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0$ (no free charge inside the wave guide):

$$0 = \boldsymbol{\nabla} \cdot \tilde{\boldsymbol{E}} = \boldsymbol{\nabla} \cdot \tilde{\boldsymbol{E}}_0 = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla}\varphi) = \nabla^2 \varphi, \tag{6.9}$$

— a Laplace equation for φ.

The boundary condition $\boldsymbol{E}^{\parallel} = \mathbf{0}$ on the inner wall (Eq. (6.5)) implies $\boldsymbol{\nabla}\varphi = \mathbf{0}$ on the wall, so φ is constant on the wall. Finally, solutions to Laplace's equation don't have local minima or maxima inside the region (the **maximum principle**) — so φ must be constant everywhere inside the wave guide too. Hence

$$\tilde{\boldsymbol{E}}_0 = \boldsymbol{\nabla}\varphi = \mathbf{0}. \tag{6.10}$$

No TEM waves are possible in a hollow wave guide. ■

Note that this argument only holds for an **empty** wave guide — it is not true for a coaxial cable, which has a second (inner) conductor, so that φ need not be constant on the (now disconnected) boundary.

6.3.1 Solving Maxwell's Equations in a hollow wave guide

Let's now use our ansatz, Eq. (6.7) and the boundary conditions Eq. (6.5) to solve Maxwell's equations.

We proceed in 3 steps.

Step 1. Let's plug our ansatz into Maxwell's equations (iii) and (iv). Using $\frac{\partial}{\partial z} \to ik$ and $\frac{\partial}{\partial t} \to -i\omega$ for our ansatz Eq. (6.6), Faraday's law, $\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$, gives three component equations:

$$\begin{aligned}
(x \text{ component}) \quad & \frac{\partial E_z}{\partial y} - \underbrace{\frac{\partial E_y}{\partial z}}_{=ikE_y} = \underbrace{-\frac{\partial B_x}{\partial t}}_{=i\omega B_x} \\
(y \text{ component}) \quad & ikE_x - \frac{\partial E_z}{\partial x} = i\omega B_y \\
(z \text{ component}) \quad & \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = i\omega B_z.
\end{aligned} \tag{6.11}$$

And Ampère's law without free current inside the wave guide, $\boldsymbol{\nabla} \times \boldsymbol{B} = (1/c^2)\frac{\partial \boldsymbol{E}}{\partial t}$, gives

$$\begin{aligned} \frac{\partial B_z}{\partial y} - ikB_y &= -i\frac{\omega}{c^2}E_x \\ ikB_x - \frac{\partial B_z}{\partial x} &= -i\frac{\omega}{c^2}E_y \\ \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} &= -i\frac{\omega}{c^2}E_z. \end{aligned} \tag{6.12}$$

Step 2. From these six equations, we can get four equations for E_x, E_y, B_x, B_y that only depend on E_z, B_z (Griffiths Problem 9.27a). As an example, let's solve for E_x and B_y: the second equation in Eq. (6.11) and the first equation in Eq. (6.12) form a linear system in E_x, B_y,

$$\begin{aligned} ikE_x - i\omega B_y &= \frac{\partial E_z}{\partial x} \\ i\frac{\omega}{c^2}E_x - ikB_y &= -\frac{\partial B_z}{\partial y}. \end{aligned} \tag{6.13}$$

Multiplying the first line by k, the second by ω, and subtracting eliminates B_y:

$$i\left(k^2 - \frac{\omega^2}{c^2}\right)E_x = k\frac{\partial E_z}{\partial x} + \omega\frac{\partial B_z}{\partial y} \quad \Rightarrow \quad E_x = \frac{i}{\left(\frac{\omega}{c}\right)^2 - k^2}\left[k\frac{\partial E_z}{\partial x} + \omega\frac{\partial B_z}{\partial y}\right] \tag{6.14}$$

Solving the same pair of equations for B_y, and the other pair (the first equation in Eq. (6.11) together with the second in Eq. (6.12)) for E_y, B_x, gives

$$\begin{aligned} E_x &= \frac{i}{\left(\frac{\omega}{c}\right)^2 - k^2}\left[k\frac{\partial E_z}{\partial x} + \omega\frac{\partial B_z}{\partial y}\right], \\ E_y &= \frac{i}{\left(\frac{\omega}{c}\right)^2 - k^2}\left[k\frac{\partial E_z}{\partial y} - \omega\frac{\partial B_z}{\partial x}\right], \\ B_x &= \frac{i}{\left(\frac{\omega}{c}\right)^2 - k^2}\left[k\frac{\partial B_z}{\partial x} - \frac{\omega}{c^2}\frac{\partial E_z}{\partial y}\right], \\ B_y &= \frac{i}{\left(\frac{\omega}{c}\right)^2 - k^2}\left[k\frac{\partial B_z}{\partial y} + \frac{\omega}{c^2}\frac{\partial E_z}{\partial x}\right]. \end{aligned} \tag{6.15}$$

Note:
1. We see explicitly that if both E_z and B_z were zero, $\tilde{\boldsymbol{E}}_0$

and $\tilde{\boldsymbol{B}}_0$ would be zero too — consistent with Eq. (6.10).
2. All that remains is to find E_z and B_z.

6.3.2 Decoupled Equations for E_z and B_z

Step 3: Use Maxwell's equations (i) and (ii) to obtain a differential equation for E_z and B_z, by plugging the expressions Eq. (6.15) for E_x, E_y, B_x, B_y into $\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0$ and $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$ (Griffiths Problem 9.27b).

This yields two **uncoupled** differential equations for E_z and B_z:

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c}\right)^2 - k^2\right] X_z = 0, \quad X_z = E_z \text{ or } B_z. \tag{6.16}$$

If $E_z = 0$, the waves are called **transverse electric** (TE) waves; if $B_z = 0$, they are **transverse magnetic** (TM) waves; and if $E_z = B_z = 0$, they would be **TEM** waves — which we have just shown cannot occur in a hollow wave guide (though EM waves in vacuum, and waves in a coaxial cable, *are* TEM waves).

In-class exercise

Try to see that Eq. (6.16) can also be obtained by plugging in Eq. (6.15) into, e.g., $\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0$.

Also, take a second and discuss with your neighbor: retrace the steps we took to get to Eq. (6.16), and see if you can understand why we had to go through all three steps. Why can't we just plug the ansatz Eq. (6.6) into Maxwell's equations and solve for E_z and B_z directly?

6.4 TE Waves in a Rectangular Wave Guide

Literature: Griffiths Ch. 9.5.2

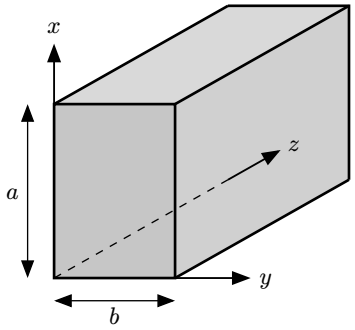


Figure 6.5: The rectangular wave guide, with side lengths a (along x) and b (along y), propagating along z.

Let's look at TE waves, i.e. $E_z = 0$. We are then left with solving Eq. (6.16) for B_z alone,

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c}\right)^2 - k^2\right] B_z = 0, \tag{6.17}$$

subject to the boundary conditions Eq. (6.5). (We could just as well work with TM modes.)

6.4.1 Separation of Variables

We proceed by **separation of variables** — very similar to how we solve the 1D or 3D Schrödinger equation in quantum mechanics. We make the ansatz

$$B_z(x, y) = X(x)Y(y). \tag{6.18}$$

Plugging this into Eq. (6.17):

$$Y\frac{\partial^2 X}{\partial x^2} + X\frac{\partial^2 Y}{\partial y^2} + \left[\left(\frac{\omega}{c}\right)^2 - k^2\right] XY = 0. \tag{6.19}$$

Dividing by XY:

$$\underbrace{\frac{1}{X}\frac{\partial^2 X}{\partial x^2}}_{\text{only depends on } x} + \underbrace{\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2}}_{\text{only depends on } y} + \left[\left(\frac{\omega}{c}\right)^2 - k^2\right] = 0. \tag{6.20}$$

Since the first term only depends on x and the second only on y (and their sum, together with the constant third term, must vanish for **all** x, y), a change in x can never be compensated by a change in y unless both terms are themselves constant. Let's call these constants $-k_x^2$ and $-k_y^2$:

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = -k_x^2, \quad \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} = -k_y^2, \tag{6.21}$$

so that, from Eq. (6.20),

$$k_x^2 + k_y^2 + k^2 = \left(\frac{\omega}{c}\right)^2. \tag{6.22}$$

The two equations of Eq. (6.21) are just harmonic oscillators, with solutions

$$X(x) = A\sin(k_x x) + B\cos(k_x x), \quad Y(y) = C\sin(k_y y) + D\cos(k_y y). \tag{6.23}$$

6.4.2 Boundary Conditions and the TE_{mn} Modes

Now we use the boundary condition $B^\perp = 0$ (Eq. (6.5)) to fix A, B, C, D, k_x, k_y, see Figure 6.6.

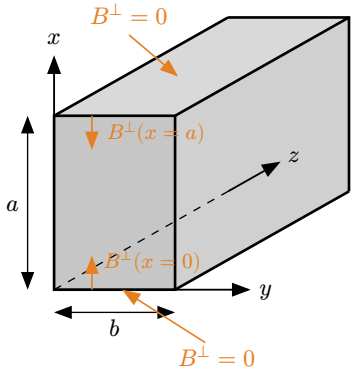


Figure 6.6: The boundary condition $B^\perp = 0$ applied to the walls $x = 0$ and $x = a$ of the rectangular wave guide.

At the walls $x = 0$ and $x = a$, the perpendicular component of $\boldsymbol{B}$ is B_x, so

$$\begin{aligned} & B_\perp(x = 0, y) = B_\perp(x = a, y) = 0. \\ \Leftrightarrow \quad & B_x(x = 0, y) = B_x(x = a, y) = 0. \end{aligned} \tag{6.24}$$

B_x is found from Eq. (6.15), with $E_z = 0$ (TE waves),

$$B_x = \frac{i}{\left(\frac{\omega}{c}\right)^2 - k^2} k \frac{\partial B_z}{\partial x} = 0, \tag{6.25}$$

for $x = 0$ and $x = a$. From this, it follows that

$$\frac{\partial B_z}{\partial x}(0, y) = \frac{\partial B_z}{\partial x}(a, y) = 0. \tag{6.26}$$

So the condition $B_x = 0$ at $x = 0, a$ is equivalent to

$$\frac{\partial X}{\partial x}(0) = \frac{\partial X}{\partial x}(a) = 0. \tag{6.27}$$

From Eq. (6.23),

$$\frac{\partial X}{\partial x} = Ak_x \cos(k_x x) - Bk_x \sin(k_x x). \tag{6.28}$$

At $x = 0$:

$$\frac{\partial X}{\partial x}(0) = Ak_x \cos(0) - Bk_x \sin(0) = Ak_x = 0 \Rightarrow A = 0, \tag{6.29}$$

assuming $k_x \neq 0$. At $x = a$, with $A = 0$ we have

$$\frac{\partial X}{\partial x}(a) = -Bk_x \sin(k_x a) = 0, \tag{6.30}$$

so (for $B \neq 0$),

$$\sin(k_x a) = 0 \quad \Rightarrow \quad k_x = \frac{m\pi}{a}, \quad m = 0, 1, 2, ... \tag{6.31}$$

and $X(x) = B\cos(m\pi x/a)$.

By the identical argument at $y = 0, b$ (using $B_y = 0$ there), we find $C = 0$ and

$$k_y = \frac{n\pi}{b}, \quad n = 0, 1, 2, ... \tag{6.32}$$

so that $Y(y) = D\cos(n\pi y/b)$.

Putting everything together, the full solution is

$$B_z(x, y) = B_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \tag{6.33}$$

— the TE_{mn} **mode**.

(By convention, the first index labels the larger dimension, $a \geq b$, and at least one of m, n must be nonzero — else Eq. (6.33) is constant and we recover the (impossible) TEM case.)

6.4.3 Dispersion Relation, Cutoff Frequency, and Wave Velocities

Plugging Eq. (6.31) and Eq. (6.32) into Eq. (6.22) and solving for k we obtain the **dispersion relation** for the TE_{mn} mode:

$$k = \frac{1}{c}\sqrt{\omega^2 - \omega_{mn}^2}, \quad \omega_{mn}^2 \equiv \underbrace{c^2\pi^2\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]}_{=c^2(k_x^2+k_y^2)}. \tag{6.34}$$

An analogy with special relativity

Compare Eq. (6.34) to the relativistic energy-momentum relation,

$$E^2 = p^2c^2 + m^2c^4 \quad \Leftrightarrow \quad p = \frac{1}{c}\sqrt{E^2 - m^2c^4}. \tag{6.35}$$

With $\omega \leftrightarrow E/\hbar$ and $k \leftrightarrow p/\hbar$, Eq. (6.34) has exactly this form, with ω_{mn} playing the role of an **effective mass**!

If $\omega < \omega_{mn}$, k becomes imaginary, and the fields are exponentially attenuated rather than propagating — ω_{mn} is called the **cutoff frequency** of the TE_{mn} mode. The lowest cutoff frequency (for $a \geq b$) is

$$\omega_{10} = \frac{c\pi}{a}, \tag{6.36}$$

and waves with $\omega < \omega_{10}$ will not propagate down the guide at all, regardless of mode.

The wave (phase) velocity is

$$v = \frac{\omega}{k} = \frac{c}{\sqrt{1 - (\omega_{mn}/\omega)^2}} > c. \tag{6.37}$$

This exceeds c — but that's fine, since it's not the speed at which energy or information travels. That is instead the **group velocity**,

$$v_g = \left(\frac{\mathrm{d}k}{\mathrm{d}\omega}\right)^{-1} = c\sqrt{1 - (\omega_{mn}/\omega)^2} < c. \tag{6.38}$$

Note that $v \cdot v_g = c^2$.

6.5 Outlook

This concludes our discussion of electromagnetic waves. We have seen how EM waves are reflected and transmitted at the boundary of linear dielectrics and conductors, studied the frequency dependence of the dielectric constant in dispersive media, and seen how waves propagate in hollow wave guides.

The formalism we developed here can also be used to study:

- **Coaxial lines**, which (unlike a hollow wave guide) also allow for TEM modes — see Griffiths Ch. 9.5.3. and the problem set 6.
- **Resonant cavities**, obtained by closing off a wave guide with a conducting cap at both ends (Griffiths Problem 9.40 and problem set 7).

Resonant cavities and axion dark matter

Resonant cavities are a key tool in the search for **axion dark matter** (recall the discussion in Chapter 2): if axions exist and constitute dark matter, they would source a tiny oscillating electric current in the presence of a strong magnetic field. Tuning a resonant cavity to match the (unknown) axion oscillation frequency can resonantly enhance this tiny signal — the basis of experiments like ADMX, and work being done here at SDU.

7 Potential Formulation of Electrodynamics

Literature: Griffiths Ch. 10.1 & 10.2

After electromagnetic waves, we are going to switch gears now. We will discuss **potentials**: the potential formulation underlies all of modern field theory (quantum field theory, used to describe all known elementary particles and their interactions). The potential formulation in electrodynamics is also very useful to describe radiation of accelerating charges, as we will see later in the lecture.

We will for now work in non-relativistic notation (3-vectors), but later realize that we can rewrite everything very compactly using relativistic 4-vector notation.

In-class exercise

How are the electric and magnetic fields derived from potentials in the electrostatic case?

Why should we bother rewriting Maxwell's equations with potentials? In the static case, for given charge and current densities, $\boldsymbol{E}$ and $\boldsymbol{B}$ are determined from Coulomb's and Biot–Savart's laws. Not anymore in electrodynamics — but we can find generalizations working with potentials.

7.1 The Vector and Scalar Potentials

Reminder: Coulomb's and Biot–Savart's laws

1. Coulomb's law: the force experienced by a test charge Q from charges q_i at rest,

$$\boldsymbol{F} = Q \cdot \boldsymbol{E}(\boldsymbol{r}), \quad \boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{N} q_i \frac{\boldsymbol{r} - \boldsymbol{r}_i}{|\boldsymbol{r} - \boldsymbol{r}_i|^3}, \tag{7.1}$$

where $\boldsymbol{r}_i$ is the position of charge q_i.

2. The Biot–Savart law determines the magnetic field of a steady (i.e. constant in time) line current I,

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} I \int \mathrm{d}\boldsymbol{l}' \times \frac{\boldsymbol{r} - \boldsymbol{r}'}{|\boldsymbol{r} - \boldsymbol{r}'|^3}, \tag{7.2}$$

where primed coordinates are the source region.

In-class exercise

Why don't Coulomb's and Biot–Savart's laws hold any longer in electrodynamics?

Coulomb's and Biot–Savart's laws silently assume **instant action at a distance**: the field "right now" is computed from the charges and currents "right now," with no time delay, no matter how far away the source is. This is fine in statics, since nothing is changing anyway, but once charges and currents move and change in time, it clashes with a basic fact of nature: no information can travel faster than the speed of light. If a charge jiggles on the far side of the room, we cannot possibly feel it instantaneously — the "news" first has to reach us, and that takes time. (Compare: we don't see what the Sun is doing **right now**, but what it was doing 8 minutes

ago, which is the time light takes to reach us.) We will make this precise with the **retarded time** later in this chapter (Eq. (7.28)), which fixes exactly this gap.

Let's start by rewriting Maxwell's equations Eq. (1.22) in terms of potentials. We start with (ii), $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$.

Just as in the electrostatic case, this equation can be automatically fulfilled by a vector field $\boldsymbol{A}$ for which

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{7.3}$$

since the divergence of a curl is always zero. We call $\boldsymbol{A}$ the **vector potential**.

With this, Faraday's law (iii) becomes

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \boldsymbol{A}) \quad \Rightarrow \quad \boldsymbol{\nabla} \times \left(\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t} \right) = \boldsymbol{0}. \tag{7.4}$$

In the parentheses (...) we have a vector field for which the curl vanishes. Such a vector field can always be written as the gradient of a scalar function (see tutorial 1). Let's call it $V(\boldsymbol{r}, t)$, the **scalar potential**.

So $\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t} = -\boldsymbol{\nabla} V$, i.e.

$$\boldsymbol{E} = -\boldsymbol{\nabla} V - \frac{\partial \boldsymbol{A}}{\partial t}. \tag{7.5}$$

What about the other two Maxwell equations, (i) and (iv)? Let's check explicitly.

From Gauss's law, $\boldsymbol{\nabla} \cdot \boldsymbol{E} = \rho/\varepsilon_0$:

$$\begin{aligned} \frac{\rho}{\varepsilon_0} &= \boldsymbol{\nabla} \cdot \left(-\boldsymbol{\nabla} V - \frac{\partial \boldsymbol{A}}{\partial t} \right) \\ &= -\nabla^2 V - \frac{\partial}{\partial t}(\boldsymbol{\nabla} \cdot \boldsymbol{A}) \end{aligned} \tag{7.6}$$

So we find

$$\nabla^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla} \cdot \boldsymbol{A}) = -\frac{\rho}{\varepsilon_0}, \tag{7.7}$$

which is the generalization of the Poisson equation $\nabla^2 V = -\rho/\varepsilon_0$ of electrostatics.

Lastly, let's check (iv), Ampère's law with Maxwell's addition. Using Eq. (7.3) and Eq. (7.5),

$$\begin{aligned} \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{A}) &= \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(-\boldsymbol{\nabla} V - \frac{\partial \boldsymbol{A}}{\partial t} \right) \\ &= \mu_0 \boldsymbol{J} + \frac{1}{c^2} \left(-\frac{\partial^2 \boldsymbol{A}}{\partial t^2} - \boldsymbol{\nabla} \frac{\partial V}{\partial t} \right), \end{aligned} \tag{7.8}$$

where we used $\mu_0 \varepsilon_0 = 1/c^2$. For the left-hand side we use the curl-of-a-curl identity from Section 3.7,

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}. \tag{7.9}$$

Collecting all terms including potentials on the right-hand side,

$$\nabla^2 \boldsymbol{A} - \frac{1}{c^2}\frac{\partial^2 \boldsymbol{A}}{\partial t^2} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \boldsymbol{A} + \frac{1}{c^2}\frac{\partial V}{\partial t}\right) = -\mu_0 \boldsymbol{J}. \tag{7.10}$$

The d'Alembert operator

Remember the d'Alembert operator (or "box" operator),

$$\Box \equiv \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}, \tag{7.11}$$

so that the first two terms of Eq. (7.10) combine into $\Box \boldsymbol{A}$.

Eq. (7.7) and Eq. (7.10) together contain **all** of Maxwell's equations.

In-class exercise

How many equations are these (counting components)?

Eq. (7.7) and Eq. (7.10) certainly don't look nice, but once $\boldsymbol{A}$ and V are determined, we can calculate the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ from Eq. (7.3) and Eq. (7.5).

7.2 Gauge Freedom

We have some **freedom** in the choice of $\boldsymbol{A}$ and V, because for a physical system with given $\boldsymbol{E}$ and $\boldsymbol{B}$, several different $\boldsymbol{A}$ and V yield the **same** field configurations $\boldsymbol{E}$ and $\boldsymbol{B}$.

This is the case because the equations for $\boldsymbol{E}$ and $\boldsymbol{B}$, Eq. (7.3) and Eq. (7.5), don't uniquely determine $\boldsymbol{A}$ and V. We can impose some extra conditions on $\boldsymbol{A}$ and V without changing the physical fields $\boldsymbol{E}$ and $\boldsymbol{B}$.

This freedom is called **gauge freedom** (pronounced "gauge" as in "cage"), and can be used to greatly simplify Eq. (7.7) and Eq. (7.10). This means that we can change $\boldsymbol{A}$ and V by so-called **gauge transformations** without changing $\boldsymbol{E}$ and $\boldsymbol{B}$.

Gauge freedom and the standard model

The idea of gauge freedom is a key building block of the standard model of particle physics.

Let's see how this works exactly. Given two vector potentials $\boldsymbol{A}$ and $\boldsymbol{A}'$, what's their connection to yield the same $\boldsymbol{B}$ field?

Take $\boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\alpha}$. What's the condition on $\boldsymbol{\alpha}$? Taking the curl,

$$\boldsymbol{\nabla} \times \boldsymbol{A}' = \boldsymbol{\nabla} \times \boldsymbol{A} + \boldsymbol{\nabla} \times \boldsymbol{\alpha}, \tag{7.12}$$

tells us that we need $\boldsymbol{\nabla} \times \boldsymbol{\alpha} = \boldsymbol{0}$ in order for $\boldsymbol{B}$ to remain unchanged. This is true if $\boldsymbol{\alpha}$ can be written as a gradient of a scalar function,

$$\boldsymbol{\alpha} = \boldsymbol{\nabla}\lambda, \quad \text{so that} \quad \boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\lambda. \tag{7.13}$$

By demanding that $\boldsymbol{E}$ is unchanged, we can find the allowed gauge transformation for the scalar potential, $V' = V + \beta$:

$$\boldsymbol{E} = -\frac{\partial \boldsymbol{A}'}{\partial t} - \boldsymbol{\nabla} V' = \underbrace{-\frac{\partial \boldsymbol{A}}{\partial t} - \boldsymbol{\nabla} V}_{=\boldsymbol{E}} - \boldsymbol{\nabla}\left(\frac{\partial \lambda}{\partial t} + \beta\right). \tag{7.14}$$

So that $\boldsymbol{E}$ is unchanged, we need $\boldsymbol{\nabla}\left(\frac{\partial \lambda}{\partial t} + \beta\right) = \boldsymbol{0}$, i.e.

$$\beta = -\frac{\partial \lambda}{\partial t}(\boldsymbol{r}, t) + k(t) \tag{7.15}$$

— it is possible to add some function that only depends on time, such that $\boldsymbol{\nabla} k(t) = \boldsymbol{0}$. But we can just absorb $k(t)$ into $\lambda(\boldsymbol{r}, t)$.

So the allowed gauge transformations are given by

$$\boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\lambda, \quad V' = V - \frac{\partial \lambda}{\partial t}, \tag{7.16}$$

which do not change the physical fields $\boldsymbol{E}, \boldsymbol{B}$.

7.3 Choice of Gauge

A **gauge condition** is a set of conditions imposed on $\boldsymbol{A}$ or λ which fixes (i.e. removes) the gauge freedom. One picks one representative from the many gauge-equivalent configurations that all encode the same physics, i.e. the same $\boldsymbol{E}$ and $\boldsymbol{B}$, as illustrated in Figure 7.1.

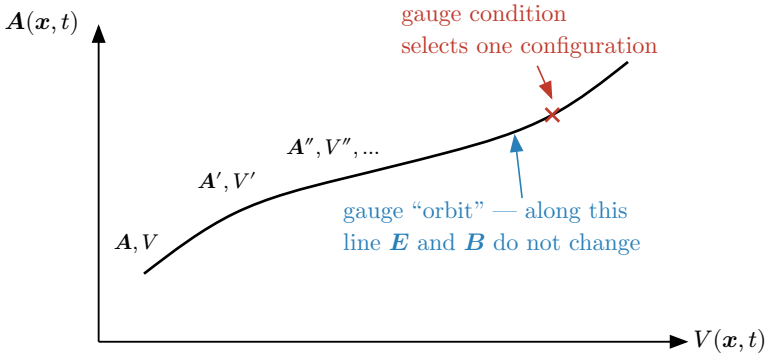


Figure 7.1: A gauge condition selects one representative $(\boldsymbol{A}, V)$ from the gauge orbit — the set of all potentials that describe the same physical fields $\boldsymbol{E}$ and $\boldsymbol{B}$.

Let's discuss two popular examples of gauge conditions.

7.3.1 Coulomb Gauge

In the **Coulomb gauge**, we pick

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0. \tag{7.17}$$

Plugging this into Eq. (7.7) and Eq. (7.10), we recover the Poisson equation from Eq. (7.7),

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}, \tag{7.18}$$

which we can solve directly using the Green's function method (see the tutorial and the next lecture),

$$V(\boldsymbol{r},t) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\boldsymbol{r}',t)}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}V'. \tag{7.19}$$

Two comments:

1. This is an example of how we can choose a gauge condition to simplify Maxwell's equations — at least Eq. (7.7). Eq. (7.10) still looks pretty daunting,

   $$\Box\boldsymbol{A} - \frac{1}{c^2}\boldsymbol{\nabla}\frac{\partial V}{\partial t} = -\mu_0\boldsymbol{J}, \tag{7.20}$$

   which is one of the disadvantages of the Coulomb gauge.

   Note: A gauge condition is not just **any** condition on $\boldsymbol{A}$ and V — $\boldsymbol{E}$ and $\boldsymbol{B}$ must remain unchanged. For example,

   $$\frac{\partial V}{\partial x} + \frac{\partial}{\partial t}A_x = 0 \tag{7.21}$$

   is in general **not** an allowed gauge condition, as it would imply $E_x = 0$, which does not describe the most general physical configuration. (More on this in the tutorial.)

2. The solution Eq. (7.19) implies that if we change the charge distribution, then the potential changes **immediately everywhere** (e.g. the potential on the Moon changes immediately when we change ρ in our lab in the basement), even for $|\boldsymbol{r}-\boldsymbol{r}'| \gg 0$.

   This seems at odds with special relativity, which states that nothing can travel faster than the speed of light. The point is: V is not a physical quantity we can measure — $\boldsymbol{E}$ is, and it depends on both $\boldsymbol{\nabla}V$ and $\frac{\partial \boldsymbol{A}}{\partial t}$ (Eq. (7.5)). $\boldsymbol{E}$ will only change after some time has passed. If V were a physical quantity, this would enable faster-than-light communication.

7.3.2 Lorenz Gauge

> **A note on terminology**
>
> "Lorenz gauge" $\neq$ "Lorentz transformation" — these are two different scientists, a Dane (Ludvig Lorenz) and a Dutchman (Hendrik Lorentz).

In this gauge, we pick

$$\boldsymbol{\nabla}\cdot\boldsymbol{A} = -\frac{1}{c^2}\frac{\partial V}{\partial t}, \tag{7.22}$$

which is a scalar condition. What happens to the differential equations for V and $\boldsymbol{A}$, Eq. (7.7) and Eq. (7.10)?

From Eq. (7.7) we see that

$$\nabla^2 V + \frac{\partial}{\partial t}\left(-\frac{1}{c^2}\frac{\partial V}{\partial t}\right) = \nabla^2 V - \frac{1}{c^2}\frac{\partial^2 V}{\partial t^2} = \Box V = -\frac{\rho}{\varepsilon_0}, \tag{7.23}$$

which is a wave equation for V with a source term.

In-class exercise

What happens to Eq. (7.10)?

For Eq. (7.10) we find that the gradient term on the left-hand side disappears (by Eq. (7.22)), and we're left with

$$\nabla^2 \boldsymbol{A} - \frac{1}{c^2}\frac{\partial^2 \boldsymbol{A}}{\partial t^2} = \Box \boldsymbol{A} = -\mu_0 \boldsymbol{J}, \tag{7.24}$$

which is also a wave equation with a source term. Thus, in Lorenz gauge we have

$$\Box V = -\frac{\rho}{\varepsilon_0}, \quad \Box \boldsymbol{A} = -\mu_0 \boldsymbol{J}. \tag{7.25}$$

A hint of relativity

This already suggests that we can use 4-vectors with $(V/c, \boldsymbol{A})$ and $(\rho c, \boldsymbol{J})$ — see a future lecture on the relativistic formulation of electrodynamics.

In Lorenz gauge, we can most easily achieve the main goal of electrodynamics: find the field configurations for a given distribution of charges and currents. We do this by solving Eq. (7.25) for V and $\boldsymbol{A}$, and then get the fields from Eq. (7.3) and Eq. (7.5).

7.4 Retarded Potentials

In the static case, Eq. (7.25) reduce to

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}, \quad \nabla^2 \boldsymbol{A} = -\mu_0 \boldsymbol{J}, \tag{7.26}$$

which are solved with the Green's function method (see the tutorial and the math interlude next lecture),

$$\begin{aligned} V(\boldsymbol{r}) &= \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}V', \\ \boldsymbol{A}(\boldsymbol{r}) &= \frac{\mu_0}{4\pi}\int_V \frac{\boldsymbol{J}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}V'. \end{aligned} \tag{7.27}$$

How does this generalize to the non-static case and make the solution time-dependent?

We know that a disturbance in the electromagnetic field travels at the speed of light, as this is the velocity of the EM waves. So, it's not the status of the source **right now** at time t that matters, but rather the state the source was in at the time when the emission was sent out by the source, at time t_r. t_r is called the **retarded time**,

$$t_r = t - \frac{|\boldsymbol{r}-\boldsymbol{r}'|}{c}. \tag{7.28}$$

time it takes for the emission to travel from source at $\boldsymbol{r}'$ to observer at $\boldsymbol{r}$

Here, t is the time now, when the emission is observed, and $\boldsymbol{r}$ ($\boldsymbol{r}'$) is the observer's (source's) position.

Figure 7.2 illustrates this for two different source points: a signal that reaches the observer at $\boldsymbol{r}$ at time t from two different source points $\boldsymbol{r}_1'$ and $\boldsymbol{r}_2'$ must have been emitted at two different retarded times $t_{r,1}$ and $t_{r,2}$.

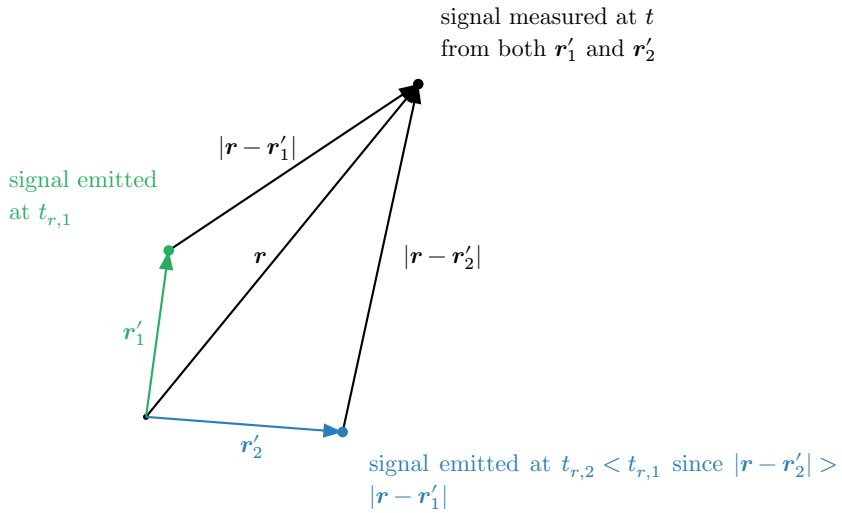


Figure 7.2: A signal measured at the observer's position $\boldsymbol{r}$ at time t, coming from two different source points $\boldsymbol{r}_1'$ and $\boldsymbol{r}_2'$, must have been emitted at different retarded times: since $|\boldsymbol{r}-\boldsymbol{r}_2'| > |\boldsymbol{r}-\boldsymbol{r}_1'|$, the signal from $\boldsymbol{r}_2'$ was emitted earlier, $t_{r,2} < t_{r,1}$.

In-class exercise

Do you understand this concept? Discuss with your neighbor.

Thus, the natural generalization of Eq. (7.27) that we could guess is

$$V(\boldsymbol{r},t) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\boldsymbol{r}', t_r(|\boldsymbol{r}-\boldsymbol{r}'|))}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}V',$$
$$\boldsymbol{A}(\boldsymbol{r},t) = \frac{\mu_0}{4\pi}\int_V \frac{\boldsymbol{J}(\boldsymbol{r}', t_r(|\boldsymbol{r}-\boldsymbol{r}'|))}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}V'. \tag{7.29}$$

Note that t_r depends on the source point $\boldsymbol{r}'$ being integrated over.

These are called **retarded potentials**, which reduce to the correct static solutions Eq. (7.27) in the static case.

7.4.1 Verifying the Retarded Potentials

These seem plausible, but we just guessed them. We need to make sure that they fulfill Maxwell's equations and the Lorenz gauge condition.

Let's check this explicitly for the scalar potential. The calculation will be the same for each component of the vector potential.

We'll need to compute $\nabla^2 V$, which involves $\nabla^2(1/|\boldsymbol{R}|)$, where we introduce the shorthand

$$\begin{aligned} \boldsymbol{R} &\equiv \boldsymbol{r} - \boldsymbol{r}' = (x - x', y - y', z - z'), \\ |\boldsymbol{R}| &= \left[(x - x')^2 + (y - y')^2 + (z - z')^2\right]^{1/2}. \end{aligned} \tag{7.30}$$

All gradients below are derivatives with respect to the **unprimed** coordinate $\boldsymbol{r}$ (the observer's position).

We will need a few identities (see also Griffiths Example 1.3 and Problem 1.13 — try to calculate these yourself, at least (i) and (iii) should be straight forward!):

$$\begin{aligned} &\text{(i)} \quad \boldsymbol{\nabla}\frac{1}{|\boldsymbol{R}|} = -\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2} = -\frac{\boldsymbol{R}}{|\boldsymbol{R}|^3}, \\ &\text{(ii)} \quad \nabla^2\left(\frac{1}{|\boldsymbol{R}|}\right) = \boldsymbol{\nabla}\cdot\left(\boldsymbol{\nabla}\frac{1}{|\boldsymbol{R}|}\right) = \boldsymbol{\nabla}\cdot\left(-\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2}\right) \\ &\qquad = 4\pi\delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}') \quad \text{(see next lecture)}, \\ &\text{(iii)} \quad \boldsymbol{\nabla}|\boldsymbol{R}| = \hat{\boldsymbol{R}}, \\ &\text{(iv)} \quad \boldsymbol{\nabla}\cdot\left(\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|}\right) = \frac{1}{|\boldsymbol{R}|^2}. \end{aligned} \tag{7.31}$$

Deriving identity (iv)

In Cartesian coordinates, using (iii),

$$\begin{aligned} \boldsymbol{\nabla}\cdot\left(\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|}\right) = \boldsymbol{\nabla}\cdot\left(\frac{\boldsymbol{R}}{|\boldsymbol{R}|^2}\right) &= \frac{\partial}{\partial x}\left(\frac{x - x'}{|\boldsymbol{R}|^2}\right) + \frac{\partial}{\partial y}\ldots \\ &= \frac{1}{|\boldsymbol{R}|^2} + (x - x')\frac{\partial}{\partial x}\frac{1}{(x - x')^2 + (y - y')^2 + (z - z')^2} + \frac{\partial}{\partial y}\ldots \\ &= \frac{1}{|\boldsymbol{R}|^2} - (x - x')\frac{2(x - x')}{\left((x - x')^2 + (y - y')^2 + (z - z')^2\right)^2} + \frac{\partial}{\partial y}\ldots \\ &= \frac{1}{|\boldsymbol{R}|^2} - (x - x')\frac{2(x - x')}{|\boldsymbol{R}|^4} + \ldots \end{aligned}$$

and summing the three (one from each of $\partial_x, \partial_y, \partial_z$),

$$\boldsymbol{\nabla}\cdot\left(\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|}\right) = \frac{3}{|\boldsymbol{R}|^2} - \boldsymbol{R}\cdot\frac{2\boldsymbol{R}}{|\boldsymbol{R}|^4} = \frac{3}{|\boldsymbol{R}|^2} - \frac{2}{|\boldsymbol{R}|^2} = \frac{1}{|\boldsymbol{R}|^2}. \quad \checkmark \tag{7.32}$$

Combining the chain rule with $t_r = t - |\boldsymbol{R}|/c$ (Eq. (7.28)) and identity (iii), we also need

$$\text{(v)} \quad \boldsymbol{\nabla}\rho(\boldsymbol{r}', t_r) = \dot{\rho}\boldsymbol{\nabla}t_r = -\frac{1}{c}\dot{\rho}\boldsymbol{\nabla}|\boldsymbol{R}| = -\frac{1}{c}\dot{\rho}\hat{\boldsymbol{R}}, \tag{7.33}$$

where the gradient only acts on $\boldsymbol{r}$ in the retarded time (not on $\boldsymbol{r}'$), and $\dot{\rho} \equiv \partial\rho/\partial t_r = \partial\rho/\partial t$ (since $\partial t_r/\partial t = 1$).

Equipped with these results, we now tackle $\nabla^2 V = \boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}V)$. Let's start with

$$\begin{aligned}\boldsymbol{\nabla} V &= \frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V' \boldsymbol{\nabla}\left(\frac{\rho(\boldsymbol{r}', t_r)}{|\boldsymbol{R}|}\right) \\ &= \frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V' \left(\frac{1}{|\boldsymbol{R}|}\boldsymbol{\nabla}\rho + \rho\boldsymbol{\nabla}\frac{1}{|\boldsymbol{R}|}\right).\end{aligned} \tag{7.34}$$

Using (v) and (i) from Eq. (7.31) and Eq. (7.33), we find

$$\boldsymbol{\nabla} V = \frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V' \left[-\frac{\dot\rho}{c}\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|} - \rho\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2}\right]. \tag{7.35}$$

So that

$$\nabla^2 V = \frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V' \left[\underbrace{\boldsymbol{\nabla}\cdot\left(-\frac{\dot\rho}{c}\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|}\right)}_{\text{term I}} + \underbrace{\boldsymbol{\nabla}\cdot\left(-\rho\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2}\right)}_{\text{term II}}\right]. \tag{7.36}$$

Using the product rule $\boldsymbol{\nabla}\cdot(f\boldsymbol{A}) = f(\boldsymbol{\nabla}\cdot\boldsymbol{A}) + \boldsymbol{A}\cdot(\boldsymbol{\nabla} f)$ term I becomes:

$$\mathrm{I} = -\frac{1}{c}\left[\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|}\cdot\boldsymbol{\nabla}\dot\rho + \dot\rho\boldsymbol{\nabla}\cdot\left(\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|}\right)\right]. \tag{7.37}$$

Using (v) (applied to $\dot\rho$) and (iv),

$$\mathrm{I} = -\frac{1}{c}\left[\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|}\cdot\underbrace{\left(-\frac{1}{c}\ddot\rho\hat{\boldsymbol{R}}\right)}_{\text{using (v)}} + \underbrace{\frac{\dot\rho}{|\boldsymbol{R}|^2}}_{\text{using (iv)}}\right] = \frac{\ddot\rho}{c^2|\boldsymbol{R}|} - \frac{\dot\rho}{c|\boldsymbol{R}|^2}, \tag{7.38}$$

where in the last step we used $\hat{\boldsymbol{R}}\cdot\hat{\boldsymbol{R}} = 1$ for the first term in the square brackets. Turning to the second term and using again the product rule, we have

$$\mathrm{II} = -\left[(\boldsymbol{\nabla}\rho)\cdot\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2} + \rho\boldsymbol{\nabla}\cdot\left(\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2}\right)\right]. \tag{7.39}$$

Using (v) and (ii) this simplifies to

$$\mathrm{II} = -\left[\underbrace{\left(-\frac{1}{c}\dot\rho\hat{\boldsymbol{R}}\right)}_{\text{using (v)}}\cdot\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2} + \rho\cdot\underbrace{4\pi\delta^{(3)}(\boldsymbol{R})}_{\text{using (ii)}}\right] = \frac{\dot\rho}{c|\boldsymbol{R}|^2} - 4\pi\rho\delta^{(3)}(\boldsymbol{R}). \tag{7.40}$$

Adding I and II, the $\dot\rho/(c|\boldsymbol{R}|^2)$ term in I and the $-\dot\rho/(c|\boldsymbol{R}|^2)$ term in II marked in pink cancel exactly, leaving

$$\mathrm{I} + \mathrm{II} = \frac{\ddot\rho}{c^2|\boldsymbol{R}|} - 4\pi\rho\delta^{(3)}(\boldsymbol{R}). \tag{7.41}$$

Collecting results we find

$$\begin{aligned}\nabla^2 V &= \frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V'\left[\frac{\ddot{\rho}}{c^2|\boldsymbol{R}|} - 4\pi\rho\delta^{(3)}(\boldsymbol{R})\right] \\ &= \underbrace{\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\left(\frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V'\frac{\rho}{|\boldsymbol{R}|}\right)}_{=\frac{1}{c^2}\frac{\partial^2 V}{\partial t^2}} - \frac{\rho(\boldsymbol{r},t)}{\varepsilon_0},\end{aligned} \tag{7.42}$$

where in the first term we pulled $1/c^2\partial_t^2$ out of the integral (the t-derivatives act only on the retarded time dependence, so they commute with $\int_V \mathrm{d}V'$), and in the second term the delta function collapses the integral, setting $\boldsymbol{r}' = \boldsymbol{r}$. Hence

$$\nabla^2 V - \frac{1}{c^2}\frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\varepsilon_0}. \tag{7.43}$$

Thus, we have shown that indeed our guess Eq. (7.29) satisfies the differential equation for the scalar potential in Lorenz gauge, Eq. (7.25). Similarly, one can show the same for the vector potential, component by component.

For some charge and current distribution, we can now perform the integrals in Lorenz gauge, Eq. (7.29), to find V, $\boldsymbol{A}$, and from them the $\boldsymbol{E}$ and $\boldsymbol{B}$ fields.

Comment: advanced potentials and causality

Electrodynamics is invariant under time reversal. Therefore, the **advanced potentials**, defined by the **advanced time**

$$t_a = t + \frac{|\boldsymbol{r} - \boldsymbol{r}'|}{c}, \tag{7.44}$$

are also a possible solution to Maxwell's equations (you will check this in the exercises). However, in this case the potentials would be determined by charge and current distributions **in the future**.

We reject this solution — which arises because the d'Alembertian Eq. (7.11) is time-symmetric — on the basis of **causality**.

8 Liénard–Wiechert Potentials

Literature: Griffiths Ch. 10.3.1 & 10.3.2, David Tong's lecture notes

Last lecture (and also in the Green's function interlude below) we saw that the solutions to the potentials in Lorenz gauge are

$$V(\boldsymbol{r},t) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\boldsymbol{r}', t_r(|\boldsymbol{r}-\boldsymbol{r}'|))}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}V',$$
$$\boldsymbol{A}(\boldsymbol{r},t) = \frac{\mu_0}{4\pi}\int_V \frac{\boldsymbol{J}(\boldsymbol{r}', t_r(|\boldsymbol{r}-\boldsymbol{r}'|))}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}V', \tag{8.1}$$

with the retarded time $t_r = t - |\boldsymbol{r}-\boldsymbol{r}'|/c$ (Eq. (7.28)).

In this lecture, we first fill in a gap from last time: we derive the Green's function for the d'Alembert operator that we used to justify Eq. (8.1), as a mathematical interlude. Then we specialize to the case of a point charge q whose position at time t is determined by some path $\boldsymbol{w}(t)$. We will see that in this case, the potentials are given by the **Liénard–Wiechert potentials**. These potentials are regularly used in high-energy (astro-)particle physics to derive the radiation emitted by point charges — we will look at this in more detail next lecture.

8.1 Mathematical Interlude: Green's Functions

Let's review the Green's function method in 1D, and then apply it to our situation.

Consider the following differential equation (e.g. the Poisson equation $\nabla^2 V = -\rho/\varepsilon_0$, in 1D),

$$\frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = s(x). \tag{8.2}$$

Given the **Green's function** $G(x,x')$ for the differential operator $\frac{\mathrm{d}^2}{\mathrm{d}x^2}$, i.e. a function that satisfies

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}G(x,x') = \delta(x-x'), \tag{8.3}$$

the solution to Eq. (8.2) is

$$f(x) = \int_{-\infty}^{\infty}\mathrm{d}x' G(x,x')s(x'). \tag{8.4}$$

Thus, the Green's function allows us to turn a **differential** equation into an **integral** equation.

> **In-class exercise**
>
> Check yourself: show that Eq. (8.4) is a solution to Eq. (8.2).

Indeed,

$$\frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = \int \mathrm{d}x' \underbrace{\frac{\mathrm{d}^2}{\mathrm{d}x^2}G(x,x')}_{=\delta(x-x') \text{ by definition}} s(x') = \int \mathrm{d}x'\delta(x-x')s(x') = s(x), \tag{8.5}$$

where we could move the second derivative into the integral, since it acts only on x, not on the integration variable x'.

8.1.1 Green's Function for the Laplacian

Now let's check if this is also true in 3D, for the Poisson equation of electrostatics, $\nabla^2 V(\boldsymbol{r}) = -\rho(\boldsymbol{r})/\varepsilon_0$. For the Green's function that satisfies

$$\nabla^2 G(\boldsymbol{r}, \boldsymbol{r}') = \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}'), \tag{8.6}$$

we have, in complete analogy with Eq. (8.4),

$$V(\boldsymbol{r}) = -\frac{1}{\varepsilon_0} \int_V \mathrm{d}V' G(\boldsymbol{r}, \boldsymbol{r}') \rho(\boldsymbol{r}'). \tag{8.7}$$

Plugging in,

$$\begin{aligned} \nabla^2 V(\boldsymbol{r}) &= -\frac{1}{\varepsilon_0} \int_V \mathrm{d}V' \underbrace{\nabla^2 G(\boldsymbol{r}, \boldsymbol{r}')}_{=\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')} \rho(\boldsymbol{r}') \\ &= -\frac{1}{\varepsilon_0} \int_V \mathrm{d}V' \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}') \rho(\boldsymbol{r}') = -\frac{\rho(\boldsymbol{r})}{\varepsilon_0}. \quad \checkmark \end{aligned} \tag{8.8}$$

This all looks fine, but in order for the Green's function to be useful, we need to know what it **is**. We will prove now that the Green's function for the Laplacian operator is given by

$$G(\boldsymbol{r}, \boldsymbol{r}') = -\frac{1}{4\pi} \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|}. \tag{8.9}$$

What we need to calculate is $\nabla^2(1/|\boldsymbol{R}|)$ (recall $\boldsymbol{R} \equiv \boldsymbol{r} - \boldsymbol{r}'$, as in the previous chapter), but we have to take care when $\boldsymbol{r} = \boldsymbol{r}'$.

Case I: $\boldsymbol{r} \neq \boldsymbol{r}'$. Using identity (i) from Eq. (7.31),

$$\nabla^2 \frac{1}{|\boldsymbol{R}|} = \boldsymbol{\nabla} \cdot \left(\boldsymbol{\nabla} \frac{1}{|\boldsymbol{R}|} \right) = \boldsymbol{\nabla} \cdot \left(-\frac{\boldsymbol{R}}{|\boldsymbol{R}|^3} \right). \tag{8.10}$$

Let's calculate the z-component (the x- and y-components work the same way):

$$\begin{aligned} \frac{\partial}{\partial z} \left[-\frac{z - z'}{|\boldsymbol{R}|^3} \right] &= -\frac{1}{|\boldsymbol{R}|^3} - (z - z') \frac{\partial}{\partial z} \frac{1}{|\boldsymbol{R}|^3} \\ &= -\frac{1}{|\boldsymbol{R}|^3} + (z - z') \frac{3(z - z')}{|\boldsymbol{R}|^5} = -\frac{1}{|\boldsymbol{R}|^3} + 3\frac{(z - z')^2}{|\boldsymbol{R}|^5}, \end{aligned} \tag{8.11}$$

using the product rule and the chain rule (feel encouraged to check this yourself if in doubt). Summing the three (one from each of $\partial_x, \partial_y, \partial_z$),

$$\begin{aligned} \nabla^2 \frac{1}{|\boldsymbol{R}|} &= -\frac{3}{|\boldsymbol{R}|^3} + 3\frac{(x - x')^2 + (y - y')^2 + (z - z')^2}{|\boldsymbol{R}|^5} \\ &= -\frac{3}{|\boldsymbol{R}|^3} + \frac{3}{|\boldsymbol{R}|^3} = 0. \end{aligned} \tag{8.12}$$

Case II: $\boldsymbol{r} = \boldsymbol{r}'$. Here $1/|\boldsymbol{R}|$ is divergent, and thus $\nabla^2(1/|\boldsymbol{R}|)$ diverges too. We therefore guess that it should be proportional to $\delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}')$, because the delta

function has exactly the property of being zero everywhere except where its argument is zero, where it diverges.

To determine the constant of proportionality C, we write

$$\begin{aligned}\int_V \mathrm{d}V \nabla^2 \frac{1}{|\boldsymbol{R}|} &= \int_V \mathrm{d}V \boldsymbol{\nabla} \cdot \left(\boldsymbol{\nabla} \frac{1}{|\boldsymbol{R}|}\right) \\ &= C \int_V \mathrm{d}V \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}') = C.\end{aligned} \tag{8.13}$$

Now use the divergence theorem to change the volume integral over a divergence to a surface integral and compute the gradient with relation (i) from Eq. (7.31),

$$\int_V \mathrm{d}V \boldsymbol{\nabla} \cdot \left(\boldsymbol{\nabla} \frac{1}{|\boldsymbol{R}|}\right) = \oint_S \left(-\frac{\boldsymbol{R}}{|\boldsymbol{R}|^3}\right) \cdot \mathrm{d}\boldsymbol{a} = C. \tag{8.14}$$

We integrate over a sphere of radius $|\boldsymbol{R}|$ around $\boldsymbol{r}'$, with $\mathrm{d}\boldsymbol{a} = |\boldsymbol{R}|^2 \hat{\boldsymbol{R}} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi$:

$$\begin{aligned}&-\int_0^{2\pi} \mathrm{d}\varphi \int_0^{\pi} \mathrm{d}\theta \left(-\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2}\right) \cdot \left(|\boldsymbol{R}|^2 \hat{\boldsymbol{R}} \sin\theta\right) \\ &\quad = -\int_0^{2\pi} \mathrm{d}\varphi \int_0^{\pi} \sin\theta \, \mathrm{d}\theta = -4\pi = C,\end{aligned} \tag{8.15}$$

using $\hat{\boldsymbol{R}} \cdot \hat{\boldsymbol{R}} = 1$. So $C = -4\pi$, and

$$\nabla^2 \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} = -4\pi \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}'). \tag{8.16}$$

which is exactly Eq. (8.6) with $G(\boldsymbol{r}, \boldsymbol{r}') = -1/(4\pi|\boldsymbol{r} - \boldsymbol{r}'|)$ — confirming Eq. (8.9).

Consistency with identity (ii)

Since $\boldsymbol{\nabla}(1/|\boldsymbol{R}|) = -\hat{\boldsymbol{R}}/|\boldsymbol{R}|^2$ (identity (i) in Eq. (7.31)), Eq. (8.16) is equivalent to

$$\boldsymbol{\nabla} \cdot \left(\frac{\hat{\boldsymbol{R}}}{|\boldsymbol{R}|^2}\right) = -\nabla^2 \frac{1}{|\boldsymbol{R}|} = 4\pi \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}'), \tag{8.17}$$

which is exactly identity (ii) in Eq. (7.31) from last lecture, which we had deferred to "next lecture" — now proven!

8.1.2 Green's Function for the d'Alembert Operator

For the retarded potentials in Lorenz gauge, we have a similar problem, but instead of the Laplacian, we need to find the Green's function of the d'Alembert operator $\Box = \nabla^2 - 1/c^2 \partial_t^2$ (Eq. (7.11)), with

$$\Box G(\boldsymbol{r}, \boldsymbol{r}'; t, t') = \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}') \delta(t - t'), \tag{8.18}$$

so that, for e.g. the scalar potential, we have

$$\Box V(\boldsymbol{r}, t) = -\frac{1}{\varepsilon_0} \int_V \mathrm{d}V' \int \mathrm{d}t' G(\boldsymbol{r}, \boldsymbol{r}'; t, t') \rho(\boldsymbol{r}', t'). \tag{8.19}$$

From what we know from the $\nabla^2 V$ case and the retarded time $t_r = t - |\boldsymbol{r} - \boldsymbol{r}'|/c$, we can guess the ansatz

$$G(\boldsymbol{r}, \boldsymbol{r}'; t, t') = \frac{f\left(t - \frac{|\boldsymbol{r}-\boldsymbol{r}'|}{c} - t'\right)}{|\boldsymbol{r} - \boldsymbol{r}'|}, \tag{8.20}$$

with a scalar function $f(q)$. One can show (see Farside's lecture notes — you will also check this yourself in the exercises) that

$$G(\boldsymbol{r}, \boldsymbol{r}'; t, t') = -\frac{1}{4\pi}\frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|}\delta\left(t - \frac{|\boldsymbol{r} - \boldsymbol{r}'|}{c} - t'\right). \tag{8.21}$$

Then we can perform the integral over t', which gives us the result for the potentials from last lecture: plugging Eq. (8.21) into Eq. (8.19),

$$\begin{aligned} V(\boldsymbol{r}, t) &= \frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V' \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|}\int \mathrm{d}t' \,\delta\left(t - \frac{|\boldsymbol{r} - \boldsymbol{r}'|}{c} - t'\right)\rho(\boldsymbol{r}', t') \\ &= \frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V' \frac{\rho\left(\boldsymbol{r}', t - \frac{|\boldsymbol{r}-\boldsymbol{r}'|}{c}\right)}{|\boldsymbol{r} - \boldsymbol{r}'|}, \qquad \uparrow = t_r \end{aligned} \tag{8.22}$$

with the delta function ensuring that we have to evaluate the charge (and current) distributions at the retarded time. This reproduces exactly the retarded potential of Eq. (8.1).

8.2 Liénard–Wiechert Potentials of Point Charges

Let's apply our formalism to the case of a point charge q whose position at time t is determined by some path $\boldsymbol{w}(t)$, as illustrated in Figure 8.1.

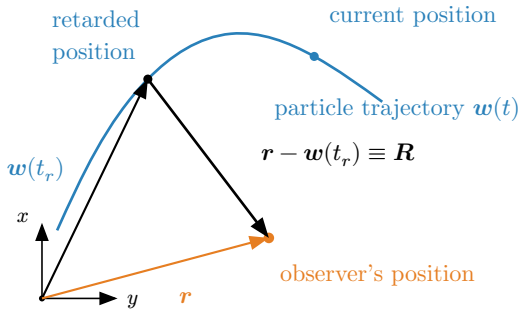


Figure 8.1: A point charge moving along the trajectory $\boldsymbol{w}(t)$. The field at the observer's position $\boldsymbol{r}$ at time t is determined by the charge's **retarded** position $\boldsymbol{w}(t_r)$, not its current position.

The retarded time is given implicitly by

$$\underbrace{|\boldsymbol{r} - \boldsymbol{w}(t_r)|}_{\substack{\text{distance "news" must} \\ \text{travel} \equiv |\boldsymbol{R}|}} = \underbrace{c\,(t - t_r)}_{\substack{\text{time it takes to} \\ \text{make the trip}}}. \tag{8.23}$$

The potential is $V(\boldsymbol{r},t) = 1/(4\pi\varepsilon_0)\int_V \rho(\boldsymbol{r}',t_r)/|\boldsymbol{r}-\boldsymbol{r}'|\,\mathrm{d}V'$ (Eq. (8.1)). For a point charge, we would expect

$$\rho(\boldsymbol{r}',t_r) \sim q\delta^{(3)}(\boldsymbol{r}'-\boldsymbol{w}(t_r)) \tag{8.24}$$

and thus

$$V(\boldsymbol{r},t) = \frac{1}{4\pi\varepsilon_0}\frac{q}{|\boldsymbol{r}-\boldsymbol{w}(t_r)|}. \tag{8.25}$$

But this is wrong! The denominator is fine, but the integral $\int_V \mathrm{d}V'\rho(\boldsymbol{r}',t_r) \neq q$, since the retarded time obliges us to integrate over the charge distribution ρ at **different times**.

Instead, we can motivate

$$\int_V \rho(\boldsymbol{r},t_r)\,\mathrm{d}V' = \frac{q}{1-\hat{\boldsymbol{R}}\cdot\boldsymbol{v}/c}, \tag{8.26}$$

where $\boldsymbol{v}$ is the particle velocity at t_r. This is a purely geometrical effect, and to understand it, consider a train of length L moving with velocity v towards the observer, Figure 8.2.

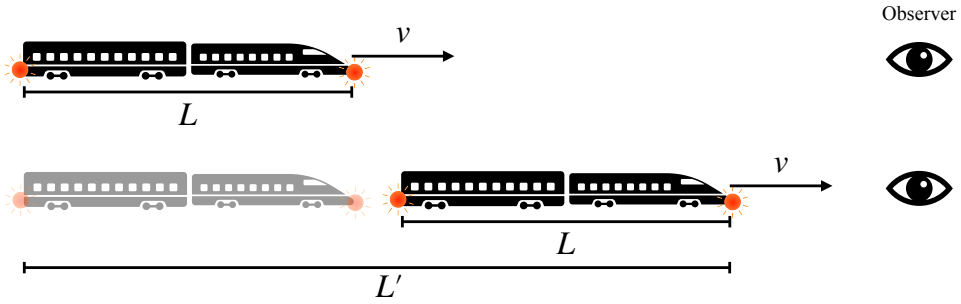


Figure 8.2: Light emitted simultaneously from the engine and the last car (as received by the observer) was not emitted at the same time, since the light from the last car takes longer to arrive. As a consequence, the approaching train appears a bit longer: $L' > L$.

The light we receive simultaneously from the last car and the engine wasn't emitted at the same time, as it took a longer time for the light from the last car to reach us. As a consequence, the approaching train will look a bit longer: light from the last car travels L' to reach the engine's (later) position. During the time it takes the light to travel L', the train itself has moved by $L'-L$. We can equate the light travel time $\frac{L'}{c}$ with the time it took the train to move by $L'-L$ to find an expression for L':

$$\begin{aligned}\frac{L'}{c} = \frac{L'-L}{v} \quad &\Rightarrow \quad L'\left(\frac{1}{v}-\frac{1}{c}\right) = \frac{L}{v} \quad \Rightarrow \quad L'\left(1-\frac{v}{c}\right) = L \\ &\Rightarrow \quad L' = \frac{L}{1-\frac{v}{c}}.\end{aligned} \tag{8.27}$$

Note: this has **nothing** to do with Lorentz contraction — it's rather reminiscent of the Doppler factor.

In general, the velocity $\boldsymbol{v}$ will make an angle θ to the line of sight with the observer, in which case

$$L' = \frac{L}{1 - \frac{v}{c}\cos\theta}, \quad \frac{v}{c}\cos\theta = \frac{\boldsymbol{v}\cdot\hat{\boldsymbol{R}}}{c}. \tag{8.28}$$

As the dimension perpendicular to the velocity of the train does not change, a volume element of the train transforms as

$$\mathrm{d}V' = \frac{\mathrm{d}V}{1 - \hat{\boldsymbol{R}}\cdot\boldsymbol{v}/c}. \tag{8.29}$$

Thus, whenever you have an integral of the type in Eq. (8.26), you pick up a factor $\left(1 - \hat{\boldsymbol{R}}\cdot\boldsymbol{v}/c\right)^{-1}$, and the final result for the potential is

$$V(\boldsymbol{r},t) = \frac{1}{4\pi\varepsilon_0}\frac{1}{|\boldsymbol{R}|}\frac{q}{1 - \hat{\boldsymbol{R}}\cdot\boldsymbol{v}/c} = \frac{1}{4\pi\varepsilon_0}\frac{qc}{|\boldsymbol{R}|c - \boldsymbol{R}\cdot\boldsymbol{v}}, \tag{8.30}$$

with $\boldsymbol{R} = \boldsymbol{r} - \boldsymbol{w}(t_r)$, all evaluated at t_r.

In-class exercise

Discuss with your neighbor: is this clear?

8.2.1 Formal Derivation

Let's derive this result more formally. For this (and also for the lectures on radiation), we will follow parts of David Tong's lecture notes (Ch. 6.4.1).

The potential at $(\boldsymbol{r},t)$ for a point charge following the path $\boldsymbol{w}(t)$ is given by

$$V(\boldsymbol{r},t) = \frac{q}{4\pi\varepsilon_0}\int_V \mathrm{d}V' \frac{\delta^{(3)}(\boldsymbol{r}' - \boldsymbol{w}(t_r))}{|\boldsymbol{r} - \boldsymbol{r}'|}. \tag{8.31}$$

This integral is cumbersome, since $\boldsymbol{r}'$ appears in the delta function in the obvious place, but also in $t_r = t - |\boldsymbol{r} - \boldsymbol{r}'|/c$.

The best way to handle this is to go a step back and use the Green's function for the d'Alembert operator, Eq. (8.21):

$$\begin{aligned} V(\boldsymbol{r},t) &= -\frac{1}{\varepsilon_0}\int \mathrm{d}t' \int_V \mathrm{d}V' \rho(\boldsymbol{r}',t') G(\boldsymbol{r},\boldsymbol{r}';t,t') \\ &= \frac{q}{4\pi\varepsilon_0}\int \mathrm{d}t' \int_V \mathrm{d}V' \frac{\delta^{(3)}(\boldsymbol{r}' - \boldsymbol{w}(t'))\delta\left(t_{r(|\boldsymbol{r}-\boldsymbol{r}'|)} - t'\right)}{|\boldsymbol{r} - \boldsymbol{r}'|}. \end{aligned} \tag{8.32}$$

Notice how $\boldsymbol{w}$ now depends on the **integration** variable t', and not on the retarded time. Now we can simply carry out the integration over $\mathrm{d}V'$, which yields (plugging in the definition of t_r)

$$V = \frac{q}{4\pi\varepsilon_0}\int \mathrm{d}t' \frac{\delta(t - |\boldsymbol{r} - \boldsymbol{w}(t')|/c - t')}{|\boldsymbol{r} - \boldsymbol{w}(t')|}. \tag{8.33}$$

Now we still have the issue that t' appears in two places in the argument of the delta function, but it's more straightforward to deal with it. Let's introduce the separation vector

$$\boldsymbol{R}(t') = \boldsymbol{r} - \boldsymbol{w}(t'), \tag{8.34}$$

and define a helper function

$$f(t') = t' + |\boldsymbol{R}(t')|/c, \tag{8.35}$$

with

$$\dot{f}(t') = \frac{\mathrm{d}f}{\mathrm{d}t'} = 1 + \frac{1}{c}\frac{\mathrm{d}}{\mathrm{d}t'}|\boldsymbol{R}(t')|. \tag{8.36}$$

You can show this explicitly with e.g. Cartesian coordinates and the chain rule — here, we just use that $\mathrm{d}\,\mathrm{d}t'|\boldsymbol{R}| = \hat{\boldsymbol{R}} \cdot \dot{\boldsymbol{R}}$ (since $|\boldsymbol{R}| = \sqrt{\boldsymbol{R} \cdot \boldsymbol{R}}$). With Eq. (8.34),

$$\dot{\boldsymbol{R}}(t') = -\dot{\boldsymbol{w}}(t') = -\boldsymbol{v}(t'), \tag{8.37}$$

so that

$$\dot{f}(t') = 1 - \frac{1}{c}\hat{\boldsymbol{R}} \cdot \boldsymbol{v} = 1 - \frac{\hat{\boldsymbol{R}}(t') \cdot \boldsymbol{v}(t')}{c}. \tag{8.38}$$

Now we can solve the integral Eq. (8.33) by doing a substitution, changing the variable of integration from t' to f:

$$\begin{aligned} V(\boldsymbol{r}, t) &= \frac{q}{4\pi\varepsilon_0} \int \mathrm{d}t' \frac{1}{|\boldsymbol{R}(t')|}\delta(t - f(t')) \\ &= \frac{q}{4\pi\varepsilon_0} \int \mathrm{d}f \left|\frac{\mathrm{d}t'}{\mathrm{d}f}\right| \frac{1}{|\boldsymbol{R}(t')|}\delta(t - f), \end{aligned} \tag{8.39}$$

with the substitution Jacobian $|\mathrm{d}t'/\,\mathrm{d}f| = \left|\dot{f}(t')\right|^{-1}$. The delta function sets $f(t') = t$ — which, by the definition of f, is exactly the implicit retarded-time condition Eq. (8.23), i.e. $t' = t_r$:

$$V(\boldsymbol{r}, t) = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\dot{f}}(t')\frac{1}{|\boldsymbol{R}(t')|}\right]_{f(t')=t} = \frac{q}{4\pi\varepsilon_0}\frac{1}{\dot{f}(t_r)|\boldsymbol{R}(t_r)|}. \tag{8.40}$$

Plugging in Eq. (8.38),

$$V(\boldsymbol{r}, t) = \frac{q}{4\pi\varepsilon_0}\frac{c}{c|\boldsymbol{R}(t_r)| - \boldsymbol{R}(t_r) \cdot \boldsymbol{v}(t_r)}, \tag{8.41}$$

which is the same result that we found above in Eq. (8.30).

We can find the vector potential in exactly the same fashion, by using

$$\boldsymbol{J}(\boldsymbol{r}, t) = q\boldsymbol{v}(t)\delta^{(3)}(\boldsymbol{r} - \boldsymbol{w}(t)), \tag{8.42}$$

which gives us (from the definition of $\boldsymbol{A}$, with $\mu_0 = 1/(\varepsilon_0 c^2)$)

$$\boldsymbol{A}(\boldsymbol{r}, t) = \frac{q\mu_0}{4\pi}\frac{c\boldsymbol{v}(t_r)}{c|\boldsymbol{R}(t_r)| - \boldsymbol{R}(t_r) \cdot \boldsymbol{v}(t_r)} = \frac{\boldsymbol{v}(t_r)}{c^2}V(\boldsymbol{r}, t). \tag{8.43}$$

Collecting Eq. (8.41) and Eq. (8.43), these are the **Liénard–Wiechert potentials** for moving charges:

$$V(\boldsymbol{r},t) = \frac{1}{4\pi\varepsilon_0}\frac{qc}{c|\boldsymbol{R}| - \boldsymbol{R}\cdot\boldsymbol{v}}, \quad \boldsymbol{A}(\boldsymbol{r},t) = \frac{\boldsymbol{v}}{c^2}V(\boldsymbol{r},t), \tag{8.44}$$

with $\boldsymbol{R} = \boldsymbol{r} - \boldsymbol{w}(t_r)$, $\boldsymbol{v} = \dot{\boldsymbol{w}}(t_r)$, all evaluated at the retarded time t_r.

From these, you can get the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ of a moving point charge in the usual way (Eq. (7.3), Eq. (7.5)), see the slides for a summary. We will look at this, and the radiation it produces, in more detail next lecture.

Note

You might argue that we haven't actually one anything: the Liénard-Wiechert potentials still depend on t_r through $\boldsymbol{R}(t_r)$ and the velocity $\boldsymbol{v}(t_r)$. To actually calculate the potentials, you have to specify the particle trajectory $\boldsymbol{w}(t)$ and then use the implicit relation $|\boldsymbol{r} - \boldsymbol{w}(t_r)| = c(t - t_r)$ to find an expression for t_r. The example of a particle at constant velocity, i.e., $\boldsymbol{w}(t) = \boldsymbol{v}t$ is worked out in Griffith's book, example 10.3.

9 Radiation

Literature: Griffiths Ch. 11.1

We've seen that Maxwell's equations have wave solutions, which correspond to light, or (more generally) electromagnetic radiation. Now we would like to understand: how do we actually produce light?

We will see that **accelerated** charges produce radiation, i.e. they emit light (we already saw that acceleration of a point charge leads to an electric field when discussing the Liénard-Wiechert potentials). This matters in a variety of settings, for example:

- Oscillating charges lead to an oscillating dipole moment and the emission of EM waves. This is how an **antenna** works!
- Relativistic charged particles, i.e. particles with a velocity close to the speed of light, that are deflected by a magnetic field will emit so-called **synchrotron radiation**.

> **Synchrotron radiation in practice**
>
> Synchrotron radiation is important in many astrophysical settings, and is one of the major energy-loss channels in circular particle accelerators like the LHC — and the reason why you want to build colliders with large radii, to reduce synchrotron losses. It's also why accelerating protons is easier than electrons, since synchrotron losses scale with $1/m^4$, where m is the mass of the particle, and $m_p/m_e \approx \frac{1}{2} \cdot 10^3$.

> **Comment: classical instability of matter**
>
> The classical description that we will deal with here would lead to an instability of matter: an electron orbiting its nucleus would lose energy continuously and plunge into the nucleus. You need quantum mechanics to explain why this does not happen — we will touch on this in the tutorials.

9.1 Radiated Power Through a Distant Surface

Here, we're interested in the radiation far away from some volume where charges and currents produce the radiation in the first place. To make this more concrete, we're often interested in the radiated power passing through a surface far away from the source, as sketched in Figure 9.1, such that $r' \equiv |\boldsymbol{r}'| \ll |\boldsymbol{r}| \equiv r$.

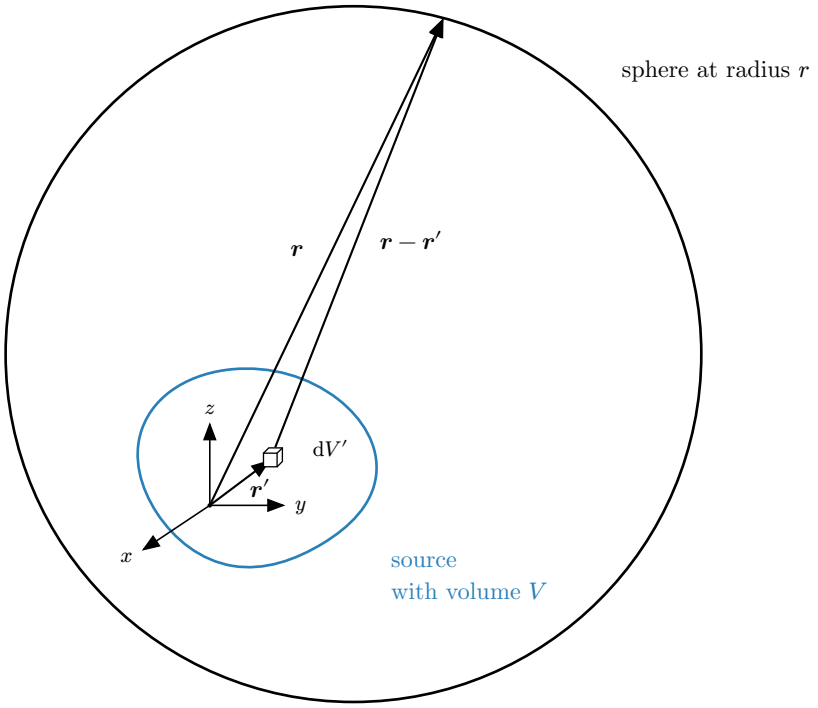


Figure 9.1: A localized source of charges and currents, confined to a volume V of size $\sim d^3$, and a distant sphere of radius r, with $|\boldsymbol{r}-\boldsymbol{r}'| \gg d$ for all $\boldsymbol{r}' \in V$.

The radiated power passing through the entire surface of the sphere is given by the integral of the Poynting vector:

$$P(r,t) = \oint \boldsymbol{S}\cdot \mathrm{d}\boldsymbol{a} = \frac{1}{\mu_0}\oint (\boldsymbol{E}\times\boldsymbol{B})\cdot \mathrm{d}\boldsymbol{a}, \tag{9.1}$$

taking into account that the radiation passing through the sphere at time t was emitted at time

$$t_r = t - \frac{|\boldsymbol{r}-\boldsymbol{r}'|}{c} \approx t - \frac{r}{c}. \tag{9.2}$$

To find the fields, we start with the retarded potentials using Lorenz gauge, see Eq. (7.29),

$$\begin{aligned} V(\boldsymbol{r},t) &= \frac{1}{4\pi\varepsilon_0}\int_V \mathrm{d}V' \frac{\rho(\boldsymbol{r}',t_r)}{|\boldsymbol{r}-\boldsymbol{r}'|}, \\ \boldsymbol{A}(\boldsymbol{r},t) &= \frac{\mu_0}{4\pi}\int_V \mathrm{d}V' \frac{\boldsymbol{J}(\boldsymbol{r}',t_r)}{|\boldsymbol{r}-\boldsymbol{r}'|}, \end{aligned} \tag{9.3}$$

where we only consider $\rho, \boldsymbol{J}$ to be nonzero for $\boldsymbol{r}' \in V$ (some finite volume). Sorry, letter V stands here both for volume and for the scalar potential, but we hope you can tell them apart from context..

9.2 The Multipole Expansion

We consider a volume of size $V \sim d^3$. If $|\boldsymbol{r}-\boldsymbol{r}'| \gg d$ for all $\boldsymbol{r}' \in V$, then we can actually approximate $|\boldsymbol{r}-\boldsymbol{r}'| \approx |\boldsymbol{r}| = r$. In fact, we will also keep the first-order

term of the **Taylor expansion** of $|\boldsymbol{r}-\boldsymbol{r}'|$ — this works similarly to the Taylor expansion you know from single-variable calculus.

Reminder: Taylor expansion of a scalar field

The absolute value can be regarded as a function $f:\mathbb{R}^3\to\mathbb{R}$ with $f(\boldsymbol{r}')=|\boldsymbol{r}-\boldsymbol{r}'|$ for fixed $\boldsymbol{r}$. Recall that for such a scalar field, the expansion around a vector $\boldsymbol{a}$ is given by

$$f(\boldsymbol{r}')\approx f(\boldsymbol{a})+((\boldsymbol{r}'-\boldsymbol{a})\cdot\boldsymbol{\nabla}')f(\boldsymbol{a})+\frac{1}{2}((\boldsymbol{r}'-\boldsymbol{a})\cdot\boldsymbol{\nabla}')^2 f(\boldsymbol{a})+\cdots, \tag{9.4}$$

where the first-order term, written out explicitly, is

$$(r'_x-a_x)\frac{\partial f}{\partial x'}\Big|_{\boldsymbol{a}}+(r'_y-a_y)\frac{\partial f}{\partial y'}\Big|_{\boldsymbol{a}}+(r'_z-a_z)\frac{\partial f}{\partial z'}\Big|_{\boldsymbol{a}}. \tag{9.5}$$

Our goal is now to expand the absolute value of the seperation vector $|\boldsymbol{r}-\boldsymbol{r}'|$ as well as the retarded time t_r and identify the leading order terms that contribute most to the radiated power. This is what we call the *multipole expansion*.

9.2.1 Expanding the Separation Vector

Expanding around $\boldsymbol{a}=\boldsymbol{0}$ and using the explicit form of $f(\boldsymbol{r}')=|\boldsymbol{r}-\boldsymbol{r}'|$,

$$|\boldsymbol{r}-\boldsymbol{r}'|=r-\left[\frac{1}{2|\boldsymbol{r}-\boldsymbol{r}'|}2(\boldsymbol{r}-\boldsymbol{r}')\right]_{\boldsymbol{r}'=0}\cdot\boldsymbol{r}'+\cdots\approx r-\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r}. \tag{9.6}$$

we differentiate w.r.t. the primed coordinates

Similarly, for $1/|\boldsymbol{r}-\boldsymbol{r}'|$ (which we also need, for the potentials Eq. (9.3)),

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}=\frac{1}{r}-\left[-\frac{1}{2}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|^3}(-2)(\boldsymbol{r}-\boldsymbol{r}')\right]_{\boldsymbol{r}'=0}\cdot\boldsymbol{r}'+\cdots\approx\frac{1}{r}+\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^3}. \tag{9.7}$$

In both cases, we only kept the zeroth and first order terms.

9.2.2 Expanding the Retarded Time Dependence

Again, we have to take care that the charge and current distributions depend on the retarded time, $\rho(\boldsymbol{r}',t_r)=\rho\left(\boldsymbol{r}',t-\frac{|\boldsymbol{r}-\boldsymbol{r}'|}{c}\right)$. In order to expand this further, we need to assume that the movement of the charges is **non-relativistic**, so that an associated current does not change considerably over the timescale $\tau\sim\frac{d}{c}$ that it would take the current to cross the volume d^3.

In this case, we can expand ρ (and $\boldsymbol{J}$) in time t around the retarded time at the origin, $t_0\equiv t-\frac{r}{c}$ (using the chain rule),

$$\rho\left(\boldsymbol{r}',t-\frac{|\boldsymbol{r}-\boldsymbol{r}'|}{c}\right)=\rho(\boldsymbol{r}',t_0)+\dot{\rho}(\boldsymbol{r}',t_0)[\boldsymbol{\nabla}'t_r]_{\boldsymbol{r}'=0}\cdot\boldsymbol{r}'+\cdots \tag{9.8}$$

The gradient of t_r with respect to the **primed** coordinates is

$$\boldsymbol{\nabla}'t_r=\boldsymbol{\nabla}'\left(t-\frac{|\boldsymbol{r}-\boldsymbol{r}'|}{c}\right)=-\frac{1}{c}\boldsymbol{\nabla}'|\boldsymbol{r}-\boldsymbol{r}'|=-\frac{1}{c}\frac{(-1)(\boldsymbol{r}-\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}=\frac{1}{c}\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|} \tag{9.9}$$

where we used that $\boldsymbol{\nabla}|\boldsymbol{r}-\boldsymbol{r}'| = \frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|}$ (with respect to the **unprimed** coordinates; the extra minus sign for $\boldsymbol{\nabla}'$ comes from differentiating $\boldsymbol{r}-\boldsymbol{r}'$ with respect to the primed variable). Evaluated at $\boldsymbol{r}' = 0$, keeping only first order,

$$[\boldsymbol{\nabla}' t_r]_{\boldsymbol{r}'=0} = \frac{1}{c}\frac{\boldsymbol{r}}{|\boldsymbol{r}|} = \frac{\hat{\boldsymbol{r}}}{c}. \tag{9.10}$$

Plugging Eq. (9.10) into Eq. (9.8), and similarly for $\boldsymbol{J}$,

$$\begin{aligned}\rho(\boldsymbol{r}', t_r) &\approx \rho(\boldsymbol{r}', t_0) + \dot{\rho}(\boldsymbol{r}', t_0)\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{rc},\\ \boldsymbol{J}(\boldsymbol{r}', t_r) &\approx \boldsymbol{J}(\boldsymbol{r}', t_0) + \dot{\boldsymbol{J}}(\boldsymbol{r}', t_0)\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{rc}.\end{aligned} \tag{9.11}$$

What does the $\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{rc}$ term mean physically? $\boldsymbol{r}\cdot\boldsymbol{r}' = rr'\cos\theta$ is the projection of $\boldsymbol{r}'$ onto $\boldsymbol{r}$, and since we're considering a volume $\propto d^3$, this is of order $\sim rd$. Since $\boldsymbol{r}\cdot\boldsymbol{r}'/(rc) \sim \frac{d}{c} = \tau$, this term is the time it takes a light signal to traverse the source volume.

9.2.3 The Dipole Term of the Vector Potential

We now expand the vector potential (we could also do this with the scalar potential), using the expansions of both $\boldsymbol{J}$ (Eq. (9.11)) and $1/|\boldsymbol{r}-\boldsymbol{r}'|$ (Eq. (9.7)). We only want to keep leading-order terms, $\mathcal{O}(d)$.

> **In-class exercise**
>
> We will show in a moment that $\dot{\boldsymbol{J}}$ itself is of order $\sim d$. With this in mind, which terms should we keep when multiplying out the two brackets?

As it turns out, this is simply the combination of the **first** term in the expansion of $\boldsymbol{J}$ and the **first** term of the expansion of $1/|\boldsymbol{r}-\boldsymbol{r}'|$: combinations with the second term in $1/|\boldsymbol{r}-\boldsymbol{r}'|$ are neglected since they fall off with an additional factor $1/r$ before the integral; combinations with the second term in $\boldsymbol{J}$ are neglected because they would be of second order in d. Using both Eq. (9.11) and Eq. (9.7),

$$\begin{aligned}\boldsymbol{A}(\boldsymbol{r}, t) &= \frac{\mu_0}{4\pi}\int_V \mathrm{d}V' \frac{\boldsymbol{J}(\boldsymbol{r}', t_r)}{|\boldsymbol{r}-\boldsymbol{r}'|}\\ &\approx \frac{\mu_0}{4\pi}\int_V \mathrm{d}V' \left[\boldsymbol{J}(\boldsymbol{r}', t_0) + \dot{\boldsymbol{J}}(\boldsymbol{r}', t_0)\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{rc}\right]\left[\frac{1}{r} + \frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^3}\right],\end{aligned} \tag{9.12}$$

so that, keeping only the leading-order term (the $\frac{1}{r}$ term can be pulled outside the integral),

$$\boldsymbol{A}(\boldsymbol{r}, t) = \frac{\mu_0}{4\pi r}\int_V \mathrm{d}V' \boldsymbol{J}(\boldsymbol{r}', t_0) + \mathcal{O}\left(d^2, \frac{1}{r^2}\right). \tag{9.13}$$

9.2.4 The Electric Dipole Moment

We can re-write

$$\int_V \mathrm{d}V' \boldsymbol{J}(\boldsymbol{r}', t_0) = \dot{\boldsymbol{p}}(t_0) \tag{9.14}$$

where

$$\boldsymbol{p} = \int_V \mathrm{d}V' \boldsymbol{r}' \rho(\boldsymbol{r}', t_0) \tag{9.15}$$

is the **electric dipole moment**.

As the dipole moment is already of order d (remember $\boldsymbol{p} = q\boldsymbol{d}$ for the configuration of a charge $+q$ at the origin and $-q$ displaced by $\boldsymbol{d}$), we can neglect any higher-order terms in the expansion of $\boldsymbol{A}$.

Let's show why Eq. (9.14) holds by considering the following divergence

$$\boldsymbol{\nabla} \cdot (\boldsymbol{J} x_i') = (\boldsymbol{\nabla} \cdot \boldsymbol{J}) x_i' + J_i, \tag{9.16}$$

where x_i' is the i-th component of $\boldsymbol{r}'$ (e.g. y'). Using the continuity equation, $\boldsymbol{\nabla} \cdot \boldsymbol{J} + \dot{\rho} = 0$ (Eq. (1.20)),

$$\boldsymbol{\nabla} \cdot (\boldsymbol{J} x_i') = -\dot{\rho} x_i' + J_i \quad \Rightarrow \quad J_i = \boldsymbol{\nabla} \cdot (\boldsymbol{J} x_i') + \dot{\rho} x_i'. \tag{9.17}$$

Plugging this in for the i-th component of the integral,

$$\int_V \mathrm{d}V' J_i = \int_V \mathrm{d}V' [\boldsymbol{\nabla} \cdot (\boldsymbol{J} x_i') + \dot{\rho} x_i']. \tag{9.18}$$

Using Gauss's theorem, the first term becomes a boundary term, and we assume $\boldsymbol{J} \to 0$ as $|\boldsymbol{r}'| \to \infty$, so that

$$\int_V \mathrm{d}V' J_i = \int_V \mathrm{d}V' \dot{\rho}(\boldsymbol{r}', t_0) x_i'. \tag{9.19}$$

Combining all three components $i = x, y, z$, and pulling the time derivative out of the integral,

$$\int_V \mathrm{d}V' \boldsymbol{J}(\boldsymbol{r}', t_0) = \frac{\mathrm{d}}{\mathrm{d}t} \int_V \mathrm{d}V' \rho(\boldsymbol{r}', t_0) \boldsymbol{r}' = \frac{\mathrm{d}}{\mathrm{d}t} \boldsymbol{p}(t_0) = \dot{\boldsymbol{p}}(t_0). \tag{9.20}$$

This confirms Eq. (9.14).

Plugging Eq. (9.14) into Eq. (9.13) we find,

$$\boldsymbol{A}(\boldsymbol{r}, t) \approx \frac{\mu_0}{4\pi r} \dot{\boldsymbol{p}}\left(t - \frac{r}{c}\right). \tag{9.21}$$

This is the **electric dipole approximation**. It says that, to leading order in $\frac{d}{r}$, for any charge distribution, it's the dipole moment that generates the vector potential far away.

In-class exercise

Take a minute and discuss Eq. (9.21) with your neighbor.

From $\boldsymbol{A}$, we can now deduce the fields $\boldsymbol{E}$, $\boldsymbol{B}$, and from them get the Poynting vector and the radiated power.

9.3 Fields in the Dipole Approximation

9.3.1 Magnetic Field

Let's start with the magnetic field,

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A} = \boldsymbol{\nabla} \times \left(\frac{\mu_0}{4\pi r} \dot{\boldsymbol{p}}\left(t - \frac{r}{c}\right)\right). \tag{9.22}$$

The curl acts on the $\frac{1}{r}$ term as well as on the argument of $\dot{\boldsymbol{p}}$. We can use the product rule

$$\boldsymbol{\nabla} \times (f(\boldsymbol{r})\boldsymbol{C}(\boldsymbol{r})) = f(\boldsymbol{\nabla} \times \boldsymbol{C}) - \boldsymbol{C} \times (\boldsymbol{\nabla} f), \tag{9.23}$$

and the fact that $\boldsymbol{\nabla} r = \hat{\boldsymbol{r}}$ and $\boldsymbol{\nabla}\left(\frac{1}{r}\right) = -\frac{\boldsymbol{r}}{r^3}$, to find

$$\begin{aligned} \boldsymbol{B} &= \frac{\mu_0}{4\pi}\left[\frac{1}{r}\boldsymbol{\nabla} \times \dot{\boldsymbol{p}} - \dot{\boldsymbol{p}} \times \left(\boldsymbol{\nabla}\frac{1}{r}\right)\right] \\ &\xrightarrow{\text{chain rule first term}} = \frac{\mu_0}{4\pi}\left[\frac{1}{rc}(-\boldsymbol{\nabla} r) \times \ddot{\boldsymbol{p}} + \dot{\boldsymbol{p}} \times \frac{\boldsymbol{r}}{r^3}\right] \\ &\xrightarrow{\text{switched order in cross product in second term}} = -\frac{\mu_0}{4\pi rc}(\hat{\boldsymbol{r}} \times \ddot{\boldsymbol{p}}) - \frac{\mu_0}{4\pi r^2}(\hat{\boldsymbol{r}} \times \dot{\boldsymbol{p}}). \end{aligned} \tag{9.24}$$

Details on the calculation using Cartesian coordinates

We can do this calculation also explicitly in Cartesian coordinates, but it's much more efficient to use the Levi-Civita tensor and component notation,

$$\frac{\mu_0}{4\pi}\begin{pmatrix}\partial_x\\ \partial_y\\ \partial_z\end{pmatrix} \times \frac{1}{r}\begin{pmatrix}\dot{p}_x\\ \dot{p}_y\\ \dot{p}_z\end{pmatrix} = \frac{\mu_0}{4\pi}\sum_{i,j,k}\varepsilon_{ijk}\partial_i\left(\frac{\dot{p}_j}{r}\right)\hat{\boldsymbol{x}}_k. \tag{9.25}$$

Let's calculate $\partial_i\left(\frac{1}{r}\right)$, $i = 1, 2, 3$:

$$\partial_i \frac{1}{r} = \partial_i \frac{1}{\sqrt{x^2 + y^2 + z^2}} = -\frac{1}{2}\frac{1}{r^3}2x_i = -\frac{x_i}{r^3}, \tag{9.26}$$

and, using the chain rule on the retarded-time argument,

$$\partial_i \dot{p}_j = \ddot{p}_j \partial_i\left(t - \frac{r}{c}\right) = \ddot{p}_j\left(-\frac{1}{c}\right)\frac{1}{2}\frac{1}{r}2x_i = -\ddot{p}_j\frac{x_i}{cr}. \tag{9.27}$$

So that

$$\begin{aligned} \frac{\mu_0}{4\pi}\left(\boldsymbol{\nabla} \times \frac{\dot{\boldsymbol{p}}}{r}\right)_k &= -\frac{\mu_0}{4\pi}\sum_{i,j}\varepsilon_{ijk}\left(x_i\frac{\dot{p}_j}{r^3} + x_i\frac{\ddot{p}_j}{r}\right) \\ &= -\frac{\mu_0}{4\pi}\left[\frac{1}{r}\left(\frac{\hat{\boldsymbol{r}}}{r} \times \dot{\boldsymbol{p}}\right)_k + (\hat{\boldsymbol{r}} \times \ddot{\boldsymbol{p}})_k\right], \end{aligned} \tag{9.28}$$

using that $\sum_{i,j}\varepsilon_{ijk}x_i\dot{p}_j = (\boldsymbol{r} \times \dot{\boldsymbol{p}})_k$ (and similarly for the second term). So,

$$\boldsymbol{\nabla} \times \dot{\boldsymbol{p}} = -\frac{1}{r^2}(\hat{\boldsymbol{r}} \times \dot{\boldsymbol{p}}) - \frac{1}{rc}(\hat{\boldsymbol{r}} \times \ddot{\boldsymbol{p}}). \tag{9.29}$$

Collecting results,

$$\boldsymbol{B}(\boldsymbol{r},t) = -\frac{\mu_0}{4\pi rc}(\hat{\boldsymbol{r}} \times \ddot{\boldsymbol{p}}) - \frac{\mu_0}{4\pi r^2}(\hat{\boldsymbol{r}} \times \dot{\boldsymbol{p}}), \tag{9.30}$$

with $\dot{\boldsymbol{p}}$ and $\ddot{\boldsymbol{p}}$ evaluated at $t - \frac{r}{c}$.

In-class exercise

From the two terms in $\boldsymbol{B}$ (Eq. (9.30)), which one do you think dominates far away?

Far away from the source, we expect it to be the first one, since the second one drops off as $\frac{1}{r^2}$. We can make this statement more precise if we assume that the dipole oscillates in time at angular frequency ω. Then, $\ddot{\boldsymbol{p}} = \omega \cdot (\boldsymbol{p})$ and we expect the dipole to generate EM waves at wavelength $\lambda \sim \frac{c}{\omega}$. As long as $r \gg \lambda$, the second term indeed dominates,

$$\boldsymbol{B}(\boldsymbol{r},t) \approx -\frac{\mu_0}{4\pi rc}\left[\hat{\boldsymbol{r}} \times \ddot{\boldsymbol{p}}\left(t - \frac{r}{c}\right)\right]. \tag{9.31}$$

The region $r \gg \lambda$ is called the **far-field zone**.

Hierarchy of scales

$$r \gg \lambda \gg d. \tag{9.32}$$

9.3.2 Electric Field

To get the $\boldsymbol{E}$ field, either you start from the potentials, or you use Maxwell's equation (iv) (Eq. (1.22)), with $\mu_0\varepsilon_0 = \frac{1}{c^2}$ and $\boldsymbol{J} = \boldsymbol{0}$ outside the source volume V:

$$\dot{\boldsymbol{E}} = c^2(\boldsymbol{\nabla} \times \boldsymbol{B}). \tag{9.33}$$

$\boldsymbol{\nabla} \times \boldsymbol{B}$ will again have two contributions, one from $\frac{1}{r}$ and the other from the argument of $\ddot{\boldsymbol{p}}$. Again, using the far-field approximation, the curl will be dominated by terms where $\boldsymbol{\nabla}$ acts on $\ddot{\boldsymbol{p}}$ (i.e. gives a $\dddot{\boldsymbol{p}}$ term with a single power of $\frac{1}{r}$, rather than a $\frac{1}{r^2}$ term). The result is

$$\dot{\boldsymbol{E}}(\boldsymbol{r},t) \approx \frac{\mu_0}{4\pi rc^2}\hat{\boldsymbol{r}} \times \left(\hat{\boldsymbol{r}} \times \dddot{\boldsymbol{p}}\left(t - \frac{r}{c}\right)\right), \tag{9.34}$$

so that, integrating in time (equivalently, this is just differentiating Eq. (9.31) once more and using $\boldsymbol{\nabla} \times \boldsymbol{B} = \frac{\dot{\boldsymbol{E}}}{c^2}$),

$$\boldsymbol{E}(\boldsymbol{r},t) \approx \frac{\mu_0}{4\pi r}\hat{\boldsymbol{r}} \times \left(\hat{\boldsymbol{r}} \times \ddot{\boldsymbol{p}}\left(t - \frac{r}{c}\right)\right) = -c(\hat{\boldsymbol{r}} \times \boldsymbol{B}), \tag{9.35}$$

which is the same relation as for plane waves (note that this only holds in the far field!).

An oscillating dipole emits **spherical waves**, and the fields fall off as $1/r$ (compare to $1/r^2$ for the $\boldsymbol{E}$ field of a static point charge). In spherical coordinates, with $\ddot{\boldsymbol{p}}$ pointing along the $\hat{\boldsymbol{z}}$ direction, the fields are

$$\boldsymbol{E}(r,\theta,t) = \frac{\mu_0}{4\pi}\ddot{p}(t_0)\frac{\sin\theta}{r}\hat{\boldsymbol{\theta}}, \quad \boldsymbol{B}(r,\theta,t) = \frac{\mu_0}{4\pi c}\ddot{p}(t_0)\frac{\sin\theta}{r}\hat{\boldsymbol{\phi}}, \tag{9.36}$$

with $t_0 = t - \frac{r}{c}$.

9.4 The Larmor Formula

Given the fields, we can now finally come back to calculate the radiated power from a source, passing through a surface.

The Poynting vector is (guess its direction first!)

$$\boldsymbol{S} = \frac{1}{\mu_0}\boldsymbol{E}\times\boldsymbol{B} = -\frac{c}{\mu_0}(\hat{\boldsymbol{r}}\times\boldsymbol{B})\times\boldsymbol{B}. \tag{9.37}$$

Using the BAC–CAB rule, $\boldsymbol{A}\times(\boldsymbol{B}\times\boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A}\cdot\boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A}\cdot\boldsymbol{B})$,

$$\boldsymbol{S} = \frac{c}{\mu_0}\boldsymbol{B}\times(\hat{\boldsymbol{r}}\times\boldsymbol{B}) = \frac{c}{\mu_0}[\hat{\boldsymbol{r}}(\boldsymbol{B}\cdot\boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{B}\cdot\hat{\boldsymbol{r}})] = \frac{c}{\mu_0}|\boldsymbol{B}|^2\hat{\boldsymbol{r}}, \tag{9.38}$$

switched order of the cross product, gives a minus sign

using $\boldsymbol{B}\cdot\hat{\boldsymbol{r}} = 0$, since $\boldsymbol{B}\propto\hat{\boldsymbol{r}}\times\ddot{\boldsymbol{p}}$, so it's perpendicular to $\hat{\boldsymbol{r}}$. Plugging in Eq. (9.31),

$$\boldsymbol{S} = \frac{\mu_0}{16\pi^2 r^2 c}|\hat{\boldsymbol{r}}\times\ddot{\boldsymbol{p}}|^2\hat{\boldsymbol{r}}. \tag{9.39}$$

In spherical coordinates, with $\ddot{\boldsymbol{p}}$ along $\hat{\boldsymbol{z}}$,

$$\boldsymbol{S}(\boldsymbol{r},t) = \frac{\mu_0}{16\pi^2 c}\ddot{p}^2\frac{\sin^2\theta}{r^2}\hat{\boldsymbol{r}}. \tag{9.40}$$

Some observations:

1. The power is emitted **radially**.
2. The emitted power falls off as $1/r^2$, as it should, to ensure energy conservation.
3. Although the power is emitted radially, it's not uniform, due to the $\sin^2\theta$ dependence in Eq. (9.40). Power is largest $\perp$ to the dipole $\boldsymbol{p}$, and 0 parallel to $\boldsymbol{p}$.

In-class exercise

Discuss: why does the $1/r^2$ drop-off of $\boldsymbol{S}$ ensure energy conservation?

The total radiated power is found by integrating over a sphere,

$$P = \oint \boldsymbol{S}\cdot\mathrm{d}\boldsymbol{a}, \tag{9.41}$$

(shown in the exercises — note that $\mathrm{d}\boldsymbol{a}$ is also in the $\hat{\boldsymbol{r}}$ direction), which gives

$$P(t) = \frac{\mu_0}{6\pi c}|\ddot{\boldsymbol{p}}|^2, \tag{9.42}$$

(note that this is still time-dependent).

For a point charge, the dipole moment is given by $\boldsymbol{p}(t) = q\boldsymbol{d}(t)$, where $\boldsymbol{d}$ is the distance to the origin. Then

$$\ddot{\boldsymbol{p}}(t) = q\ddot{\boldsymbol{d}}(t) = q\boldsymbol{a}(t), \tag{9.43}$$

where $\boldsymbol{a}$ is the acceleration vector. Thus, the radiated power is (using $\mu_0 = 1/(\varepsilon_0 c^2)$)

$$P = q^2 \frac{|\boldsymbol{a}|^2}{6\pi\varepsilon_0 c^3}. \tag{9.44}$$

This is the famous **Larmor formula** for the radiated power of an accelerated point charge. It's also possible to arrive at this formula by starting from the Liénard–Wiechert potentials (Eq. (8.44)), deriving the $\boldsymbol{E}$ and $\boldsymbol{B}$ fields from them, and calculating $\boldsymbol{S}$ (again dropping higher-order terms in d and using the far-field approximation).

Time-averaged radiated power (shown in the exercises)

If the dipole is oscillating in time, i.e. $\boldsymbol{p} = \boldsymbol{p}_0 \cos(\omega t)$, then $|\ddot{\boldsymbol{p}}|^2 = |\boldsymbol{p}_0|^2 \omega^4 \cos^2(\omega t)$, and it is common practice to average over one period $T = 2\pi/\omega$, which gives another factor of $\frac{1}{2}$. In this case, the average radiated power reads

$$\langle P \rangle = \frac{\mu_0}{12\pi c} |\boldsymbol{p}_0|^2 \omega^4. \tag{9.45}$$

You will verify this in the exercises.